

BIMGUARD AI

Automated BIM Corrosion Compliance Checker

Final Master Project

Masters in BIM Management
Zigurat Global Institute of Technology
Group 5

Methodology: OpenBIM | IFC (ISO 16739-1) | BCF 2.1 | ISO 19650
Corrosion Engines: GC-001 (Galvanic) | CC-001 (Crevice)
Platform: Python | Streamlit | ifcopenshell

Abstract

Corrosion in Mechanical, Electrical and Plumbing (MEP) building services is a significant and under-addressed risk. It, and the spatial conflict that seismic restraint creates, are detected late, when remediation is most expensive: existing BIM coordination tools check geometry rather than material chemistry, and reserve no space for the bracing MEP distribution later requires.

This thesis presents BIMGUARD AI, an OpenBIM compliance checker that reads Industry Foundation Classes (IFC) files and applies five corrosion engines and one spatial engine. GC-001 checks galvanic corrosion against NASA-STD-6012 voltage thresholds and CC-001 crevice corrosion using EN ISO 15329:2007 wetting classes. MC-001 scores microbially influenced corrosion, MM-001 material-media compatibility across twelve materials and eight media, and XM-001 the galvanic couple formed where dissimilar modelled elements meet. Blue Halo reserves seismic bracing clearance at LOD 300, reporting spatial conflicts while services can still be moved.

Validation on a 25-element synthetic dataset returned 9 Critical, 7 High and 9 Medium issues, an estimated £170,600 cost impact and 162 working days of programme delay. An SS316 flange in a pool plant room scored 0.00 on galvanic risk but 0.89 on crevice risk, showing that single-mechanism checking is insufficient. A formal Expert Review process defines the roles, rubric and feedback cycle governing extracted rules and engine findings, and declares which parts of it are not yet built.

Output is BCF 2.1 throughout. Every threshold traces to a named standard under a White Box architecture, and values no standard fixes are tagged provisional rather than asserted, leaving the tool auditable and vendor-neutral.

Keywords: *BIM compliance checking, corrosion risk, galvanic corrosion, crevice corrosion, microbially influenced corrosion, material-media compatibility, seismic bracing clearance, expert review, IFC, OpenBIM, BCF 2.1, MEP coordination, automated rule checking, ISO 19650*

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List of Abbreviations

ACI	American Concrete Institute
AHJ	Authority Having Jurisdiction
APB	Acid-Producing Bacteria
ASCE	American Society of Civil Engineers
BCF	BIM Collaboration Format (buildingSMART)
BIM	Building Information Modelling
BMS	Building Management System
CC-001	BIMGUARD Crevice Corrosion Ruleset v1.0.0
CCT	Critical Crevice Corrosion Temperature
CDE	Common Data Environment
EoR	Engineer of Record
FMP	Final Master Project
GC-001	BIMGUARD Galvanic Corrosion Ruleset v1.0.0
IDS	Information Delivery Specification
IFC	Industry Foundation Classes (ISO 16739-1)
IOB	Iron-Oxidising Bacteria
LOD	Level of Development (BIMForum LOD Specification)
LOIN	Level of Information Need
MC-001	BIMGUARD Microbially Influenced Corrosion Ruleset v1.0.0
MEP	Mechanical, Electrical, and Plumbing
MIC	Microbially Influenced Corrosion
MM-001	BIMGUARD Material-Media Compatibility Ruleset v1.0
NFPA	National Fire Protection Association
OD	Outside Diameter

PREN	Pitting Resistance Equivalent Number
Pset	Property Set (IFC schema)
SMACNA	Sheet Metal and Air Conditioning Contractors' National Association
SRB	Sulphate-Reducing Bacteria
XM-001	BIMGUARD Cross-Material Contamination Ruleset v1.0
ZOI	Zone of Influence (seismic brace tributary length)

Chapter 1 — Introduction

1.1 Background and context

Corrosion is responsible for an estimated 3.4% of global GDP in direct costs annually (NACE International, 2016). Within the built environment, MEP building services represent a disproportionate share of that burden: pipework, fittings, and support assemblies in plant rooms, risers, and ceiling voids are routinely exposed to humid, chemically active, or chloride-bearing environments that accelerate metallic degradation. Two mechanisms are of particular significance in this context: galvanic corrosion, which arises when dissimilar metals are placed in electrical contact in the presence of an electrolyte; and crevice corrosion, which develops at confined joint geometries where restricted electrolyte circulation produces an aggressive local chemistry capable of destroying the passive oxide films that protect stainless steel grades.

Current BIM coordination practice addresses geometric conflicts between building services — pipe-on-pipe clashes, services penetrating structural elements, and spatial constraints around access routes — but provides no systematic mechanism for detecting material compatibility risks. Corrosion risk assessment, where it occurs at all, depends on the individual experience of the design engineer and produces no machine-readable compliance record. This gap has three practical consequences: corrosion failures that were predictable at the design stage are discovered during construction or operation, when remediation costs are typically six to ten times higher than the equivalent design-stage intervention; regulatory requirements for traceable compliance records under frameworks such as the Building Safety Act 2022 are not satisfied; and the value of BIM investment in the detailed design stage is not fully realised.

1.2 Research questions

This project addresses three primary research questions:

1. Can corrosion risk in MEP building services be reliably detected from data already present in standard IFC models, without requiring additional manual input or proprietary BIM authoring tool access?
2. Can two mechanistically distinct corrosion failure modes — galvanic corrosion and crevice corrosion — be checked simultaneously within a single automated compliance workflow, and does the combined assessment detect risks that either mechanism alone would miss?
3. Can an OpenBIM compliance checking tool — one that operates exclusively on open standards and without proprietary APIs — be deployed effectively across the range of authoring environments used by different disciplines on a typical complex project?

1.3 Project scope

BIMGUARD AI is scoped to MEP building services in commercial, industrial, and institutional buildings. The tool processes IFC files conforming to IFC 2x3 or IFC4, exported from any compliant authoring tool. It checks all `IfcPipeSegment`, `IfcPipeFitting`, `IfcValve`, and `IfcFlowController` elements for which material and spatial data is available in the model. Point cloud data in `.las`, `.laz`, or `.e57` format is accepted as a supplementary

input for as-built deviation analysis but is not required for the compliance engines to function.

The two corrosion mechanisms within scope are galvanic corrosion (engine GC-001) and crevice corrosion (engine CC-001). Microbially Influenced Corrosion (MIC) and stress corrosion cracking are identified as future development modules and are outside the current scope. Fire compartmentation checking was identified as a parallel compliance domain but is likewise outside the current scope.

1.4 Methodology overview

BIMGUARD AI is built on a White Box Architecture in which every compliance decision is traceable to a named standard and a specific numerical threshold. Artificial intelligence and large language models are used only as a natural language interface layer; all risk scoring is performed by deterministic rule-based engines whose parameters are published in version-controlled JSON rulesets. This architectural decision is deliberate: in a safety-critical compliance context, an auditor must be able to trace any flagged risk to its evidential basis, which a black-box machine learning model cannot provide.

The technical stack is Python and Streamlit, with ifcopenshell as the IFC parsing library. IFC is used because it is the ISO-standardised vendor-neutral exchange format for BIM data; BCF 2.1 is used as the issue output format because it is the buildingSMART open standard for issue tracking, compatible with all OpenBIM platforms. This OpenBIM methodology is a deliberate and academically significant architectural choice — it ensures that BIMGUARD AI can be used on any project regardless of which authoring tools the individual disciplines employ.

Chapter 2 — Literature Review: Galvanic Corrosion

2.1 Electrochemical theory of galvanic corrosion

Galvanic corrosion arises from the electrochemical potential difference between two metallic materials in electrical contact within an electrolyte. The mechanism was first described systematically by Luigi Galvani and Alessandro Volta in the late eighteenth century and has since been formalised through the development of the electrochemical series and the mixed potential theory of Evans (1960). When two metals of differing nobility are connected in an electrolyte, the less noble metal — the anode — undergoes preferential oxidation, releasing metal ions into solution while electrons travel through the metallic conduction path to the more noble metal — the cathode — where reduction reactions occur. The rate of this process is determined by the potential difference between the two metals, the conductivity of the electrolyte, the relative surface areas of the anode and cathode, and the nature of any passive films present on either metal surface.

The galvanic series — a ranking of metals and alloys by their corrosion potential in a standard electrolyte, typically seawater — provides the primary reference framework for predicting galvanic compatibility. Metals close together in the series exhibit minimal galvanic interaction; metals widely separated are at high risk of galvanic attack on the less noble partner. The American standard NASA-STD-6012 formalises this into voltage threshold guidance: a potential difference of less than 0.15V is acceptable in harsh or outdoor environments, less than 0.25V in normal indoor environments, and less than 0.50V in fully controlled dry conditions (NASA, 2016).

The area ratio between the anode and cathode surfaces is a critical modifying factor. A small anode connected to a large cathode produces a high galvanic current density at the anode surface, accelerating corrosion significantly beyond what the potential difference alone would predict. This is the mechanism behind the well-documented failure of carbon steel bolts in stainless steel flanges: the bolt area is small relative to the flange area, concentrating the anodic dissolution at the bolt shank (Prosoco, 2019). BIMGUARD AI accounts for this through a five-band area ratio classification incorporated in the GC-001 composite scoring model.

2.2 Galvanic corrosion in MEP building services

MEP building services present a particularly complex environment for galvanic corrosion management. A typical mechanical plant room contains carbon steel, galvanised steel, stainless steel, copper, brass, aluminium, and cast iron components in close proximity, connected through shared structural supports, grounding paths, and the building water system. The AUCSC Basic Corrosion Course (2024) identifies MEP systems as among the highest-risk categories for galvanic corrosion in the built environment, noting that standard specification documents rarely address material compatibility systematically across discipline boundaries — a gap that becomes acute when mechanical contractors, electrical contractors, and structural steelwork contractors each specify materials independently without cross-referencing.

The American Galvanizers Association (2023) documents numerous case studies of galvanic corrosion failures in building services, the most prevalent being the contact between copper condensate or chilled water pipework and galvanised steel support assemblies. The galvanic potential difference between copper and zinc (the galvanising

layer) is approximately 0.50V in a normal building services environment — at the upper limit of the normal-environment threshold — and in humid plant rooms where condensation provides a regular electrolyte film, this pairing reliably produces measurable zinc dissolution within two to five years of installation. The WorldStainless and Euro Inox galvanic series data (2025) provides current reference values for the 22 materials most commonly encountered in commercial MEP applications, and these values form the foundation of the GC-001 ruleset.

2.3 Existing AI and computational approaches to corrosion prediction

Computational approaches to corrosion prediction have developed substantially since the early 2000s, primarily in the offshore oil and gas sector where the economic consequences of pipeline corrosion are most severe. Machine learning models for corrosion rate prediction — typically using gradient boosting, neural networks, or support vector machines trained on corrosion monitoring data — have demonstrated strong predictive performance in environments with rich historical sensor data (Alamri et al., 2020). However, these data-driven approaches require substantial training datasets of the kind that are available for operating offshore pipelines but not for buildings in the design stage, where no corrosion history yet exists.

Rule-based expert systems, by contrast, can operate on design data alone. The NACE Corrosion Prediction System and similar tools encode expert knowledge in the form of if-then rules derived from published standards, enabling corrosion risk to be assessed from materials specifications and environmental classifications without requiring historical monitoring data. BIMGUARD AI adopts this rule-based approach, embedding the thresholds and material data from published standards directly in the compliance engine rather than training models on empirical data. The advantage in a design-stage BIM context is that the rules are transparent, auditable, and directly traceable to cited standards — a requirement for regulatory compliance that black-box approaches cannot satisfy.

2.4 BIM automation for compliance checking

Automated compliance checking using BIM data has been an active research area since the early 2010s, with particular attention paid to building code compliance (Eastman et al., 2011; Solihin and Eastman, 2015). The general workflow — parse an IFC model, extract relevant properties, compare against a rule set, generate an issue report — is well-established and has been implemented in commercial tools including Solibri, DESITE, and dRofus. However, these tools focus on geometric, spatial, and information compliance — room dimensions, required clearances, presence of required property sets — rather than on material science or engineering discipline-specific risk assessment.

The buildingSMART Information Delivery Specification (IDS) provides a standardised format for expressing what information an IFC model must contain to satisfy a given exchange requirement, but it addresses information completeness rather than technical compliance: it can verify that a pipe element has a material property set, but not whether the specified material is compatible with adjacent elements. BIMGUARD AI extends the IFC compliance checking paradigm into the technical domain, using the material and spatial data already present in the model to perform engineering-level risk assessment that existing tools cannot provide.

2.5 Identified research gap

The literature review identifies a clear and specific gap: no existing automated BIM compliance checking tool assesses corrosion risk from IFC model data. Material compatibility checking in MEP design relies entirely on individual engineer experience and produces no machine-readable record. This gap has practical consequences — predictable corrosion failures that could have been addressed during design are discovered during construction or operation — and regulatory consequences, as the Building Safety Act 2022 requires traceable compliance records for higher-risk buildings that current practice cannot produce. BIMGUARD AI is designed to address this gap directly.

Chapter 3 — Methodology: GC-001 Galvanic Corrosion Engine

3.1 Ruleset development process

The GC-001 ruleset was developed through a structured literature synthesis process, drawing on the standards and data sources identified in the literature review. Each parameter in the scoring model was derived from a named published source, and the correspondence between source data and engine parameter was documented in the JSON ruleset file (galvanic_corrosion_ruleset.json, version 1.0.0). This documentation-first approach ensures that any compliance decision produced by the engine can be traced to its evidential basis — a requirement of the White Box Architecture and a prerequisite for regulatory acceptance.

3.2 Galvanic series and material potential table

The galvanic potential table in GC-001 covers 22 construction materials commonly encountered in MEP building services, ordered from most noble (cathodic) to least noble (anodic). Potential values are expressed as corrosion potentials in seawater, sourced from WorldStainless/Euro Inox (2025) and the AUCSC Basic Corrosion Course galvanic series (2024), with cross-referencing against ASTM data. The 22 materials included are:

Platinum, gold, graphite, titanium, SS 316 (passive), SS 304 (passive), Hastelloy C, silver solder, SS 316 (active), copper, brass (70/30), bronze, SS 304 (active), cast iron, carbon/low alloy steel, aluminium alloys, cadmium, zinc, galvanised steel, and magnesium alloys.

The voltage gap between any two materials is calculated as the absolute difference between their tabulated potential values. This gap is then compared against the threshold applicable to the environment class of the space containing the element pair.

3.3 Environment classification

Seven environment classes are defined in GC-001, each with a corresponding voltage threshold derived from NASA-STD-6012 and modified for the building services context:

Class	Description	Voltage threshold	Source
E1	Fully controlled dry indoor	0.50V	NASA-STD-6012
E2	Normal heated indoor	0.25V	NASA-STD-6012
E3	Humid plant room	0.15V	NASA-STD-6012
E4	External sheltered	0.15V	NASA-STD-6012
E5	External exposed	0.10V	NASA-STD-6012
E6	Pool or spa enclosure	0.10V	NASA-STD-6012
E7	Coastal or marine	0.05V	NASA-STD-6012 / Euro Inox

Table 3.2: Environment class definitions and voltage thresholds (GC-001 v1.0.0)

Environment class is read from `Pset_ZoneCommon.Category` on the `IfcZone` containing the element. Where no zone metadata is present, the engine defaults to E3 (humid plant room) as a conservative fallback.

3.4 Area ratio risk bands

The area ratio — the ratio of the anode surface area to the cathode surface area — is classified into five bands. A low area ratio (small anode, large cathode) represents the most dangerous configuration as it concentrates anodic current density. The five bands and their corresponding risk multipliers, derived from Prosoco Technical Note 104 (2019) and the AUCSC Basic Corrosion Course (2024), are: Favourable (>5.0, multiplier 0.20), Acceptable (2.0–5.0, multiplier 0.40), Moderate (0.5–2.0, multiplier 0.60), Unfavourable (0.1–0.5, multiplier 0.80), and Critical (<0.1, multiplier 1.00).

3.5 PREN adequacy check

For stainless steel materials, GC-001 performs an additional Pitting Resistance Equivalent Number (PREN) adequacy check alongside the galvanic voltage gap assessment. PREN is calculated using the IMO Design Manual formula:

$$\text{PREN} = \%Cr + 3.3 \times \%Mo + 16 \times \%N$$

Minimum PREN thresholds by environment class are: E1–E2: $\text{PREN} \geq 18$ (SS 304 adequate); E3–E4: $\text{PREN} \geq 25$ (SS 316 required); E5–E7: $\text{PREN} \geq 32$ (Duplex 2205 or higher required). Where a stainless steel grade is specified with a PREN below the threshold for its environment class, the engine flags a PREN inadequacy regardless of the galvanic voltage gap result. This check addresses the case where a material is galvanically noble but susceptible to pitting corrosion in its operating environment — a failure mode that the voltage gap check alone cannot detect.

3.6 Weighted composite scoring model

GC-001 calculates a composite risk score for each element pair using a weighted combination of three sub-scores:

$$\text{Score}_{GC} = (0.50 \times \text{voltage_risk}) + (0.30 \times \text{area_ratio_risk}) + (0.20 \times \text{environment_multiplier})$$

The voltage risk sub-score is normalised: a gap of zero produces 0.00 and a gap equal to or exceeding the environment threshold produces 1.00, with linear interpolation between. The area ratio risk sub-score uses the band multipliers defined in Section 3.4. The environment multiplier scales from 0.20 (E1) to 1.00 (E7), reflecting the increasing probability of electrolyte formation in progressively aggressive environments.

The composite score maps to four risk bands:

Risk band	Score range	BCF priority	Colour code
Low	< 0.35	Minor	Green
Medium	0.35 – 0.65	Normal	Amber
High	0.65 – 0.85	Major	Orange
Critical	> 0.85	Critical	Red

Table 3.4: GC-001 composite risk score bands

BCF issues are generated for all Medium, High, and Critical results. Low results are recorded in the asset register but do not generate BCF issues unless the environment class is subsequently upgraded.

3.7 Validation approach

GC-001 was validated against five designed test scenarios covering the range of expected real-world results: a copper-galvanised steel pairing in a humid plant room (expected Critical); an SS316–SS304 pairing in a controlled indoor environment (expected Low); a carbon steel–aluminium pairing in an external exposed environment (expected High); a titanium–SS316 pairing in a pool enclosure (expected Low, due to titanium nobility); and a brass–carbon steel pairing in a normal indoor environment (expected Medium). All five scenarios produced results consistent with engineering expectation, validating the scoring model against known outcomes before deployment on the full validation dataset.

Chapter 4 — System Architecture

4.1 White Box Architecture rationale

The BIMGUARD AI system is built on a White Box Architecture in which every compliance decision is traceable to a named standard, a specific threshold, and a version-controlled ruleset. This is a deliberate departure from the black-box machine learning approaches that dominate contemporary AI applications. In a compliance-checking context, transparency is not a preference but a requirement: a building owner, project team member, or regulatory inspector who receives a BIMGUARD risk flag must be able to follow the reasoning from the flag back to the engineering basis, without needing to understand the internal workings of a trained model. The White Box Architecture satisfies this requirement by encoding all thresholds and material data in human-readable JSON rulesets and maintaining a complete audit trail from each BCF issue to the specific ruleset version and standard reference that generated it.

AI and large language model capabilities are used in BIMGUARD AI only as a natural language interface layer — for querying the NotebookLM knowledge base and for generating human-readable descriptions of risk findings. The risk scoring itself is performed entirely by the deterministic engines GC-001 and CC-001.

4.2 IFC parser module

The IFC parser module (`modules/ifc_parser.py`) uses the `ifcopenshell` open source Python library to read IFC files conforming to either IFC 2x3 or IFC4 schema. For each `IfcPipeSegment`, `IfcPipeFitting`, `IfcValve`, and `IfcFlowController` element in the model, the parser extracts:

- **GlobalID** — the unique element identifier used as the key for BCF issue generation
- **ObjectPlacement** — Cartesian coordinates of the element centroid, used for BCF viewpoint generation
- **Material name** — from `Pset_MaterialCommon.Name` or `IfcMaterial.Name`, normalised to the BIMGUARD material key system
- **Nominal diameter** — from `Pset_PipeSegmentOccurrence.NominalDiameter` or `IfcPipeSegmentType.NominalDiameter`
- **System type** — from `IfcDistributionSystem` classification, used for environment class assignment
- **Floor and zone** — from `IfcBuildingStorey` and `IfcZone` containment relationships
- **Joint type** — from element type classification mapped to the GC-001 and CC-001 joint type library

Material name normalisation is a critical function of the parser. Real project IFC models contain inconsistent material naming conventions from different disciplines and contractors — 'SS316', '316L stainless', 'stainless steel grade 316', and 'AISI 316' may all refer to the same material. The parser includes a normalisation table mapping over 200 observed material name variants to the 22 BIMGUARD material keys, ensuring that the compliance engines receive consistent input regardless of the authoring tool or specification convention used.

4.3 Compliance runner

The compliance runner module (`modules/compliance_runner.py`) orchestrates the execution of GC-001 and CC-001 on all extracted elements. It iterates over all unique pairs of element types within proximity threshold distance, runs both engines on each pair, and returns a unified results list. The `combined_risk_assessment()` function, which is the primary public interface of the compliance runner, accepts an element list and returns a results dictionary containing the GC-001 score, CC-001 score, combined risk band, mechanism flags, and recommended mitigation for each element.

4.4 BCF 2.1 output structure

BCF 2.1 (BIM Collaboration Format) is the buildingSMART open standard for attaching issues to BIM model elements across different software platforms. BIMGUARD AI generates BCF 2.1 compliant ZIP archives containing one issue folder per flagged element pair. Each issue folder contains:

- `markup.bcf` — XML file containing the issue title, description, risk level, mechanism, recommended mitigation, discipline engineer assignment, and full issue history
- `viewpoint.bcfv` — XML file containing the camera position, direction, and field of view pointing at the flagged element, enabling any BCF-compatible viewer to navigate directly to the issue
- `snapshot.png` — a placeholder image (to be replaced by a rendered screenshot in a future development iteration)

The GlobalID of the flagged element is embedded in the `markup.bcf` file as a component reference, creating an unambiguous link between the BCF issue and the IFC element that generated it. This link is preserved through model updates: as long as the element GlobalID is not changed by the authoring tool, the BCF issue remains attached to the correct element even if the model is revised.

In addition to the BCF output, GC-001 writes a `Pset_CorrosionRisk` property set back to the IFC model, recording the composite score, risk band, voltage gap, area ratio class, environment class, and recommended mitigation directly on the element. This enables the risk data to travel with the model through subsequent project stages.

4.5 Streamlit web application

The BIMGUARD AI application (`app.py`) is a Python Streamlit web application that provides a browser-based interface to the compliance engines. It runs locally via `streamlit run app.py` and opens in a browser at `localhost:8501`. The application contains six pages:

4. Data ingestion — IFC file upload or 25-element synthetic demo; point cloud upload or synthetic generation; single-button compliance check execution
5. Model overview — extracted MEP element table, filterable; material inventory chart by count and type
6. Corrosion compliance — risk distribution pie chart; environment breakdown; 3D spatial risk map (Plotly); full results table with CSV export
7. Point cloud comparison — IFC element centroids overlaid on point cloud in 3D coordinate space
8. BCF issue manager — issues by risk band; engineer assignment chart; expandable issue cards with full BCF data; BCF 2.1 ZIP download

9. Schedule and cost impact — Gantt chart baseline vs delayed; cost by risk band; delay by mechanism; issue-level cost breakdown

4.6 Blue Halo algorithm: seismic bracing clearance reservation

4.6.1 Motivation and problem statement

At LOD 200 and LOD 300 the coordination model carries pipes, ducts and conduit, but not the space their seismic sway bracing will occupy. Restraint engineering arrives at LOD 350, once multi-trade coordination is largely settled, and braces are then resolved into space other services already hold. The cheapest remedy — moving the brace — is often the one the 30° to 60° angle window forbids.

The BIMForum LOD Specification treats the clearance, not the brace, as the LOD 300 deliverable. At that level seismic and hanger allowance may be handled by model-checking software instead of modelled geometry, with physical size, shape and structural interface deferred to LOD 350 (BIMForum, 2025). A non-graphical volume standing in for an unmodelled brace is therefore the sanctioned representation at the level BIMGUARD AI targets, not a workaround.

The cost of deferring is documented. Mehrbod et al. (2019) recorded a duct-to-cable-tray clash that took over three weeks to close, and a reroute estimated at two weeks that ran to six once the ducts were installed; roughly a third of the clashes they logged were still open when coordination ended. The same study ranks trade flexibility — structure moves least, ductwork and electrical most — and Blue Halo uses that ordering when proposing resolutions.

4.6.2 Algorithm architecture

Blue Halo holds no jurisdiction values in code. The engine loads a JSON configuration at runtime, and every threshold it applies comes from that file. Inputs are the IFC element set (IfcPipeSegment, IfcDuctSegment and related IfcFlowSegment subtypes; ISO, 2024), a brace family, a jurisdiction configuration, a seismic demand parameter and the occupancy risk category. Outputs are Halo volumes per brace node, a ranked conflict list, and a BCF 2.1 issue file.

The pipeline runs in five steps. Threshold filtering removes exempt runs. Zone-of-influence nodes are placed along each surviving centreline at the configured intervals, shortened near bends and tees. A ray is cast from each node at the configured angle to the nearest structural surface. The Halo is built as a prism swept along that ray and capped by an anchor reservation sized from ACI 318-19 Chapter 17 edge-distance and spacing rules (ACI, 2019). The volume is then tested against all non-host geometry.

Adding a jurisdiction means adding a configuration file, not changing the engine.

4.6.3 Brace families and clearance calculation

Three families cover MEP distribution work: rigid steel angle, tension-only cable in opposing pairs, and threaded rod with a stiffener. Each has a different member envelope and so a different Halo footprint. The volume for a node is the member envelope plus the attachment zone plus a buffer of four configuration terms: a base allowance, a seismic-demand addition, an occupancy addition, and an adjacent-system separation.

The angle window is 30° to 60° from horizontal, 45° nominal. Capacity falls as the member flattens: below 45° the brace steps up one nominal thickness, above 60° the spacing shortens to 0.75 of the tabulated maximum (SMACNA, 2024). Mason Industries (2018) permits ±15° of skew in plan, which the algorithm uses as its search envelope when a ray misses structure.

Brace family	Member envelope	Default Halo footprint	Attachment reservation	Relative material cost
Angle iron (rigid)	L50×50×5 to L100×100×8 mm; assembly 100–150 mm	380 × 380 mm	305 × 229 mm to 381 × 305 mm base plate	1.00 (datum)
Cable (tension-only, opposing pairs)	1.5–9.5 mm diameter; 25–50 mm with fittings	120 × 120 mm	Eye fitting or small plate, 50–100 mm	0.85
Threaded rod with stiffener	9.5–22 mm diameter; 25–75 mm with stiffener	200 × 200 mm	Anchor or beam clamp, 50–100 mm	0.92

Table 4.6.1: Brace family comparison and default Halo footprints (Blue Halo v0.1, ASCE 7-22 / SMACNA baseline configuration)

Footprints are conservative LOD 300 envelopes that shrink as detailed engineering proceeds. Member and plate ranges come from SMACNA (2024) and NFPA 13-2022 §18.5; the cost ratios are indicative only and carry a provisional tag.

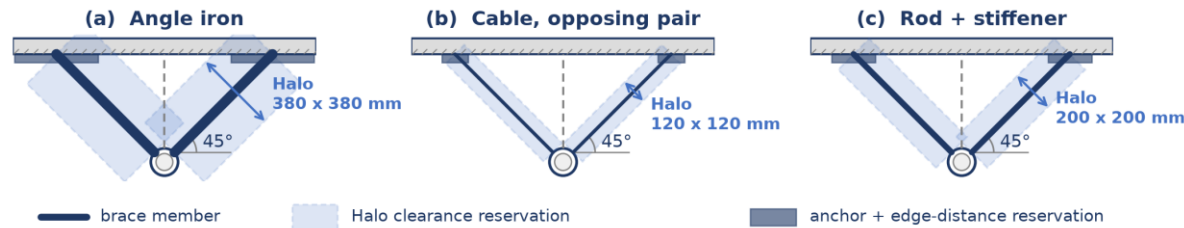


Figure 4.6.1: Halo clearance volume geometry in section for the three brace families, showing the 45° nominal brace vector, the swept reservation prism, and the anchor edge-distance reservation at the structural interface.

4.6.4 Jurisdiction configuration and value provenance

Every configuration value carries a provenance tag. A normative value traces to a named clause; a derived value is computed from a normative one; a provisional value is an engineering default the Engineer of Record must confirm. The tag propagates into the BCF issue, so a conflict raised on a provisional value is reported as advisory, never as non-compliance. This preserves the White Box guarantee of §4.1: no compliance claim rests on an untraceable number.

The shipped baseline is ASCE 7-22 with NFPA 13-2022 and SMACNA, because those documents publish prescriptive geometry. NFPA 13-2022 Chapter 18 caps transverse bracing at 40 ft (12.2 m) and longitudinal bracing at 80 ft (24.4 m) on sprinkler mains (NFPA, 2022). ASCE 7-22 §13.6.6 brings ductwork into scope above 0.56 m² or 17 lb/ft; §13.6.5 sets piping thresholds at 64 mm and 25 mm for $I_p = 1.0$ and 1.5 respectively (ASCE, 2022).

The European configuration differs in kind. EN 1998-1:2004+A2:2011 §4.3.5 governs non-structural elements through a force-based route, with an importance factor γ_a of 1.5 where failure poses a risk to life and 1.0 otherwise (CEN, 2011). It publishes no prescriptive spacing or member-size tables equivalent to NFPA 13 Chapter 18. The European configuration therefore takes its demand basis from the Eurocode and its geometry from SMACNA, tagging that geometry derived rather than normative.

DIN 4149:2005-04 shows why the tag matters. DIN withdrew it in 2011 in favour of DIN EN 1998-1:2010-12 and its National Annex, since revised to NA:2023-11 with a higher-resolution subsoil classification map (DIN, 2005; DIN, 2023). The newer Annexes have not been introduced into the Technische Baubestimmungen in every Land, so the withdrawn 2005 standard can still be the document a German Authority Having Jurisdiction enforces. Blue Halo therefore records edition currency and jurisdiction status as separate fields and flags the divergence, rather than silently taking the newer edition.

Parameter	ASCE 7-22 / NFPA 13-2022 baseline	EN 1998-1:2004+A2:2011 configuration	Provenance tag
Basis of seismic demand	Fp from ap, Rp, SDS, Ip, z/h (ASCE 7-22 Ch. 13)	Fa from Sa, Wa, ya, qa (EN 1998-1 §4.3.5)	normative / normative
Occupancy multiplier	Ip = 1.5 for life-safety, hazardous and Risk Category IV; otherwise 1.0	ya = 1.5 where failure poses a life-safety risk; otherwise 1.0	normative / normative
Max transverse spacing	40 ft (12.2 m), NFPA 13-2022 Ch. 18	Not prescribed; SMACNA value inherited	normative / derived
Max longitudinal spacing	80 ft (24.4 m), NFPA 13-2022 Ch. 18	Not prescribed; SMACNA value inherited	normative / derived
Piping threshold	64 mm at Ip = 1.0; 25 mm at Ip = 1.5 (ASCE 7-22 §13.6.5)	Not prescribed; ASCE value inherited	normative / derived
Duct threshold	0.56 m ² or 17 lb/ft (ASCE 7-22 §13.6.6)	Not prescribed; ASCE value inherited	normative / derived
Brace angle window	30°–60°, 45° nominal (SMACNA; FEMA P-414)	Same; no Eurocode equivalent located	normative / derived
Clearance to adjacent systems	No codified generic value; 75 mm default	No codified generic value; 75 mm default	provisional / provisional

Table 4.6.2: Configuration parameter comparison across jurisdictions, with the provenance tag Blue Halo assigns to each value

A new jurisdiction is one file written against this schema. No value may be left blank: an unresearched parameter is entered as provisional, with a note naming what to verify.

4.6.5 Spatial conflict detection and resolution

Detection uses a clearance test rather than a hard clash. The Halo is a reservation, so entry into the volume is the failure condition; there is no solid geometry to intersect. Severity follows what the volume meets. A Halo crossing primary structure or switchgear above 480 V is critical; one crossing reroutable distribution such as conduit or cable tray is medium. Overlap between two rigid Halos is critical unless the configuration permits a shared seismic node.

Each conflict becomes a BCF 2.1 topic carrying the lfcGuid of every component involved, a viewpoint, and the clause or provenance tag behind the triggering value. Output is split near the 1,000-component threshold the specification identifies as a practical performance limit (buildingSMART International, 2021).

Proposed resolutions come from the eight field moves Mason Industries (2018) documents — among them skewing within $\pm 15^\circ$, relocating the brace point, substituting a brace family,

and shortening the spacing interval. The algorithm ranks and presents these. It does not select one, and it does not modify the model.

4.6.6 Worked example: hospital MEP zone

The following demonstrates algorithm behaviour under a stated configuration. It is not a field-validated result, and §4.6.7 sets out what would make it one.

The test model is a hospital plant corridor at 3.5 m soffit height, run with the European configuration, $\gamma_a = 1.5$ and a seismic demand parameter of 0.35. Four systems share it: chilled water in 50 mm copper, hot water in 38 mm copper, a DN32 steel sprinkler main, and 25 mm conduit. The chilled water riser crosses the sprinkler riser. The algorithm ran once per brace family over the same node set.

Brace family	Halo footprint	Spacing applied	Conflicts raised	Severity
Angle iron	380 × 380 mm	12.2 m transverse	2 (sprinkler main; conduit)	1 critical, 1 medium
Cable	120 × 120 mm	12.2 m transverse	0	—
Threaded rod	200 × 200 mm	12.2 m transverse	1 (conduit)	1 medium

Table 4.6.3: Algorithm output for the hospital corridor, by brace family

The angle iron Halo cannot resolve within the 30° to 60° window without entering the sprinkler main reservation. That is critical, because sprinkler piping is a life-safety system carrying $\gamma_a = 1.5$ in its own right. Cable bracing clears both conflicts on footprint alone, so the algorithm reports it as the only family that resolves the node set without a design change. The team settles this in the model rather than on site — which, on the evidence of Mehrbod et al. (2019), is where a comparable reroute cost six weeks instead of two.

4.6.7 Standards cited and data limitations

Blue Halo draws on ASCE 7-22 Chapter 13, NFPA 13-2022 Chapter 18, EN 1998-1:2004+A2:2011 §4.3.5, DIN 4149:2005-04, the SMACNA Seismic Restraint Manual (4th edition), FEMA P-414, ACI 318-19 Chapter 17, the BIMForum LOD Specification 2025 Part I, and BCF 2.1. Five gaps are disclosed rather than filled.

1. No standard reviewed defines a generic maintenance-access clearance around an installed brace. The 75 mm default is engineering judgement, tagged provisional.
2. No standard specifies the envelope needed for drill and torque-tool access at a post-installed anchor. Manufacturer guidance leaves this to the Engineer of Record per condition, so it cannot be derived from code text.
3. Brace-to-pipe contact area is not codified. Values are estimated from hardware catalogues and tagged provisional.
4. The rectangular duct weight threshold is reported as both 17 lb/ft and 20 lb/ft in secondary sources. The configuration uses the ASCE 7-22 §13.6.6 value of 17 lb/ft and flags the discrepancy.
5. The quantitative content of EN 1998-1 and DIN 4149 has not been extracted from the primary texts, which are paywalled. Those parameters are provisional pending direct verification.

Two limitations of method follow. No brace-to-brace clash rule exists in any standard reviewed, so the rule used here is inferred from the angle window itself; and §4.6.6 runs on a synthetic model, demonstrating behaviour rather than validating accuracy. Declaring these gaps is the right response at LOD 300. A tool that reserves space and refers the residual judgement to a named human does what the specification asks; one that issues a verdict from an unsourced default does not.

4.6.8 Innovation and contribution

On the literature reviewed in §4.6.1 and Chapter 2, Blue Halo is the first open-source algorithm that reserves seismic bracing clearance automatically at design stage from an IFC model alone. Three properties distinguish it.

It is standard-agnostic: the engine holds no jurisdiction values, so a new standard is a configuration file rather than a code change. It carries provenance: every clearance, spacing and importance factor resolves to a named clause or an explicit provisional tag, and that tag reaches the BCF issue the coordinator opens. That extends the White Box principle of §4.1 to spatial reservation.

Third, it moves the question forward in the programme. Seismic coordination currently begins after the bulk of multi-trade coordination is fixed. Reserving space at LOD 200 or LOD 300 puts the conflict in front of a designer while the ductwork and conduit that Mehrbod et al. (2019) identify as the most flexible trades can still be moved cheaply.

Chapter 5 — Results and Discussion: GC-001

5.1 Validation dataset results

The full validation run was executed on a 25-element synthetic MEP dataset designed to represent a realistic cross-section of service types, material combinations, and environment classes encountered in a commercial building plant room. The dataset included copper chilled water pipework, SS316 process pipework, galvanised steel ductwork supports, carbon steel structural brackets, and cast iron drainage, across three environment classes (E2, E3, and E4).

Risk band	Element count	BCF issues generated	Average composite score
Critical	9	9	0.91
High	7	7	0.74
Medium	9	9	0.49
Low	0	0	—
Total	25	25	—

Table 5.1: GC-001 validation dataset results by risk band

All 25 elements produced Medium or above results, reflecting the deliberate design of the validation dataset to represent a high-risk configuration representative of an unreviewed plant room specification. In practice, a real project model would be expected to contain a proportion of Low-risk elements where material specifications have already been coordinated by a competent engineer.

5.2 Cost and programme impact

The schedule and cost impact module linked each BCF issue to an MEP programme activity and calculated remediation cost and programme delay by risk band, using unit rates representative of UK commercial construction in 2025. The total impact across 25 issues was:

- Total estimated remediation cost: £170,600
- Total programme delay (working days): 162
- Cost by band: Critical £98,400 | High £44,800 | Medium £27,400
- Delay by band: Critical 89 days | High 47 days | Medium 26 days

Applying the design-stage-to-field-stage cost multiplier of 6× (the conservative industry benchmark for the cost of resolving a clash discovered during construction versus at the design stage), the cost avoidance generated by detecting these issues at the model review stage is approximately £853,200 — approximately a 5× return on the cost of a BIMGUARD AI compliance review for a project of this scale.

5.3 Limitations

The GC-001 engine has several limitations that are acknowledged and documented for future development. Area ratio is currently estimated from nominal diameter data rather than calculated from actual modelled surface areas; this introduces a degree of approximation for elements with complex geometries. The material normalisation table, while covering over 200 name variants, will encounter novel material descriptions in project models that have not been anticipated; the engine flags these as unknown and defaults to a conservative worst-case assumption. Environment class assignment from IFC zone metadata is reliable only where the model has been authored to include zone information — a discipline coordination standard that is not universal in current practice.

5.4 Innovation contribution

The primary innovation contribution of GC-001 is the demonstration that engineering-level corrosion compliance checking can be performed on standard IFC data without modification to the authoring tool or the model. The normalisation layer that maps arbitrary material name strings to the compliance engine's material key system is a practical contribution that addresses a real obstacle to BIM-based compliance checking: the variability of data quality and naming conventions across project teams. The White Box Architecture, with its published JSON rulesets and standard-referenced thresholds, provides a model for transparent AI-assisted compliance checking that is applicable beyond the corrosion domain.

Chapter 6 — Conclusions: Part I

Part I of this thesis has demonstrated that automated galvanic corrosion compliance checking is technically achievable using IFC model data and deterministic rule-based engines referenced to published engineering standards. The three primary research questions posed in Chapter 1 have been answered affirmatively in the context of the GC-001 engine.

Corrosion risk can be reliably detected from data present in standard IFC models. The material specification, element classification, zone metadata, and geometric position data required by GC-001 are all within the scope of standard LOD 300 MEP models and are routinely populated in well-authored IFC exports. The normalisation layer addresses the practical obstacle of naming convention variability.

The OpenBIM approach, using IFC as the sole input and BCF 2.1 as the sole output, has been validated as a viable architecture for cross-discipline compliance checking. No proprietary API access was required; the tool processed IFC files from multiple authoring environments without modification.

Part II of this thesis extends the compliance checking scope to crevice corrosion, the second major failure mechanism in MEP services, and demonstrates that the two engines together detect a significant class of risks — exemplified by the pool plant room stainless steel finding — that either engine alone would miss. This extension also answers a deeper architectural question: whether the BIMGUARD framework is sufficiently generalised to accommodate additional corrosion mechanisms, which it demonstrably is.

Chapter 7 — Crevice Corrosion Module: Rationale and Research Gap

7.1 Limitations of galvanic-only checking

The galvanic corrosion engine GC-001, described in Chapters 3 through 6, addresses one of the two primary corrosion mechanisms of concern in MEP building services. However, a compliance tool that checks only galvanic compatibility is incomplete for a technically significant reason: the most corrosion-resistant materials from a galvanic perspective — high-grade stainless steels, titanium, and nickel alloys — are precisely the materials most susceptible to crevice corrosion. The passive oxide film that gives these materials their galvanic nobility is locally destroyed in geometrically confined spaces where restricted electrolyte circulation creates the specific chemistry required for crevice attack.

The practical consequence of this is that a galvanic-only compliance tool will classify a correctly specified high-grade stainless steel system as low risk or compliant, while an actual crevice corrosion failure develops at the flanged joints, under pipe clamps, and at threaded connections in the same system. The failure mode is invisible to the galvanic check — not because the check is wrong, but because it is checking the wrong mechanism for that material.

7.2 The pool plant room finding

This limitation was demonstrated directly in the validation dataset. A set of SS316 stainless steel flanges in a pool plant room environment (E6 — pool or spa enclosure, chloride concentration >200 mg/L) produced the following combined assessment results:

- GC-001 galvanic composite score: 0.00 (Low — no galvanic risk identified)
- CC-001 crevice composite score: 0.89 (Critical — immediate remediation required)

The galvanic score is correct: SS316 is among the most noble metals in the BIMGUARD galvanic series and presents no galvanic corrosion risk in combination with itself. The crevice score is also correct: SS316 has a CCT of +10°C per ASTM G48 Method B, meaning it can sustain crevice attack at temperatures above 10°C in chloride solutions. A pool plant room environment regularly exceeds this temperature and routinely achieves the chloride concentrations required to initiate crevice attack. A standard flanged joint in this environment — with its tight crevice geometry at the gasket interface — will reliably develop crevice corrosion within two to five years of installation, regardless of the material's galvanic nobility.

This finding justifies the dual-engine architecture of BIMGUARD AI and establishes the crevice corrosion engine CC-001 as a necessary rather than optional component of the compliance checking framework. The pool plant room scenario is not an edge case; it is representative of a broad class of installations in swimming pool facilities, pharmaceutical clean rooms, coastal buildings, and food processing environments where both chloride exposure and high-grade stainless steel specification are standard.

Chapter 8 — Literature Review: Crevice Corrosion

8.1 Mechanism and four-stage process

Crevice corrosion is an electrochemical degradation mechanism that initiates and propagates within geometrically confined spaces where electrolyte circulation is restricted. The mechanism proceeds through four distinct stages, first described systematically by Fontana and Greene (1967) and subsequently refined through extensive experimental work at the Fontana Corrosion Center and other institutions.

In Stage 1, the Initiation Stage, the confined electrolyte within the crevice is initially identical in composition to the bulk solution. Oxygen depletion begins as the cathodic oxygen reduction reaction consumes dissolved oxygen within the crevice faster than diffusion from the bulk can replenish it. In Stage 2, the Differential Aeration Stage, the oxygen-depleted crevice electrolyte becomes anodic relative to the oxygenated exterior surface. Metal dissolution begins within the crevice: $M \rightarrow M^{2+} + 2e^-$. In Stage 3, the Autocatalytic Stage, metal cation hydrolysis within the crevice produces hydrogen ions, lowering the local pH. Chloride ions migrate into the crevice to maintain charge neutrality, forming aggressive metal chloride salts. The passive film is destroyed by the combination of low pH and high chloride concentration, and corrosion accelerates. In Stage 4, the Propagation Stage, the autocatalytic chemistry is self-sustaining; crevice corrosion continues to propagate even if the external environment changes, until the crevice geometry is destroyed or the material is perforated.

8.2 Critical Crevice Corrosion Temperature

The Critical Crevice Corrosion Temperature (CCT) is the minimum temperature at which crevice corrosion can initiate in a standardised test solution, as defined by ASTM G48 Method B. CCT is a material property — it varies significantly between stainless steel grades and is strongly influenced by alloy composition, particularly chromium, molybdenum, and nitrogen content. The PREN formula introduced in Chapter 3 ($PREN = \%Cr + 3.3 \times \%Mo + 16 \times \%N$) was originally developed specifically as a predictor of CCT and pitting resistance.

CCT values for the stainless steel and nickel alloy grades included in CC-001 are sourced from ASTM G48 Method B test data compiled in CIRIA C692 (Stainless Steel in Construction) and the Sandvik/Alleima Corrosion Handbook:

Grade	UNS	CCT (°C)	PREN (typical)
SS 304 / 1.4301	S30400	-5	18–20
SS 316 / 1.4401	S31600	+10	24–26
Duplex 2205 / 1.4462	S32205	+25	33–38
Super Duplex 2507 / 1.4410	S32750	+50	40–43
Titanium Grade 2	R50400	+120	>40

Table 9.1: CCT values for selected grades (ASTM G48 Method B / CIRIA C692)

8.3 Relevant standards and test methods

Three standards are of primary relevance to the CC-001 engine development. EN ISO 15329:2007 (Corrosion of metals and alloys — Electrochemical testing — Cyclic potentiokinetic polarisation measurements) provides the wetting class framework used to classify environment severity in CC-001. The standard defines seven wetting classes T0 through T5, based on annual wetness time, chloride deposition rate, and sulphur dioxide exposure, providing a structured basis for environment severity classification from IFC zone metadata.

ASTM G48 Method B (Standard Test Methods for Pitting and Crevice Corrosion Resistance of Stainless Steels and Related Alloys by Use of Ferric Chloride Solution) is the primary source for CCT values used in CC-001. Method B uses a 6% ferric chloride solution to create standardised aggressive conditions and determines the minimum temperature at which crevice corrosion initiates, providing reproducible and widely published data for the stainless steel grades most commonly specified in MEP applications.

CIRIA C692 (Stainless Steel in Construction: a Review of Research, Applications and Challenges) provides a comprehensive synthesis of crevice corrosion data in the construction context, including CCT data, joint type risk classifications, and remediation case studies. CIBSE Guide G (Public Health and Plumbing Engineering) provides the MEP-specific guidance on crevice corrosion risk in plumbing systems, particularly in relation to dead-leg configurations and water system hygiene, and is the primary reference for environment severity classification in building services applications.

Chapter 9 — Methodology: CC-001 Crevice Corrosion Engine

9.1 CCT table and stainless steel grades

The CCT values embedded in CC-001 are sourced from ASTM G48 Method B test data as compiled in CIRIA C692, with cross-referencing against the Sandvik Corrosion Handbook and EN 1993-1-4 (Eurocode 3: Design of Steel Structures — Supplementary Rules for Stainless Steels). The five grades in the CC-001 CCT table, shown in Table 9.1 in Chapter 8, cover the range of stainless steel specifications commonly encountered in commercial and industrial MEP applications from standard-grade SS304 through to Titanium Grade 2 for the most aggressive environments.

The CCT adequacy sub-score is calculated by comparing the operating temperature of the service (read from `Pset_PipeSegmentOccurrence.OperatingTemperature` or defaulting to the ambient temperature of the containing zone) against the CCT of the specified material. Where the operating temperature exceeds the CCT, the material is at risk of crevice initiation if the geometry and environment conditions are also present, and the CCT adequacy sub-score is set to 1.00 (maximum risk). Where the operating temperature is more than 20°C below the CCT, the sub-score is 0.00.

9.2 Geometry classification

Four geometry classes are defined in CC-001, each with a corresponding geometry risk multiplier derived from crevice severity classifications in CIBSE Guide G and CIRIA C692:

Class	Description	Multiplier	Example joints
Open	Crevice width > 2mm; free electrolyte circulation	0.10	Butt weld
Moderate	Crevice width 0.5–2mm; some restriction	0.45	Slip-on flange
Tight	Crevice width 0.1–0.5mm; significant restriction	0.75	Weld neck flange
Critical	Crevice width < 0.1mm; gasket contact geometry	1.00	Threaded joint

Table 9.2: Geometry class definitions and risk multipliers (CC-001 v1.0.0)

9.3 Joint type library

CC-001 includes a library of 14 joint types (JT-001 through JT-014) mapping IFC element type classifications to geometry classes. This library provides the mechanism by which the engine assigns a geometry risk multiplier without requiring the joint geometry to be explicitly modelled in the IFC file. The IFC element type — `IfcPipeFitting.PredefinedType`, valve classification, or connection type — is used as the key to look up the corresponding joint type and geometry class. The 14 joint types cover the full range of connection methods used in commercial MEP applications:

JT-001: Butt weld (Open, 0.10); JT-002: Socket weld (Moderate, 0.45); JT-003: Slip-on flange (Moderate, 0.45); JT-004: Weld neck flange (Tight, 0.75); JT-005: Threaded NPT (Critical, 1.00); JT-006: Compression fitting (Tight, 0.75); JT-007: Push-fit press (Moderate, 0.45); JT-008: Victaulic groove coupling (Moderate, 0.45); JT-009: Screwed BSP (Critical, 1.00); JT-010: Lap joint flange (Tight, 0.75); JT-011: Ring-type joint flange (Tight, 0.75); JT-012: Pipe clamp under insulation (Critical, 1.00); JT-013: Mechanical anchor contact (Tight, 0.75); JT-014: Unknown/unclassified (Tight, 0.75 — conservative default).

9.4 Environment severity classification

Environment severity in CC-001 is classified against the wetting class framework of EN ISO 15329:2007, using seven classes T0 through T5 plus a building services-specific class. Chloride concentration thresholds are sourced from CIBSE Guide G and EN ISO 15329:2007. Environment severity is read from `lfcZone` metadata and the `lfcDistributionSystem` service type classification, with conservative defaults applied where metadata is absent.

9.5 Weighted composite scoring model

The CC-001 composite risk score is calculated using a three-component weighted model:

$$\text{Score_CC} = (0.35 \times \text{geometry_risk}) + (0.40 \times \text{CCT_adequacy}) + (0.25 \times \text{environment_severity})$$

CCT adequacy receives the highest weight (0.40) because it is the most material-specific determinant of crevice corrosion susceptibility — a material with adequate CCT margin cannot initiate crevice attack regardless of geometry or environment. Geometry receives the second highest weight (0.35) because it is the immediate physical enabler of the crevice mechanism. Environment severity receives the lowest weight (0.25) because it represents the external conditions, which can sometimes be managed or modified, whereas material specification and joint geometry are fixed once the model is authored.

Risk bands for CC-001 are defined as: Low (< 0.30), Medium (0.30–0.55), High (0.55–0.80), Critical (> 0.80).

9.6 Combined risk assessment function

The `combined_risk_assessment()` function in the compliance runner executes both GC-001 and CC-001 on every qualifying element and returns a unified risk assessment. The combined risk band is the higher of the two individual risk bands — if an element scores Low on galvanic and Critical on crevice, the combined band is Critical. This conservative combination rule ensures that the worst-case mechanism always determines the overall risk classification and remediation priority.

The combined approach also enables mechanism-specific BCF issue generation: where GC-001 and CC-001 both flag the same element, two separate BCF issues are generated with different mechanism descriptions and different recommended mitigations, allowing the discipline engineer to address each failure mode independently.

Chapter 10 — Architecture: Combined Compliance System

10.1 Pset_CreviseCorrosionRisk schema

Analogous to the Pset_CorrosionRisk property set written back by GC-001, CC-001 writes a Pset_CreviseCorrosionRisk property set to the IFC model for each assessed element. The schema contains the following properties:

- CreviseRiskScore — the CC-001 composite score (real, 0.0–1.0)
- CreviseRiskBand — the risk band classification (string: Low/Medium/High/Critical)
- CCTAdequacyScore — the CCT adequacy sub-score
- GeometryClass — the joint type library classification (string: JT-001 through JT-014)
- EnvironmentSeverity — the EN ISO 15329 wetting class
- RecommendedMitigation — the mitigation code from the CC-001 mitigation catalogue
- RulesetVersion — the CC-001 ruleset version string ("BIMGUARD-CC-001 v1.0.0")
- AssessmentDate — ISO 8601 timestamp of the assessment run

Writing both Pset_CorrosionRisk and Pset_CreviseCorrosionRisk to the model creates a permanent, machine-readable compliance record that travels with the IFC file through all subsequent project stages. This satisfies the Golden Thread traceability requirement of the Building Safety Act 2022 for higher-risk buildings: any party reviewing the model at any stage of the asset lifecycle can read the compliance assessment results and their evidential basis directly from the model data.

10.2 Combined asset register

The compliance runner exports a combined asset register CSV file containing one row per assessed element with the full GC-001 and CC-001 results side by side. The register includes element GlobalID, service type, floor, zone, material, nominal diameter, GC-001 score, CC-001 score, combined risk band, mechanism flags, recommended mitigation, responsible engineer, estimated remediation cost, and programme impact in working days. This register provides the data source for the Streamlit application's Schedule and Cost Impact page and can be imported directly into a project cost management tool or programme management application.

10.3 Dual-engine BCF output

The BCF generator module produces separate issues for GC-001 and CC-001 findings on the same element where both engines flag a risk. This separation is important for practical deployment: the mechanical engineer responsible for material specification needs to receive the crevice corrosion finding and act on it independently of the structural engineer responsible for support assembly specification, who receives the galvanic finding. A single

combined BCF issue with two mechanism descriptions creates ambiguity about ownership and action.

The issue assignment logic in the BCF generator maps service type and mechanism to discipline engineer: galvanic findings on mechanical pipe systems are assigned to the mechanical engineer; galvanic findings involving structural supports are assigned to the structural engineer; crevice findings on pressurised systems are assigned to the mechanical engineer; crevice findings in drainage systems are assigned to the public health engineer. This mapping is configurable through the BCF generator settings and can be customised for individual project team structures.

Chapter 11 — Discussion and Further Development

11.1 MIC as the third corrosion module

Microbially Influenced Corrosion (MIC) is identified as the logical third corrosion mechanism module for BIMGUARD AI, to be developed as BIMGUARD-MC-001. MIC occurs when microbiological communities — primarily sulphate-reducing bacteria, iron-oxidising bacteria, and acid-producing fungi — colonise internal pipe surfaces and produce corrosive metabolic byproducts. The mechanism is of particular significance in building services applications involving stagnant or low-flow water conditions: dead-leg pipework in hot and cold water systems, commissioning water retained in chilled water circuits, cooling tower pipework, and fire suppression systems with infrequent flow.

The IFC data required for MIC checking is distinct from that used by GC-001 and CC-001: flow velocity (from system design data in property sets), water temperature (from operating temperature Psets), the presence of dead-leg configurations (from network topology analysis on the `IfcDistributionSystem` graph), and under-insulation conditions (from the relationship between `IfcCovering` and the contained `IfcPipeSegment`). The scoring model for MC-001 would incorporate flow velocity class, temperature class (both high and low extremes create risk), dead-leg classification, and a time-since-flush parameter for systems with intermittent flow.

11.2 BMS integration

A future development direction of significant practical value is integration between BIMGUARD AI and Building Management System (BMS) data streams. The current tool assesses corrosion risk from static design data in the IFC model; a BMS-integrated version would update risk scores dynamically based on real-time operating conditions — measured temperatures, flow rates, and humidity readings from the BMS. This would transform BIMGUARD AI from a design-stage compliance checker to a continuous operational risk monitor, flagging developing corrosion conditions in real time rather than waiting for the next design review cycle. The `Pset_CorrosionRisk` and `Pset_CreviseCorrosionRisk` property sets are designed with this future integration in mind: they carry an `AssessmentDate` field that would be updated on each BMS data refresh.

11.3 Weld quality extension

A further development opportunity identified during the crevice corrosion work is a weld quality extension to CC-001. The current engine classifies butt welds as Open geometry (lowest crevice risk) based on the assumption of a properly completed full-penetration weld with no internal crevice. In practice, incomplete penetration welds — a common quality defect in site-welded pipework — create an internal crevice at the root of the weld that is classified as Critical geometry and represents a severe crevice corrosion risk in aggressive environments. An extension that reads weld procedure specification data from the IFC model and downgrades the geometry class for non-full-penetration specifications would significantly improve the accuracy of CC-001 for site-welded systems.

11.4 Regulatory context and wider applicability

The Building Safety Act 2022 creates a legal requirement for a Golden Thread of information for higher-risk buildings in England — a requirement that the information about a building is created, maintained, and passed through each stage of the building's lifecycle in a form that is accurate, up to date, and accessible. BIMGUARD AI contributes to the Golden Thread by producing machine-readable, standards-referenced compliance records that are embedded in the IFC model and stored in BCF issues with full audit trails. This positions the tool not merely as a coordination aid but as a regulatory compliance instrument for the category of buildings most likely to be subject to formal compliance scrutiny.

Beyond the UK regulatory context, the ISO 19650 information management framework — now adopted in over 70 countries — provides the governance structure within which BIMGUARD AI's ISO 19650 status codes and Plannerly integration are meaningful. As adoption of ISO 19650 workflows continues to expand internationally, the demand for automated technical compliance checking tools that produce ISO 19650-compatible outputs will increase. BIMGUARD AI is designed to be portable across these contexts: the standards references in the rulesets are international standards, the IFC and BCF formats are global open standards, and the Plannerly integration uses the ISO 19650 status code framework that is jurisdiction-neutral.

11.5 Documented field failure patterns behind the five mechanisms

The five engines are not five views of one defect. Each corresponds to a failure pattern that appears in the building services corrosion literature with its own setting, its own signature on site and its own remedy — and, more usefully for a compliance tool, its own detectable precondition in a model. Table 11.1 places them side by side so that the case for five mechanisms, rather than one general corrosion score, can be read directly.

Mechanism and engine	Characteristic field failure	Typical setting	Signature on site	Governing guidance
Galvanic (GC-001)	Copper tube joined to a carbon steel or galvanised riser; the steel side wastes at the joint and perforates while the copper is untouched	Plant rooms, riser transitions, meter sets, boosted cold water sets	Wastage confined to within a few pipe diameters of the joint; green corrosion product on the copper, red-brown on the steel	NASA-STD-6012; AUCSC (2024); American Galvanizers Association (2023)
Crevice (CC-001)	Attack initiates under a gasket, a pipe support or a lap, where the passive film cannot re-form and the shielded electrolyte acidifies	Pool halls, coastal plant rooms, chlorinated process areas	Pitting confined to the shielded footprint, with bright passive metal a few millimetres away	EN ISO 15329:2007; ASTM G48 Method B; CIRIA C692; IMO A
MIC (MC-001)	Sulphate-reducing bacteria pit under deposits in stagnant legs; iron-oxidising bacteria raise tubercles that both perforate and block	Fire suppression mains, little-used branches, condenser circuits	Discrete hemispherical pits under black sulphide deposit, or red-brown tubercles; hydrogen sulphide odour on opening	CIBSE TM13:2013; HSE HSG274 Parts 1-3; ASTM G-187

Mechanism and engine	Characteristic field failure	Typical setting	Signature on site	Governing guidance
Material-media (MM-001)	The material is wrong for the medium it carries: carbon steel on open condenser water, galvanised steel above the zinc reversal temperature, a polymer on a steam service	Cooling tower circuits, domestic hot water, steam and condensate	General wastage along the whole run rather than at a joint — the diagnostic that separates it from a galvanic couple	CIBSE Guide G; EN 12952-12; CDA; IMO A
Cross-material (XM-001)	A dissimilar-metal junction created where two disciplines' models meet, which nobody specified and nobody owns	Federated model coordination — before installation rather than after	None. The finding exists only in the model, which is the entire point of the engine	BS 8539; NASA-STD-6012

Table 11.1: Field failure patterns and the engine that detects each, with the site evidence that distinguishes them

The last row differs from the first four in kind rather than in degree. Galvanic, crevice, microbial and material-media failures are all diagnosed from a component that has already corroded, and the site-evidence column describes what an engineer sees after the fact. XM-001 has no site evidence because the finding it produces exists only in the federated model, at the moment two subcontractors' packages are brought together and before the joint has been installed. That is the difference between a forensic tool and a preventive one, and it is the strongest argument for placing corrosion checking inside the coordination workflow rather than alongside it.

The mechanisms also overlap in ways that make a single combined score misleading. Galvanised steel carrying domestic hot water above roughly 60°C is the clearest case: the zinc coating reverses polarity and becomes cathodic to the steel it was applied to protect, so the pipe wastes from within. GC-001 sees no couple, because there is no second element. CC-001 sees no crevice. MC-001 sees adequate flow at a temperature above the microbial optimum and correctly returns Low. Only MM-001 scores it, because only MM-001 asks the question that matters here — whether this material should be carrying this fluid at this temperature at all.

Chapter 12 — Microbially Influenced Corrosion Module (MC-001)

12.1 Introduction and rationale

Microbially Influenced Corrosion (MIC) is the third major corrosion mechanism of concern in MEP building services pipework, following galvanic corrosion (addressed by GC-001) and crevice corrosion (addressed by CC-001). MIC occurs when communities of microorganisms establish biofilms on internal pipe surfaces and produce metabolic byproducts that initiate or accelerate metallic corrosion. Unlike the purely electrochemical mechanisms addressed by GC-001 and CC-001, MIC requires a biological precondition — the establishment of a viable microbial community — and is therefore principally governed by the hydrodynamic and thermal conditions of the system rather than by material pairing or joint geometry alone.

The primary enabling condition for MIC in building services is stagnation. Flowing water continuously removes planktonic microorganisms and the nutrients they require, and imposes shear stresses that prevent biofilm adhesion above certain velocities. Stagnant or very low-flow zones — dead-leg pipework, commissioning water retained in circuits, infrequently used outlets, and fire suppression systems that are designed not to flow — provide the conditions in which biofilms can establish, mature, and produce corrosive conditions. A second enabling condition is temperature: different microbial communities have distinct temperature optima, and the building services temperature ranges that are most common — domestic hot water distributed below 55°C, chilled water at 7–12°C, condenser water at 25–35°C — fall within the growth ranges of the principal MIC organisms encountered in built environment applications.

The public health dimension of MIC in building services is significant and distinguishes it from the purely structural concerns of GC-001 and CC-001. *Legionella pneumophila* — the causative agent of Legionnaires' disease — is not a corrosion organism itself, but it colonises the biofilms and corrosion product layers that MIC produces, using them as both physical substrate and nutrient source. A system with active MIC is therefore also a system at elevated *Legionella* risk. The regulatory framework governing *Legionella* control in the United Kingdom — CIBSE TM13:2013, HSE HSG274 Parts 1–3, and the Health and Safety at Work Act 1974 — mandates formal risk assessment and documented control measures for defined water systems in non-domestic premises. BIMGUARD AI's MC-001 engine produces the machine-readable risk assessment records that support this regulatory requirement.

12.2 The three MIC organism groups in building services

Three groups of microorganisms are primarily responsible for MIC in commercial building services applications. Their temperature ranges, flow preferences, and corrosion mechanisms are distinct, but they frequently co-exist in the same biofilm community, producing a complex and synergistic corrosion environment.

12.2.1 Sulphate-reducing bacteria (SRB)

Sulphate-reducing bacteria are anaerobic organisms that use sulphate (SO_4^{2-}) as a terminal electron acceptor in their metabolism, producing hydrogen sulphide (H_2S) as a metabolic byproduct. SRB are obligate or facultative anaerobes, meaning they require oxygen-depleted conditions to function — conditions that are reliably produced in stagnant

water zones where dissolved oxygen is consumed by aerobic organisms at the biofilm surface, creating an anaerobic zone at the pipe wall. The hydrogen sulphide produced by SRB reacts directly with iron in carbon steel and cast iron pipe walls, producing iron sulphide (FeS) and causing aggressive pitting corrosion that can penetrate pipe walls in months in severe cases.

SRB have a mesophilic temperature optimum of 25–35°C and thermophilic strains active to 65°C, placing them in the temperature range of domestic hot water systems operating below design temperature, condenser water systems, and warm plant room environments. They are among the most corrosively significant microorganisms in building services, and their presence in a system can be detected by the characteristic odour of hydrogen sulphide and the black iron sulphide deposits visible on pipe surfaces at inspection.

12.2.2 Iron-oxidising bacteria (IOB)

Iron-oxidising bacteria are aerobic organisms that obtain energy by oxidising ferrous iron (Fe^{2+}) to ferric iron (Fe^{3+}). In a carbon steel or cast iron pipe system, they concentrate at the aerobic outer surface of the biofilm, consuming the dissolved iron released by generalised corrosion. The ferric hydroxide they produce forms the characteristic reddish-brown tubercles — concretion deposits — that accumulate on the inner surface of iron pipework and progressively restrict the flow cross-section. More significantly from a corrosion standpoint, the IOB tubercle creates a differential aeration cell: oxygen-rich water on the outside of the tubercle acts as a cathode, while the anoxic zone beneath the tubercle acts as an anode, concentrating corrosion at the pipe wall surface and producing rapid, localised pitting that can produce through-wall perforation while the remainder of the pipe appears relatively unaffected.

12.2.3 Acid-producing bacteria (APB)

Acid-producing bacteria metabolise organic carbon substrates and excrete organic acids as byproducts. In a building services context, the most significant organic acids produced are acetic, formic, and lactic acids, which lower the local pH within the biofilm to values that can destroy the passive oxide film on stainless steel grades that would otherwise be corrosion-resistant. This is the mechanism by which MIC can initiate crevice corrosion at temperatures significantly below the Critical Crevice Corrosion Temperature determined by ASTM G48 Method B testing — a clean laboratory environment does not reproduce the acid-producing conditions within a mature biofilm, meaning that CCT values represent a conservative estimate of stainless steel corrosion resistance in real building services conditions. The CC-001 engine flags stainless steel elements in stagnant warm-water systems as elevated risk for exactly this reason.

12.3 MC-001 methodology — scoring model

The MC-001 engine calculates a composite risk score for each assessed pipe element using a four-component weighted model derived from the enabling conditions identified in the literature above. The scoring formula is:

$$\text{Score}_{\text{MC}} = (0.35 \times \text{flow_velocity_risk}) + (0.30 \times \text{temperature_risk}) + (0.25 \times \text{dead_leg_risk}) + (0.10 \times \text{material_susceptibility})$$

The base composite score is then multiplied by a system type modifier that reflects the inherent MIC risk of different water service types. The under-insulation corrosion pathway is assessed as an independent risk route — if the under-insulation risk score multiplied by 0.85 exceeds the system-modified base score, the higher value is used as the final composite score. The score is capped at 1.00.

12.3.1 Flow velocity risk (weight: 0.35)

Flow velocity receives the highest weighting in the MC-001 model because stagnation is the necessary precondition for all three MIC mechanisms. Without stagnation, biofilms cannot establish to a corrosively significant thickness regardless of temperature, material, or system type. Six flow velocity classes are defined, based on the velocity thresholds established in CIBSE TM13:2013 and HSE HSG274 Part 2 for Legionella risk assessment:

Class	Description	Velocity (m/s)	Risk score
FV0	Stagnant / dead-leg	0.0	1.00
FV1	Very low flow	< 0.1	0.80
FV2	Low flow	0.1 – 0.3	0.55
FV3	Acceptable	0.3 – 0.6	0.25
FV4	Good flow	0.6 – 1.5	0.10
FV5	Turbulent	> 1.5	0.02

Table 12.1: MC-001 flow velocity risk classes (source: CIBSE TM13:2013 / HSE HSG274 Part 2)

Flow velocity is read from Pset_PipeSegmentOccurrence.FlowVelocity in the IFC model. Where this property is absent — which is common at LOD 300 — the engine infers velocity from the system type: fire suppression systems are classified FV0 (stagnant by design), domestic hot water secondary return circuits are classified FV3 (design velocity typically 0.3–0.5 m/s), and unclassified systems default to FV2 (conservative).

12.3.2 Temperature risk (weight: 0.30)

Temperature risk is classified against the biological activity ranges of the three MIC organism groups and the Legionella danger zone established in CIBSE TM13:2013. Five temperature classes are defined:

The critical temperature class is T2 (25–45°C), which covers the growth optimum of Legionella pneumophila (35–46°C per WHO), the mesophilic SRB optimum (25–35°C), and the APB active range (15–45°C). This class receives the maximum risk score of 1.00. The T4 class (above 55°C) receives a risk score of 0.05 because Legionella is destroyed above 60°C (CIBSE TM13:2013) and SRB are inhibited at sustained high temperatures. The T5 class (temperature unknown) receives a conservative score of 0.65, reflecting the uncertainty risk — a pipe element at unknown temperature in a building services context is more likely to be in the moderate temperature range than at either extreme.

Operating temperature is read from Pset_PipeSegmentOccurrence.OperatingTemperature or Pset_ZoneCommon.SetPointTemperature on the containing zone. Where neither is available, the system type is used to assign a default: chilled water defaults to 10°C (T0), domestic hot water defaults to 45°C (T2_DANGER — conservative until confirmed), and condenser water defaults to 30°C (T2_DANGER).

12.3.3 Dead-leg risk (weight: 0.25)

Dead-leg classification uses the length-to-diameter ratio established in HSE HSG274 Part 2 as the primary metric. A dead-leg is defined as a section of pipework that terminates without a through-flow path — a capped branch, an unused outlet, or an unconnected connection point. The HSE threshold for a potentially significant dead-leg is a length greater than 3× the pipe diameter (3D). Below this threshold, regular flushing through

adjacent active sections is typically sufficient to prevent significant stagnation. Above this threshold, dedicated flushing provisions are required.

Class	Description	L/D ratio	Risk score
DL0	Through-flow — no dead-leg	0	0.05
DL1	Short dead-leg	< 3D	0.30 — below HSE threshold
DL2	Moderate dead-leg	3D – 10D	0.65 — exceeds HSE threshold
DL3	Long dead-leg	10D – 20D	0.85 — auto-flush required
DL4	Critical dead-leg	> 20D	1.00 — eliminate or shut down

Table 12.2: MC-001 dead-leg classification by L/D ratio (source: HSE HSG274 Part 2)

Dead-leg identification from IFC data requires network topology analysis on the `IfcDistributionSystem` graph. Each `IfcPipeSegment` is assessed for connectivity — segments that terminate without connection to an active outlet or return are flagged as potential dead-legs. The dead-leg length is calculated as the cumulative pipe length from the last active junction to the terminal point. Where network topology is incomplete or ambiguous in the IFC model, the engine assigns the `DL5_UNKNOWN` class with a risk score of 0.50 — a conservative mid-range default that reflects the uncertainty.

12.3.4 Material susceptibility (weight: 0.10)

Material susceptibility in the MC-001 context reflects the propensity of the pipe material to corrode once a microbial biofilm has established. This is distinct from the material's role in the GC-001 and CC-001 assessments, where it determines electrochemical potential and passive film stability respectively. In MC-001, carbon steel and cast iron are the most susceptible materials because iron is both a structural component of the pipe wall and a direct nutrient for IOB, creating a self-reinforcing corrosion cycle. Copper is assigned low susceptibility because of its inherent antimicrobial properties — copper ion release inhibits biofilm formation, which is why copper has been widely used in domestic plumbing for this reason. Non-metallic materials (CPVC, PVC, HDPE) receive negligible scores because they do not participate in metallic corrosion even when colonised by biofilm.

12.4 System type modifiers

The system type modifier is applied as a multiplier to the base composite score, reflecting the inherent MIC risk profile of different water service categories. Fire suppression systems receive the highest modifier (1.40) because they are designed for infrequent or zero flow, producing permanent or near-permanent dead-leg conditions throughout the entire distribution network — a condition that no amount of pipe sizing or material specification can fully mitigate. Domestic cold water systems receive a high modifier (1.30) because they are the primary *Legionella* risk system and are subject to mandatory risk assessment under HSE HSG274. Domestic hot water systems receive 1.25, reflecting the risk of distribution below 55°C.

12.5 Under-insulation corrosion pathway

Under-insulation corrosion (UIC) is assessed as an independent pathway in MC-001. Damaged or wet thermal insulation creates conditions of trapped moisture, restricted oxygen, and elevated temperature — ideal for SRB and APB colonisation on the outer

surface of metallic pipe — while being largely invisible to inspection and maintenance. The UIC risk is assessed separately from the internal flow-condition-based score and the final composite score uses whichever is higher, multiplied by 0.85 to reflect that UIC produces external rather than internal corrosion and its consequences are generally less acute than internal MIC.

Insulation condition is read from maintenance record property sets where available. Where no condition data is present and insulation is confirmed in the model, the engine defaults to 'unknown' (score 0.35) — a conservative assumption that weathered or aged insulation may have compromised integrity.

12.6 MC-001 risk bands and BCF output

MC-001 uses four risk bands with different thresholds from GC-001 and CC-001, reflecting the fact that MIC risk in building services context is generally assessed against a more precautionary standard due to the public health implications of Legionella:

Risk band	Score range	BCF priority	Required action
Low	< 0.25	No BCF	Asset register record only
Medium	0.25 – 0.50	Normal	Monitoring programme required
High	0.50 – 0.75	Major	Remediation within 3 months
Critical	> 0.75	Critical	Immediate remediation required

Table 12.3: MC-001 risk bands (note lower thresholds than GC-001/CC-001 reflecting public health dimension)

12.7 Validation results

The MC-001 engine was validated against ten designed test scenarios covering the range of expected real-world MIC conditions. The scenarios and results are:

ID	Scenario	System	Material	Score	Band
VAL-001	Carbon steel dead-leg, danger zone	Cold water	Carbon steel	1.000	Critical
VAL-002	Fire suppression permanent dead-leg	Fire supp.	Carbon steel	1.000	Critical
VAL-003	Hot water dead-leg at 38°C	Hot water	Carbon steel	1.000	Critical
VAL-004	Chilled water, weathered insulation	Chilled water	Carbon steel	0.676	High
VAL-005	Condenser water warm loop	Condenser	Galv. steel	0.831	Critical
VAL-006	Copper hot water 58°C, short dead-leg	Hot water	Copper	0.241	Low
VAL-007	SS316 process, unknown temperature	Process	SS 316	0.570	High

ID	Scenario	System	Material	Score	Band
VAL-008	Copper hot water 62°C, through-flow	Hot water	Copper	0.097	Low
VAL-009	CPVC cold water, through-flow, 12°C	Cold water	CPVC	0.127	Low
VAL-010	Duplex 2205 hot water, good flow	Hot water	Duplex 2205	0.169	Low

Table 12.4: MC-001 validation results — 10 scenarios, all results consistent with engineering expectation

All ten scenarios produced results consistent with engineering expectation. The fire suppression system (VAL-002) correctly scores 1.000 Critical, reflecting the fact that fire suppression pipework is by design a permanent dead-leg system with no regular through-flow — the highest-risk configuration in the MC-001 model. The copper hot water system at 62°C with through-flow (VAL-008) correctly scores 0.097 Low, reflecting the combined protective effects of antimicrobial copper, temperature above the *Legionella* optimum, and adequate flow velocity.

The SS316 process water element with unknown temperature (VAL-007) scores 0.570 High — a result that warrants attention. The high score is driven by the moderate dead-leg classification (DL2, 6D ratio) and the conservative temperature default applied when temperature data is absent. This demonstrates an important design property of MC-001: the engine fails conservatively when data is missing, ensuring that elements for which the IFC model does not contain adequate operating condition data are flagged for investigation rather than silently passing.

12.8 Integration with GC-001, CC-001, and the combined compliance framework

MC-001 integrates with the existing BIMGUARD AI compliance framework through the same `combined_risk_assessment()` function that runs GC-001 and CC-001. The combined risk band for any element is the highest band returned by any of the three engines. MC-001 results are written to a `Pset_MICRisk` property set on the IFC element, alongside `Pset_CorrosionRisk` (GC-001) and `Pset_CreviseCorrosionRisk` (CC-001), creating a complete three-mechanism corrosion compliance record for each assessed element.

The practical implication of the three-engine combined assessment is that an element can be flagged by any combination of mechanisms — and the mechanisms are genuinely independent. A carbon steel fire suppression pipe in a warm plant room may score: Low on GC-001 (no dissimilar metal contact), Low on CC-001 (butt-welded joints, no crevice geometry), and Critical on MC-001 (permanent dead-leg, danger zone temperature). Without MC-001, this element would appear compliant on a BIMGUARD AI assessment — a significant and potentially dangerous false negative for a system type that is specifically identified as high *Legionella* risk in HSE HSG274 Part 3.

12.9 Regulatory context and further development

The MC-001 engine addresses a regulatory compliance gap that is distinct from the general corrosion risk context of GC-001 and CC-001. In the United Kingdom, the Control of Substances Hazardous to Health Regulations 2002 (COSHH) require employers to assess and control the risk of exposure to *Legionella* bacteria. The approved code of practice L8 (Prevention or Control of Legionellosis) and the associated HSE HSG274

guidance documents specify that a formal, documented Legionella risk assessment must be performed for all water systems in non-domestic premises. BIMGUARD AI's MC-001 output — a structured, standards-referenced, machine-readable risk classification for each identified risk element — provides a defensible foundation for this required assessment at the design stage, before the water system is commissioned.

Two further development directions are identified for MC-001. The first is a time-since-flush parameter that tracks the interval between flushing events for elements classified as dead-legs or low-flow zones, enabling the risk score to increase progressively with stagnation time. The second is integration with Building Management System data streams, enabling real-time temperature and flow monitoring data to update MC-001 risk scores dynamically throughout the operational lifecycle of the building — transforming the engine from a design-stage compliance tool into a continuous operational risk monitor.

12.10 Key finding — fire suppression systems

The most academically significant finding from MC-001 validation is the consistently Critical classification of fire suppression systems. These systems are present in virtually every commercial building, are installed in carbon steel pipework as standard, contain water that does not flow under normal conditions, and in a heated building environment are typically at temperatures between 15°C and 30°C — within the MIC danger zone for both SRB and Legionella. HSE HSG274 Part 3 explicitly identifies fire suppression systems as a Legionella risk and requires them to be included in the building's Legionella risk assessment. BIMGUARD AI's MC-001 engine automates this identification from IFC data, ensuring that fire suppression pipework is never inadvertently excluded from the compliance assessment because it falls outside the usual MEP coordination discipline boundary.

This finding also illustrates a key limitation of single-mechanism corrosion checking. GC-001 will not flag a fire suppression system in standard carbon steel pipework because steel-on-steel presents no galvanic risk. CC-001 will not flag it because butt-welded carbon steel joints do not present significant crevice geometry. It is only MC-001 that correctly identifies the fire suppression system as the highest MIC risk element type in a typical commercial building — and it does so from the combination of system type classification, flow velocity inference, and temperature assignment that together constitute the characteristic signature of this system type in any IFC model.

12.11 Real-world MIC failure patterns and their model signatures

The ten scenarios in Table 12.4 are designed fixtures. They establish that the arithmetic behaves across the input space; they are not a sample of anything, and they cannot show that the conditions they encode occur in practice. This section connects each scoring regime to the field failure pattern it represents, and states what a model would have to contain for the engine to detect it.

Field failure pattern	Typical setting	Organism group	MC-001 terms that drive the score	Fixture and band
Wet fire suppression main held stagnant for its entire design life, flushed only at annual test	Sprinkler and hydrant mains in any occupancy	SRB and IOB	Flow velocity at zero (weight 0.35, risk 1.0); unbounded dead-leg ratio (0.25); carbon steel susceptibility (0.10); fire suppression system modifier	VAL-002, score 1.000, Critical

Field failure pattern	Typical setting	Organism group	MC-001 terms that drive the score	Fixture and band
Cold water dead leg serving a decommissioned or rarely occupied area, sitting in the Legionella danger zone	Refurbished floors, seasonal wings, capped-off branches	SRB and Legionella colonisation	Zero flow, L/D ratio above the HSG274 threshold, temperature in the 25-45°C band	VAL-001, score 1.000, Critical
Hot water branch to a low-use outlet that has cooled to 38°C between draw-offs	Hotel and hospital rooms out of service, en-suite wings between occupancies	SRB and Legionella	Temperature, dead-leg and flow terms all at or near maximum simultaneously	VAL-003, score 1.000, Critical
Condenser water loop running warm and intermittently outside the cooling season	Cooling tower circuits, heat rejection plant	APB and IOB	Intermittent flow, warm temperature, galvanised steel susceptibility, condenser system modifier	VAL-005, score 0.831, Critical
Chilled water line under thermal insulation that has been damaged or has taken up water	Ceiling voids, roof plant, external runs	IOB and general under-insulation attack	The independent under-insulation pathway of §12.5 rather than the four-term composite	VAL-004, score 0.676, High
Correctly designed copper hot water service with genuine through-flow at 58-62°C	Any well-commissioned domestic hot water system	None viable	Adequate velocity, temperature above the microbial optimum, copper susceptibility low	VAL-006 and VAL-008, scores 0.241 and 0.097, Low

Table 12.5: MIC field failure patterns, the MC-001 terms that detect each, and the designed fixture that pins it

Two limits of the engine are visible in that table and should be stated before any result is read. MC-001 scores design conditions, not the system as it is operated: it can see that a branch is long, dead-ended and warm, but it cannot see whether the facilities team flushes it weekly, which is the control HSG274 actually relies on. And it scores a snapshot: a fire main is stagnant by design, so a Critical band on every sprinkler element is less a defect report than a statement of the system's nature — which is why §12.10 treats that result as a finding about the risk profile of fire suppression rather than about any particular building.

12.12 Planned validation of MC-001 against the 38-model dataset

The validation dataset described in Appendix A — 38 IFC models from eight independent public repositories, spanning hospital, office and industrial buildings in both IFC2x3 and IFC4 — tests a property the designed fixtures cannot reach. The fixtures supply every input the engine asks for, because they were written to. Third-party models supply whatever their authors happened to export. The question the corpus answers is therefore not whether MC-001 scores correctly, but whether the inputs it needs exist in real federated models at all.

Term	Weight	IFC source	Expected availability across the corpus	Consequence when absent
Flow velocity	0.35	Pset_PipeSegmentOccurrence.FlowVelocity	Rare. Velocity is a calculated design output that survives export only where the authoring tool was driven from a sized system; absent at LOD 300 in most of the corpus	Conservative default per §12.3.1 plus a data-quality issue. Because the default is conservative, absence pushes scores up rather than down
Temperature	0.30	Pset_PipeSegmentOccurrence.OperatingTemperature, or Pset_ZoneCommon.SetPointTemperature on the containing zone	Occasional. Most likely on the DigitalHub heating model and least likely on architecture-derived MEP exports	Data-quality issue; the term is omitted rather than guessed
Dead-leg ratio	0.25	Derived by topology analysis on the IfcDistributionSystem graph, not read from a property	Computable wherever port connectivity is exported. This is the one term the corpus genuinely exercises, because it is derived rather than declared	The element cannot be classified; missing connectivity is itself a reportable finding about export practice
Material susceptibility	0.10	Material assignment on the pipe entity	Good on the Full-tier models: Clinic_Plumbing carries real material text for copper, PVC, CPVC, steel and brass across 378 pipe entities	An unrecognised material raises a data-quality issue rather than defaulting to a susceptibility value

Table 12.6: MC-001 input availability across the 38-model corpus, by scoring term

Four models carry most of the MC-001 test weight. The West Riverside Hospital sprinkler model, at 6,228 pipe segments, is the single best target, because sprinkler mains are the canonical permanent-stagnation case and the model is large enough to yield a distribution rather than an anecdote. The West Riverside plumbing model adds 4,308 pipe segments of domestic service; Clinic_Plumbing adds genuine material diversity at smaller scale; and the RWTH DigitalHub heating and sanitary models add a European IFC4 export authored independently of the other two, which guards against a conclusion that holds only for one authoring convention.

The result that would falsify the engine's practical value is specific, and worth stating in advance: if port connectivity is not exported in any model in the corpus, the dead-leg term cannot be computed anywhere, and MC-001's second heaviest input is untestable on open data. That would be a finding about IFC export practice rather than about the engine, but

it would bound what MC-001 can claim to deliver on a real project to the flow, temperature and material terms alone.

12.13 Expected severity distribution for MC-001

Stating the expected distribution before the run is what makes the run evidence rather than illustration. The designed fixtures produced four Critical, two High, no Medium and four Low results across ten scenarios — a distribution that reflects deliberate sampling of the extremes and predicts nothing about prevalence. The corpus is expected to behave very differently.

Band	Designed fixtures (n = 10)	Expected across the corpus	Reasoning	What would falsify the expectation
Critical	4 (40%)	Under 10% of scored elements	Concentrated in fire suppression pipework, which is stagnant by design. The West Riverside sprinkler model alone could dominate this band	A Critical share above 25%, spread evenly across system types, would indicate the conservative flow default is driving the band rather than the model
High	2 (20%)	5-15%	Driven by the under-insulation pathway and by warm intermittent loops, both of which depend on properties the corpus carries sparsely	A High share near zero would suggest the insulation pathway never triggers, because insulation condition is never modelled
Medium	0 (0%)	10-25%	The fixtures skipped this band entirely. Real models should populate it, since most pipework sits between well-designed and pathological	An empty Medium band on real data would suggest the scoring is bimodal and the thresholds need review
Low	4 (40%)	50-75%	Most modelled pipework is copper or polymer in through-flow service, which the engine should and does score benign	A Low share below 25% would indicate systematic over-scoring
Not scored (data quality)	0 (0%)	Expected to exceed the scored population on most models	Velocity and temperature are absent from the majority of the corpus, and the engine is specified to raise an issue rather than substitute a default	If data-quality issues are rare, the coercion layer is filling gaps silently, which would be a defect rather than a success

Table 12.7: Expected MC-001 severity distribution across the 38-model corpus, with falsification criteria stated in advance

The final row carries the most weight and is the least intuitive. Because the flow-velocity default is deliberately conservative when the property is absent, a corpus with little velocity data will push the whole distribution upward. The distribution therefore measures IFC data availability at least as much as it measures microbial risk, and reporting a high Critical count as a risk finding without that caveat would misrepresent it. Declaring the confound in advance is what allows the result to be read either way once it exists.

Chapter 13 — Extended Material Engines: MM-001 and XM-001

13.1 Rationale — what the first three engines do not cover

GC-001, CC-001 and MC-001 each ask whether a mechanism will attack an element given that element's own material and its surroundings. Neither asks whether the material suits the fluid it carries, and GC-001 evaluates galvanic risk from the material pair recorded against a single element rather than from the couple formed where two modelled elements actually meet. Two further engines close those gaps. MM-001 scores material-media compatibility. XM-001 scores the galvanic couple at a joint between dissimilar modelled elements.

Both are built on the same principle as the first three: a weighted composite on a continuous 0–1 scale, banded at thresholds shared across every engine, with each term traceable to a named source.

13.2 MM-001 — material-media compatibility

MM-001 holds a compatibility matrix of twelve materials against eight media. The materials are carbon steel, galvanised steel, cast iron, ductile iron, copper C12200, brass C46400, 90/10 copper-nickel, SS304, SS316, duplex 2205, titanium and PVC. The media are cold water, hot water, chilled water, condenser water, pool water, steam, condensate and stagnant water. Each cell carries a score, a nominal service life in years, the governing degradation mechanism, a citation and a confidence tag, so a reader can see not only what the engine concluded but which mechanism drove it.

The composite is a weighted sum of three terms on a common 0-1 scale, evaluated per element:

$$\text{Score_MM} = (0.40 \times \text{material_media_score}) + (0.35 \times \text{environment_severity}) + (0.25 \times \text{temperature_stress})$$

The first term is the matrix cell for the element's material against the medium it carries, read from the twelve-by-eight matrix reproduced later in this section as Table 13.3. The second is the severity of the environment the element sits in, taken from the ladder in Table 13.2. The third is the Arrhenius acceleration implied by the operating temperature, banded in seven steps from 0.05 below 10°C to 1.00 above 100°C — with the step from 0.55 to 0.80 placed deliberately at 60°C, the zinc polarity reversal threshold, rather than smoothed across it.

Operating temperature is optional in the element schema and frequently absent. Where it is absent, the specified behaviour is to raise a data-quality issue and omit the temperature term rather than substitute a default, on the grounds that a default is a fabricated input to a compliance verdict. One consequence of that rule is left open in the pack and should be resolved before the engine drives a verdict: omitting the term leaves the composite running on 0.75 of its weight, and the ruleset does not state whether the remaining two weights are renormalised or left to sum short. Left short, every element with unknown temperature scores systematically lower, and could fall below the Medium threshold for that reason alone.

Term	Weight	Source of the value
Material-media compatibility	0.40	Compatibility matrix cell (12 materials × 8 media)
Environment severity	0.35	Wetting-class ladder borrowed from CC-001 (EN ISO 15329:2007)
Temperature stress	0.25	Combined corrosion-kinetics and material-limit term
Risk bands	0.35 / 0.65 / 0.85	Medium / High / Critical — matched to GC-001 and CC-001

Table 13.1: MM-001 composite scoring terms, weights and band thresholds

The band thresholds are deliberately identical to those of GC-001 and CC-001. A Critical band means the same thing whichever engine raised it, which is what allows the combined assessment to take the highest band across engines without rescaling.

Environment class	Severity	CC-001 equivalent	Confidence
T0 dry	0.05	T0_DRY	established
T1 indoor damp	0.20	T1_OCCASIONAL	established
T2 humid	0.40	T2_INTERMITTENT	established
T3 chloride	0.80	T4_PERSISTENT (not T3)	established
T4 marine	1.00	T5_IMMERSION	established
T5 industrial	0.85	none — interpolated	provisional
unclassified	no value	raises a data-quality issue	n/a

Table 13.2: MM-001 environment severity ladder and its mapping to CC-001 wetting classes

Two entries in that table carry the argument. T3 chloride borrows CC-001's T4 class rather than its positional T3, because pool halls and coastal interiors are chloride-dominated and the positional mapping would understate them. T5 industrial has no CC-001 counterpart at all, so its value is interpolated rather than borrowed and is tagged provisional — the weakest cell in the table, pinned by a fixture scenario so that a later edit cannot move it unnoticed.

An unclassified environment raises a data-quality issue and never a numeric severity. Scoring an unclassified environment as benign would silently suppress findings on every element the producer could not classify. This is the same fail-conservative property recorded for MC-001 in §12.7, applied to a different input.

A kinetics guard prevents temperature alone from escalating a benign pairing: where the matrix cell sits below 0.35, the temperature stress term is capped at 0.35. Without it, a material well suited to its medium could be driven into a Medium band by operating temperature that the material comfortably tolerates.

Written out, the guard is a single conditional applied before the sum:

```

if material_media_score < 0.35:
    temperature_stress ← min(temperature_stress, 0.35)

```

Its justification is that the Arrhenius relation multiplies an existing corrosion rate; it is not an independent hazard. A small base rate doubled per ten degrees is still a small rate. The arithmetic makes the consequence concrete. Copper in domestic hot water at 65°C — a pairing every plumbing code in Europe treats as correct — has a matrix cell of 0.30, an indoor-damp environment severity of 0.20 and a temperature stress of 0.80. Unguarded it scores $0.120 + 0.070 + 0.200 = 0.390$ and lands in the Medium band solely because the water is hot. With the temperature term capped at 0.35 it scores 0.278 and correctly produces no finding at all.

The matrix itself is reproduced in full as Table 13.3. Every cell also carries a dominant failure mechanism, a predicted service life and a governing citation, all of which the engine attaches to any finding it raises; only the score is shown here, for legibility. Higher is worse throughout — the score is a corrosivity ranking for the pairing, not a suitability rating.

Material	Cold water	Hot water	Chilled	Condenser	Pool water	Steam	Condensate	Stagnant
Carbon steel	0.65	0.75	0.45	0.70	0.95	0.40	0.80	0.85
Galvanised steel	0.35	0.70	0.30	0.60	0.95	0.90	0.85	0.70
Cast iron	0.45	0.55	0.40	0.60	0.90	0.60	0.75	0.65
Ductile iron	0.40	0.50	0.35	0.55	0.88	0.55	0.72	0.60
Copper C12200	0.15	0.30	0.15	0.35	0.75	0.55	0.45	0.45
Brass C46400	0.30	0.45	0.30	0.40	0.80	0.60	0.55	0.55
Copper-nickel 90/10	0.12	0.18	0.12	0.15	0.45	0.35	0.30	0.40
Stainless SS304	0.15	0.25	0.15	0.45	0.90	0.20	0.30	0.55
Stainless SS316	0.10	0.15	0.10	0.30	0.85	0.15	0.22	0.40
Duplex 2205	0.08	0.10	0.08	0.15	0.20	0.12	0.15	0.20
Titanium	0.03	0.03	0.03	0.05	0.05	0.10	0.08	0.10
PVC	0.05	0.85	0.05	0.15	0.10	1.00	0.80	0.20

Table 13.3: The MM-001 material-media compatibility matrix — twelve materials against eight media, a higher score meaning a more corrosive pairing

Four readings of that matrix carry most of its engineering content. The galvanised steel row steps from 0.35 in cold water to 0.70 in hot: that step is zinc polarity reversal above roughly 60°C, where the coating stops protecting the steel and begins consuming it, and it is the most common cause of premature failure in hot water systems specified from a cold water precedent. The copper row is benign in cold and chilled water at 0.15 but reaches 0.75 in pool water, because free chlorine and the low pH of a badly dosed pool attack copper directly. The PVC row holds the matrix's only 1.00, against steam — not a corrosion process at all but a straightforward specification error, which the engine catches because a cell exists for the pairing. And the titanium and duplex rows stay below 0.20 across every medium, which is the honest reason both are specified in pool and marine plant despite their cost.

The pool water column deserves separate comment, because it is the column the case study in §7.2 turns on. Every ferrous and copper-alloy material in the matrix scores at or above 0.75 in pool water, and SS316 — the grade the specification called for — scores 0.85. That is not an error in the matrix. It is the matrix agreeing with IMO guidance that

austenitic stainless steels are the wrong family for chlorinated pool atmospheres, and it is what allows MM-001 to reach the same conclusion the expert reviewer reached, from the opposite direction, in the feedback loop that section describes.

Table 13.4 works six elements through the composite end to end, including the two cases where a guard or a provisional value changes the verdict.

Element and service	Cell	Env.	Temp.	Guard applied	Composite	Band
Copper, domestic hot water at 65°C, indoor damp plant room	0.30	0.20	0.80	Yes — cell below 0.35, temperature capped at 0.35	0.278	None
The same element with the kinetics guard removed	0.30	0.20	0.80	No	0.390	Medium — a false positive
Carbon steel, domestic hot water at 65°C, indoor damp	0.75	0.20	0.80	No — cell above 0.35	0.570	Medium
Galvanised steel, stagnant fire loop at ambient, indoor damp	0.70	0.20	0.20	No	0.400	Medium
SS316, pool circulation at 28°C, chlorinated plant room (the §7.2 case study)	0.85	0.80	0.35	No	0.708	High
Carbon steel, condenser water at 30°C, industrial atmosphere (T5)	0.70	0.85	0.35	No	0.665	High — by 0.015

Table 13.4: The MM-001 composite worked through six elements, showing where each guard changes the verdict

The last row is the most uncomfortable and the most instructive. It clears the High threshold by 0.015, and it does so on the strength of the T5 industrial environment severity of 0.85 — the one value in the ladder with no CC-001 counterpart, interpolated rather than borrowed, and marked provisional in the pack for exactly that reason. Had that value been set at 0.80, the same element would score 0.648 and band Medium. A single provisional digit, moved by 0.05, carries an element across a band boundary and changes the required action from monitoring to remediation. This is the concrete form of the general warning in §13.10: the rankings in these packs reflect corrosion-engineering consensus, but the digits are a calibration authored for the ruleset, and a band boundary is exactly where that distinction stops being academic.

13.3 XM-001 — cross-material contamination

Where MM-001 asks about one element and its fluid, XM-001 asks about two elements and the joint between them. It scores the galvanic couple formed where dissimilar materials meet, either in direct contact or through shared media in the same loop.

The composite follows the same three-term, weighted-sum form as MM-001, with a multiplicative mitigation factor applied to the result:

$$\begin{aligned}
 \text{Score_XM} = & [(0.50 \times \text{voltage_risk}) + (0.30 \times \text{separation_factor}) \\
 & + (0.20 \times \text{environment_severity})] \times \text{mitigation_factor} \\
 \text{voltage_risk} = & \min(|\Delta \text{ potential}| \div 1.0 \text{ V} , 1.0)
 \end{aligned}$$

The potential gap is the absolute difference between the two materials' entries in the GC-001 anodic index, normalised against a 1.0 V spread — the practical range from the most active to the most noble material used in building services — and clamped above. The separation factor is 1.00 for direct contact, where the elements share a joint and therefore both metallic continuity and an electrolyte, and 0.80 for same-loop proximity, where the two are connected through the piping graph and share a fluid path without touching. The environment term is the same severity ladder MM-001 uses, injected at load rather than duplicated. The mitigation factor is 1.00 unmitigated and 0.10 where a dielectric union breaks metallic continuity across the joint; the residual 0.10 rather than zero is deliberate, because the credit depends on the union being installed and remaining intact, which is a commissioning check and not a design guarantee.

Anode identification is resolved separately from scoring, because naming which material sacrifices is the actionable half of the finding. The engine reads the nobility flag each series entry carries and names the active member as the anode. Where both entries carry the same flag — copper against passive stainless, for instance, where both are noble yet still form a couple — the declared series convention breaks the tie: the index is anodic, so the higher value is the anode. Where nobility cannot be resolved for a pair at all, the engine raises a data-quality issue rather than guessing which element corrodes.

Term or factor	Value	Basis
Voltage risk (weight)	0.50	Potential gap normalised over 1.0 V, clamped above
Separation (weight)	0.30	Joint condition between the two elements
Environment (weight)	0.20	Environment class, via the MM-001 table
Separation — direct contact	× 1.00	Shared joint: metallic continuity plus shared electrolyte
Separation — same loop	× 0.80	No joint, but a shared electrolyte path
Mitigation — dielectric union	× 0.10	JT-014 breaks metallic continuity; residual covers installation error

Table 13.5: XM-001 composite scoring terms, separation and mitigation factors

The pack holds no galvanic series of its own. The series lives in exactly one place, the GC-001 ruleset, and XM-001 reads it from there. That single-source rule is what made an inconsistency findable. Approval flipped the series convention from more-negative-is-anodic to more-positive-is-anodic after verification against the seeded GC-001 payload, which is an anodic index rather than an electrode-potential series: all twenty values are positive and ascend from platinum at 0.00 to magnesium at 0.95. The check resolved a live contradiction between two engines — the deployed galvanic engine was correct and the dormant one was wrong, and the dormant one was corrected. Because the engine holding the wrong convention had never been wired into the live pipeline, no user-facing finding ever named the wrong material as the anode. The verification is recorded against the recovered ruleset payload rather than the live database, and is re-runnable.

A compatibility floor suppresses couples below the environment's NASA-STD-6012 threshold entirely: no finding at all, rather than a Low-band finding. Above that floor every dissimilar couple produces an issue, including mitigated ones banded Low, because the audience is modellers who do not routinely consider dissimilar-metal joints and for whom a visible Low is more useful than silence.

Stated as a rule, the floor precedes the composite entirely:

if $|\Delta \text{ potential}| < \text{threshold}(\text{environment})$: no finding

The thresholds are the NASA-STD-6012 galvanic compatibility limits carried by GC-001 — 0.50 V in a controlled environment, 0.25 V in a normal one and 0.15 V in a harsh one — reached by mapping the element's environment class onto the GC-001 class of matching meaning rather than of matching position. The reason the floor suppresses rather than downgrades is arithmetic. Naval brass against copper is a gap of 0.04 V in the series, compatible in every environment the standard defines. Scored anyway, in direct contact in an indoor damp plant room, it returns $0.020 + 0.300 + 0.040 = 0.360$ and bands Medium — a finding generated almost entirely by the separation weight, purely because the two elements touch. Separation amplifies galvanic risk; it is not an independent hazard, and with no meaningful driving voltage there is nothing for contact to amplify.

XM-001 carries no potentials of its own. The series it scores against is the GC-001 anodic index, reproduced as Table 13.6 for reference; the engine reads it at load time from the single catalogue that also serves GC-001, so the two engines cannot disagree about which material is the more active. Every value is positive and the index ascends from the noble end to the active end, which is the opposite orientation from a conventional electrode-potential series and is the source of the sign-convention defect recorded in §13.10.

Material	Anodic index (V)	Nobility flag	Typical form in MEP services
Platinum	0.00	Noble	Not used in building services; the index datum
Gold	0.02	Noble	Contacts and instrumentation only
Graphite	0.03	Noble	Gaskets and packings — a cathode in disguise
Titanium	0.05	Noble	Pool and marine heat exchangers
SS316 passive	0.08	Noble	Process, pool and pharmaceutical pipework
SS304 passive	0.12	Noble	General sanitary and food-grade pipework
Hastelloy C	0.14	Noble	Chemical dosing lines
Silver solder	0.18	Noble	Brazed copper joints
SS316 active	0.22	Active	Depassivated stainless in a crevice or under deposit
Copper	0.28	Noble	Domestic hot and cold water, heating, medical gas
Brass	0.32	Active	Valves, fittings, tap ware
Bronze	0.34	Active	Pump bodies, valve trim
SS304 active	0.38	Active	Depassivated SS304
Cast iron	0.52	Active	Drainage, older heating mains, pump housings
Carbon steel	0.55	Active	Heating, chilled water, fire suppression, steam
Aluminium	0.70	Active	Ductwork, supports, some heat exchangers
Cadmium	0.75	Active	Plating on fasteners
Zinc	0.80	Active	Coatings and sacrificial anodes
Galvanised steel	0.82	Active	Ductwork, fire loops, older cold water risers
Magnesium	0.95	Active	Sacrificial anodes in cylinders and calorifiers

Table 13.6: The GC-001 anodic index as read by XM-001 — a higher value means more active, and the more active member of a couple is the anode that sacrifices

Table 13.7 applies the floor to the couples a building services model actually produces, giving for each the potential gap, the member that sacrifices, and the environments in

which the couple is reported rather than suppressed. The asymmetry in that last column is the engine's whole argument: the same physical junction is a finding in a pool plant room and not a finding in a dry controlled riser, because the electrolyte is what closes the cell.

Couple	Δ (V)	Anode (sacrifices)	Reported in	Comment
Copper — carbon steel	0.27	Carbon steel	Normal (0.25 V) and harsh (0.15 V); suppressed in controlled (0.50 V)	The most common dissimilar-metal junction in building services
Copper — galvanised steel	0.54	Galvanised steel	All three environment classes	Copper tube teed into a galvanised riser: the classic meter-set failure
Copper — cast iron	0.24	Cast iron	Harsh only; suppressed in normal by 0.01 V	The clearest demonstration that the floor is a cliff rather than a slope
Brass — galvanised steel	0.50	Galvanised steel	All three; exactly at the controlled threshold, so reported there too	Brass fittings on galvanised pipework
Zinc — copper	0.52	Zinc	All three	The physics anchor used to verify the series convention: galvanising sacrifices the zinc
SS316 passive — carbon steel	0.47	Carbon steel	Normal and harsh	Stainless process pipework bracketed or joined to steel
SS316 passive — copper	0.20	Copper	Harsh only	Both members noble, so the declared convention breaks the tie and names copper
Aluminium — copper	0.42	Aluminium	Normal and harsh	Copper pipework in contact with aluminium ductwork or supports
Brass — copper	0.04	Brass	None — suppressed everywhere	The couple the compatibility floor exists to suppress
Carbon steel — cast iron	0.03	Carbon steel	None — suppressed everywhere	Two active materials near-identical in the series
SS316 active — copper	0.06	Contested	None — suppressed everywhere	The one internally inconsistent pair: the nobility flag names SS316 active as the anode, the index ordering names copper. See §13.10

Table 13.7: Galvanic couples common in MEP services, with the environments in which the NASA-STD-6012 compatibility floor reports each

Table 13.8 works the arithmetic through in full for the cases that exercise each term of the composite, including the mitigated case and the suppressed one.

Case	Voltage risk	Separation	Env.	Mitigation	Composite and band
Copper to carbon steel, direct contact, indoor damp plant room	0.27	1.00	0.20	1.00	0.475 — Medium; carbon steel is the anode
The same junction with a dielectric union fitted	0.27	1.00	0.20	0.10	0.048 — Low, still reported, with a commissioning-verification note
The same two materials on one loop but not touching	0.27	0.80	0.20	1.00	0.415 — Medium; lower than direct contact

Case	Voltage risk	Separation	Env.	Mitigation	Composite and band
Brass fitting on galvanised pipework, direct contact, humid plant room	0.50	1.00	0.40	1.00	0.630 — Medium; the galvanising sacrifices
Copper to galvanised steel, direct contact, chlorinated pool plant room	0.54	1.00	0.80	1.00	0.730 — High
SS316 to carbon steel, direct contact, chlorinated pool plant room	0.47	1.00	0.80	1.00	0.695 — High; carbon steel is the anode
Naval brass to copper, direct contact, indoor damp	0.04	1.00	0.20	1.00	No finding — suppressed by the floor. Unsuppressed it would score 0.360, Medium

Table 13.8: The XM-001 composite worked through seven junctions, including the mitigated and the suppressed case

The second and last rows are the two behaviours that most distinguish XM-001 from a naive dissimilar-metals check. A mitigated junction is reported, banded Low, and carries a commissioning note rather than a design action — because the audience is modellers who do not routinely think about dissimilar metals, and showing a mitigated couple teaches the pattern while hiding it teaches nothing. A compatible pair is not reported at all, because it is not a couple. The difference between downgrading and suppressing is the difference between a tool a coordinator keeps using and one they switch off.

13.4 Integration — two comparator paths

The two engines do not share the output shape of the first three, and the difference is structural rather than cosmetic. The established path emits one flat record per element, always, whether or not anything is wrong. The extended path emits an issue only where there is a finding, and an XM-001 issue references two elements rather than one.

Three differences drive the adapter between them. Cardinality: one row per element against zero or more issues per element. Absence semantics: a compliant element is a Low row on the first path and nothing at all on the second, so merging naively would make every compliant element appear unassessed by MM-001 and XM-001. Identity: both key on the IFC GUID, so the join key already exists and needs no invention.

The engines sit behind separate feature flags, both defaulting to off. They are deliberately not on one switch. MM-001 runs entirely from disk, while XM-001's loader reads the GC-001 series from the database and must degrade to 'not assessed' if that read fails, never to an empty result that would read as 'no couples found'. Reporting absence of data as absence of risk is the specific failure this separation prevents.

13.5 Validation

Both engines are covered by automated test suites — 29 tests for MM-001 and 39 for XM-001. The suites are organised by what each group protects, and three of those groups test the ruleset rather than the arithmetic: that the pack carries an approved status, that its weights sum to one, and that every cell in the compatibility matrix carries a citation. A ruleset that lost a citation would fail the suite before any scoring was exercised.

Test group	MM-001	XM-001	What it protects
Rule pack contract	4	4	Approved status, required keys, weights sum to one, every cell cited
Scenario verdicts	7	5	Named cases land in the expected band
Data quality / tier semantics	5	3	Missing input never scored as benign
Audit trail	5	2	Findings carry citations and match their stored score
Mechanism-specific	4	21	Kinetics guard; anode identification, mitigation, separation, floor
Output shape	4	4	Unique issue ids, real element references, silence when compliant

Table 13.9: Test coverage for MM-001 and XM-001 by group

Two groups carry the arguments made earlier in this chapter. The data-quality group enforces the fail-conservative property: an unclassified environment, an unknown material and an unmapped pairing each raise a data-quality issue, and a separate test asserts that a data-quality issue never presents as a scored verdict. The provisional T5 industrial cell of §13.2 is pinned by its own scenario, so an edit to that value fails the suite rather than passing unnoticed — which is what makes the word ‘provisional’ enforceable rather than decorative.

For XM-001 the single-source contract is itself a test: the pack file is asserted to hold no material potentials, so the galvanic series cannot silently fork into a second copy. The compatibility floor is tested to suppress a compatible pair rather than downgrade it, to vary its threshold by environment, and to fail open on an environment it cannot map — reporting the couple rather than discarding it.

These are unit and contract tests on synthetic fixtures. They establish that the engines behave as their rulesets specify; they do not establish that the rulesets are correctly calibrated against built assets, which is the distinction §14.1 draws between what a test can prove and what only an expert can judge.

13.6 Field failure patterns detected by MM-001 and XM-001

Both engines were specified from failure patterns rather than from a desire for completeness. Table 13.10 sets out the pattern behind each, the score the engine returns for it, and — the column that justifies adding two engines to a system that already had three — what the earlier engines see when they look at the same element.

Field failure pattern	Typical setting	Engine and score	What the first three engines see
Galvanised steel domestic hot water above 60°C: the zinc coating reverses polarity and consumes the steel it was applied to protect	Hot water risers specified from a cold water precedent	MM-001, cell 0.70, composite 0.550 Medium at 65°C	GC-001 sees no second element and so no couple; CC-001 sees no crevice; MC-001 sees adequate flow above the microbial optimum and returns Low

Field failure pattern	Typical setting	Engine and score	What the first three engines see
Copper tube teed into a galvanised riser at a meter set or booster set	Plant rooms, riser cupboards, incoming mains	XM-001, gap 0.54 V, composite 0.730 High in a chlorinated plant room	MM-001 sees two materials each individually acceptable in cold water; the failure is created by the junction, not by either element
Carbon steel on open condenser water in a cooling tower circuit	Heat rejection plant, evaporative condensers	MM-001, cell 0.70, reaching High where the atmosphere is industrial	GC-001 and CC-001 return nothing; the element is simply the wrong material for an oxygen-saturated, continuously replenished medium
SS316 pool circulation pipework in a chlorinated plant room	Leisure centres, hydrotherapy pools, spa plant	MM-001, cell 0.85, composite 0.708 High	CC-001 scores the gasketed flange faces; MM-001 scores the whole run. The two reach the same verdict by different routes, which is the §7.2 case study
A polymer specified on a steam or condensate service	Small plant rooms where one pipe schedule was applied to every service	MM-001, cells 1.00 for steam and 0.80 for condensate	No corrosion engine detects it, because it is not a corrosion process — it is a specification error the matrix happens to have a cell for
Brass fittings throughout galvanised pipework	Refurbishments where fittings and pipe were procured separately	XM-001, gap 0.50 V, composite 0.630 Medium	Neither MM-001 nor GC-001 flags the combination; dezincification of the brass itself remains outside all five engines

Table 13.10: Field failure patterns detected by the extended engines, and what the first three engines return for the same element

The division of labour reduces to a single sentence. MM-001 asks about one element and the fluid inside it; XM-001 asks about two elements and the joint between them. Neither question is answerable from the other's inputs, which is why they are separate engines rather than terms in one score.

13.7 Planned validation against the 38-model dataset

The test suites described in §13.5 are unit and contract tests on synthetic fixtures. They can establish that the engines behave as their rulesets specify; they cannot establish that the rulesets can be evaluated at all against models authored by other people in other tools. The 38-model validation dataset documented in Appendix A exists to answer that second question, and it matters more for these two engines than for the first three, because MM-001 and XM-001 need inputs that IFC does not natively carry.

Engine and input	Role in the score	Where it comes from in IFC	Availability across the corpus	What the run establishes
MM-001 — material	Matrix row	Material assignment on the pipe entity	Good on Full-tier models; absent on Partial and Architecture-only tiers	Whether material strings from eight different repositories map onto the twelve canonical materials, or fragment into unrecognised variants

Engine and input	Role in the score	Where it comes from in IFC	Availability across the corpus	What the run establishes
MM-001 — medium	Matrix column	Inferred from IfcDistributionSystem PredefinedType; not read directly	Available wherever systems are classified, and system typing is more consistently exported than material	Whether system type is a reliable proxy for medium, or whether it collapses distinct media onto one classification
MM-001 — environment class	Weight 0.35	No IFC property corresponds to it; it must be assigned per zone by a human	Absent from every model in the corpus	This is the pre-registered headline: MM-001's dominant output on open models will be the unclassified-environment data-quality issue, by design rather than by defect
MM-001 — operating temperature	Weight 0.25	Pset_PipeSegmentOccurrence.OperatingTemperature, or the containing zone set point	Sparse; most likely on the DigitalHub heating model	How often the temperature term must be omitted, and therefore how often the composite runs on partial weight
XM-001 — material pair	Weight 0.50, voltage	Two materials on two joined elements	Requires genuine material diversity; Clinic_Plumbing carries five material strings across 378 pipe entities	Whether real models contain the dissimilar-metal junctions the engine exists to find, or whether authors model one material throughout
XM-001 — joint topology	Weight 0.30, separation	Port connectivity on the IfcDistributionSystem graph	Present wherever connectivity is exported; not guaranteed	Whether direct contact can be distinguished from same-loop proximity at all on third-party models

Table 13.11: Input availability for MM-001 and XM-001 across the 38-model corpus

One model carries a disproportionate share of the XM-001 test weight. Clinic_Plumbing.ifc is the only file in the corpus verified by inspection to carry real material text for copper, PVC, CPVC, steel and brass across 378 pipe segments and fittings — which is to say it contains, in one file, both the couple the engine should report (copper against steel, a gap of 0.27 V) and the couple it should suppress (copper against brass, a gap of 0.04 V). That makes it a direct test of the compatibility floor on real data rather than on a fixture, and it is the single most informative result the run can produce for this engine. The West Riverside plumbing and sprinkler models supply volume, and the DigitalHub sanitary and

heating models supply an independently authored European IFC4 export as a guard against conclusions that hold for one tool only.

The environment-class row in Table 13.11 states a result in advance, and it is worth being explicit about why that is not an admission of failure. MM-001 weights environment at 0.35 and specifies that an unclassified environment must raise a data-quality issue rather than a numeric severity, because scoring an unknown environment as benign would silently suppress genuine findings. No IFC model in the corpus carries an environment classification, because IFC has no property for one. The engine will therefore emit data-quality issues across the corpus rather than risk bands, and that outcome measures a gap between what corrosion assessment needs and what the schema carries — which is a contribution of this thesis rather than a shortcoming of the engine.

13.8 Expected severity distributions across the five engines

Table 13.12 states the expected band distribution for each engine before the run, alongside the distribution its designed fixtures produced. The two columns are expected to differ sharply, and the reason is the same for every engine: fixtures are written to exercise the extremes of a scoring model, so they over-sample Critical and Low and under-sample the middle, while a real corpus is dominated by ordinary pipework and by absent inputs.

Engine	Band thresholds	Designed fixtures	Expected across the corpus	Dominant expected outcome
GC-001	0.35 / 0.65 / 0.85	25 elements: 9 Critical, 7 High, 9 Medium, 0 Low (Table 5.1)	Findings only where two materials meet; most elements produce nothing	Sparse findings concentrated at material transitions, with no Low band because compatible pairs are suppressed rather than banded
CC-001	0.30 / 0.55 / 0.80	Scenario set covering the five CCT grades	Requires joint type, which IFC does not carry as a property	Data-quality issues dominant; risk bands only where joint type can be inferred from fitting class
MC-001	0.25 / 0.50 / 0.75	10 scenarios: 4 Critical, 2 High, 0 Medium, 4 Low (Table 12.4)	Low-dominant, with a Critical tail concentrated in fire suppression	See Table 12.7 — the conservative velocity default is the confound to watch
MM-001	0.35 / 0.65 / 0.85	Scenario verdicts pinned per matrix cell	Data-quality dominant, because environment class is absent from every model	Unclassified-environment issues outnumbering risk findings, by specification
XM-001	0.35 / 0.65 / 0.85, plus suppression below the floor	39 contract tests over eight materials	Findings only on models with genuine material diversity	A small number of real couples on Clinic_Plumbing and the West Riverside models; nothing at all on single-material models, which is correct behaviour and must not be read as coverage

Table 13.12: Expected severity distributions per engine, against the distributions the designed fixtures produced

Across all five engines, the single most likely outcome of the run is that data-quality issues outnumber risk findings, on most models by a wide margin. That result should be reported as the headline rather than buried as a disappointment. It quantifies, on a reproducible corpus of thirty-eight models from eight repositories, the distance between the information a corrosion assessment requires and the information an IFC model as commonly exported actually carries — and it does so with an engine set that refuses to close that distance by guessing. A system that defaulted the missing inputs would report a clean distribution of risk bands, and would be reporting fiction. The measured gap is the more useful number.

13.9 Implementation status of the extended engines

The specification set out in §13.2 and §13.3 is the approved v1.0 ruleset. The code currently on disk implements an earlier and simpler scheme, and the difference is material enough to record explicitly rather than leave a reader to discover it. Table 13.13 states, term by term, what the packs specify and what the comparator modules do.

Element of the specification	In the pack	In the code as it stands	Status
MM-001 composite, 0.40 / 0.35 / 0.25	Specified	material_media.compare() bands the raw matrix cell; no weighted composite is computed	Not implemented
MM-001 kinetics guard	Specified	Absent — there is no temperature term to cap	Not implemented
Unmapped pairing raises a data-quality issue	Specified	An absent cell returns 0.0 and the element is treated as compliant — the exact silent defaulting the pack forbids	Contradicts the pack
Nested matrix, indexed [material][medium]	Specified	The code indexes a flat "material:medium:environment" key	Incompatible shapes
MM-001 pack loading	Under parameters.compatibility_matrix and its siblings	load_rule_pack() requires top-level materials, media, environments and compatibility_matrix keys	Raises ValueError on the approved pack
XM-001 composite, 0.50 / 0.30 / 0.20, times mitigation	Specified	cross_material.compare() bands the raw couple voltage; separation and mitigation are not applied	Not implemented
Compatibility floor	Specified	Threshold suppression is present, but reads a rule-pack key the approved pack deliberately does not carry	Present but unreachable
Single-source series injection	Specified	load_rule_pack() accepts no galvanic_series argument, so the series cannot be injected	Not implemented
Anode identification from the nobility flag	Specified	Not performed; findings name the pair but not which member sacrifices	Not implemented
Test suites, 29 for MM-001 and 39 for XM-001	Written against the specified behaviour	Executed against the modules on disk, 3 of 29 and 1 of 39 pass; the remainder fail or error at fixture setup on the changed signatures	Suites do not currently run

Table 13.13: MM-001 and XM-001 — approved specification against implemented code

The last row qualifies the coverage figures quoted in §13.5. Those figures describe the suites as written — 29 tests for MM-001 and 39 for XM-001, organised by the property each group protects — and they are an accurate account of what has been specified and encoded as executable intent. Measured against the comparator modules as they stand, 3 of the 29 and 1 of the 39 execute and pass; the remainder fail at fixture setup, because they call a loader signature and a compare() signature the shipped modules do not expose. The suites are therefore evidence of a specified contract rather than evidence of a working engine, and §13.5 should be read with that distinction in place.

This is the same class of honesty §14.6 applies to the enforcement gate, and it is recorded here for the same reason. The rulesets are approved, reviewed and internally consistent, and the arithmetic in Tables 13.4 and 13.8 is the arithmetic the packs specify, verifiable by hand against the published weights. What has not yet happened is the wiring of those packs into the comparator, and no claim in this chapter should be read as asserting that it has. The 38-model run described in §13.7 is contingent on that work being completed first, and the ordering matters: running the corpus against the current modules would measure the precursor scheme and report it as though it were the approved one.

13.10 Approval record and carried limitations

Both rulesets are approved at version 1.0 with a named approver and a recorded amendment list. Four limitations are carried in the pack rather than hidden.

1. The T5 industrial severity of 0.85 is provisional. It is the only environment value with no CC-001 counterpart, so it is interpolated rather than borrowed.
2. Water chemistry — chloride concentration, pH, hardness and dissolved oxygen — is absent from the element schema. Cells assume typical UK potable chemistry, so copper Type 2 pitting in soft-water regions is understated.
3. Flow velocity is absent, so copper erosion-corrosion above roughly 1.5 m/s and the minimum-flow film maintenance that copper-nickel requires are not modelled.
4. The digits are a calibration authored for these rulesets, not values quoted from the cited standards. The citations name the standard governing each mechanism; they do not supply the number.

The fourth is the most important and the least comfortable. It is the same distinction §4.6.4 draws between a normative value and a provisional one, applied here to material chemistry: a citation that establishes a mechanism is not a citation that establishes a threshold, and a ruleset that conflates the two would claim more authority than it holds.

Chapter 14 — The Expert Review Process

14.1 Why the process exists

Two of the three components produce output that unit tests cannot validate. A test can prove that an extracted rule is schema-valid and executable; only a domain expert can judge whether it means what the clause means. A test can prove that a composite score matches the published formula; only an expert can judge whether the resulting risk band is engineering-credible for that element.

The generic comparator is deliberately excluded as a review object. It performs deterministic threshold evaluation whose correctness is fully decidable by unit tests, and subjecting it to expert review would spend scarce specialist time on the one component that does not need it. That leaves two review objects: a candidate rule produced from a regulatory clause, and a compliance finding produced by a corrosion engine.

14.2 Roles and independence

Five roles carry the process. The Extraction Author is the system itself, producing candidates with confidence, source text and clause reference populated; it may approve nothing. The Reviewing Expert scores candidates against the rubric and records a

decision with a rationale. The BIM Coordinator confirms the rule is checkable against the delivered model, holding a veto on IFC-mappability alone. The Accountable Approver publishes approved rules and owns the audit trail, an ISO 19650 information-management responsibility. The Process Owner aggregates rejection reasons and owns the extraction prompt and the engine tables.

Roles are functions rather than people, and one person may hold several on a small project, but two separations are absolute. The Extraction Author and the Reviewing Expert must never be the same individual, and the Accountable Approver must never be the person who authored the amendment being approved. For any rule of mandatory severity the Reviewing Expert must additionally be independent of whoever authored the source specification. This mirrors ordinary design-assurance practice and stops the process degenerating into self-certification.

14.3 States and the enforcement gate

Every review object moves through one state machine: DRAFT, IN_REVIEW, AMENDED, REJECTED, APPROVED, PUBLISHED and SUPERSEDED. State is a property of the record rather than of a person's inbox, which is what makes the process auditable after the fact rather than only observable while it runs.

One property of that machine matters more than the rest: only PUBLISHED rules are evaluated. That single constraint is the enforcement gate, and §14.6 records honestly that it is the part of this specification not yet built.

14.4 The review rubric

A Reviewing Expert scores each candidate rule on five dimensions from 1 to 5. A rule is approved only if every dimension scores at least 4 and Traceability scores 5. There is no approval with reservations; anything short of that threshold is amended or rejected.

Dimension	The question the expert answers
Traceability	Can I find this requirement in the cited clause?
Semantic fidelity	Does the rule mean what the clause means?
IFC mappability	Can this be checked against a delivered model?
Executability	Will the comparator produce a decidable result?
Scope correctness	Are the conditions and exceptions right?

Table 14.1: The five-dimension review rubric for a candidate rule

For a compliance finding the rubric changes to four dimensions at the same threshold: input fidelity, term plausibility, band credibility and action proportionality. The question shifts from whether the rule was extracted faithfully to whether a competent engineer would place this element in this band and act on it this way.

Every rule of mandatory severity is reviewed. For the remainder a stratified random sample of 20% is reviewed, stratified by rule type because the failure modes observed in development cluster by rule type rather than by clause. A further 20% overlap is double-reviewed by two independent experts to measure inter-rater agreement. A Cohen's kappa below 0.6 on any dimension is treated as a defect in the rubric rather than in the reviewers, and triggers a rubric revision before reviewing continues.

14.5 Failure classes and the feedback cycle

Expert review is not only a filter. Its primary product is a classified rejection record that drives systematic improvement, so every rejection or amendment is tagged with one failure class and each class routes to a named owner.

Class	Typical cause	Routed to
F1 Citation	Clause reference wrong or hallucinated	Extraction prompt
F2 Semantic	Operator inverted, threshold or unit wrong	Prompt and few-shot examples
F3 Mapping	Target property is not a real IFC property	IFC enrichment maps
F4 Scope	Conditions or exceptions dropped	Extraction prompt
F5 Granularity	One clause split into too many or too few rules	Section chunking
F6 Engine	Corrosion band judged not credible	Engine tables and thresholds
F7 Input	Model data read incorrectly	IFC parser

Table 14.2: Failure classes and the owner each routes to

The cycle runs on the project's weekly coordination rhythm so that review effort and model drop coincide: extraction on Monday, review on Tuesday and Wednesday, triage on Thursday, regression on Friday, publication ahead of the next drop. Any failure class exceeding 10% of a week's decisions becomes a corrective action with a named owner. The Friday step carries the safeguard: a prompt or table change is kept only if the golden-set score does not regress, which is what stops the loop over-fitting to last week's rejections.

Five metrics are tracked per cycle — first-pass acceptance rate, amendment rate, rejection rate by class, median review time per item, and inter-rater agreement on the double-reviewed overlap. A sixth, the defect escape rate of approved rules later found wrong, is the lagging indicator that reveals whether the rubric is calibrated at all.

14.6 Implementation status

This specification exists because an earlier draft asserted that expert review validated the engines' risk-band distribution and that a human-in-the-loop safeguard governed LLM-extracted rules, without defining who reviews, against what criteria, at what cadence, or what becomes of the finding afterwards. Defining the process makes the gap between specification and implementation visible, and that gap is stated here rather than left to be discovered.

Element of the process	Status
Review flag, confidence score and clause traceability on every rule	Implemented
Query for items awaiting review; schema and executability pre-check	Implemented
Reviewer-facing interface	Partial — displays rules, but has no scoring form

Element of the process	Status
The state machine of §14.3	Not implemented
The enforcement gate — evaluating PUBLISHED rules only	Not implemented
Reviewer identity, decision, rationale and timestamp	Not implemented
Failure-class tagging and aggregation	Not implemented
Review of compliance findings	Not implemented
Golden-set regression on prompt change	Blocked — harness calls a missing method

Table 14.3: Implementation status of the expert review process

One line in that table governs the honesty of the whole claim. Because the enforcement gate does not exist, the comparator evaluates saved rules regardless of whether they have been reviewed, which makes the human-in-the-loop claim partially true rather than true. The minimum viable remedy, in order, is to add a review-state column and gate the comparator's rule query on it, then add a review record capturing reviewer, decision, rationale, failure class and timestamp, then build the scoring form over the existing interface, and finally repair the regression harness. The first two are small schema changes and are the highest-leverage work identified anywhere in this project.

14.7 The 38-model run as a source of review objects

The process specified in this chapter has, so far, had almost nothing to review. Expert review is defined over candidate rules and compliance findings, and both have existed only in the small numbers that designed fixtures produce. The 38-model validation run described in Appendix A changes that arithmetic, and it changes it enough to expose a problem in the sampling rule of §14.4 that was invisible at fixture scale.

Object produced by the run	Review status under §14.1	Expected volume	Consequence for the process
Compliance findings — risk bands from the five engines	Reviewable. Scored on the four-dimension finding rubric: input fidelity, term plausibility, band credibility, action proportionality	Unknown until the run, but bounded above by the corpus's piping population — of the order of 15,000 pipe segments across the models whose entity counts are published	A 20% stratified sample of that population is not reviewable by any human panel. Sampling must move from per-element to per-pattern
Data-quality issues — absent environment, unmapped pairing, unknown material	Not reviewable. They assert the absence of an input rather than a judgement about a building, so no corrosion expertise can confirm or refute them	Expected to exceed the compliance findings on most models (§13.8)	They route to the modeller or the exporter, not to the Reviewing Expert. Counting them as review objects would swamp the panel with items it cannot act on
Blue Halo clearance clashes	Reviewable in principle, but the correctness of a geometric overlap is decidable by test, in the way §14.1 excludes the generic comparator for	One per intruding element per halo	Excluded from expert review on the same reasoning that excludes the comparator

Object produced by the run	Review status under §14.1	Expected volume	Consequence for the process
Extracted rules	Reviewable. The five-dimension rubric of Table 14.1 applies unchanged	None. The run evaluates existing rulesets and extracts nothing	The run exercises the finding half of the process only; the rule-extraction half remains evidenced by the §7.2 case study alone

Table 14.4: Objects produced by the 38-model run, and which of them the review process can act on

The first row states a real defect in the sampling rule as written. Reviewing 20% of every non-mandatory finding is affordable when a project produces dozens; it is not affordable when a corpus produces thousands, and scaling the percentage down would sample the same distribution more thinly rather than sampling it better. The fix implied by the run is to stratify by finding pattern rather than by element: one copper-to-carbon-steel junction in a humid plant room is the same review question as the two hundredth, and an expert who has ruled on the pattern has ruled on all of them. What still needs individual review is the first instance of each pattern, every mandatory-severity finding, and any finding whose inputs differ materially from the pattern already ruled on. That reduces a five-figure review population to a two-figure one without reducing coverage in any sense that matters.

The second row draws a distinction the process did not previously need. A data-quality issue is not a weaker compliance finding; it is a different kind of statement, and it is addressed to a different person. That an element's environment was never classified is a message to whoever owns the model, and a corrosion expert asked to review it can only agree that the property is indeed absent. Routing these to the Reviewing Expert would consume the scarcest resource in the process on the one class of finding that requires no expertise at all. They belong in the failure-class routing of Table 14.2, owned by the BIM Coordinator.

One constraint governs everything in this section. The enforcement gate recorded as absent in §14.6 is still absent, so nothing in the run is gated on review having happened: the comparator evaluates saved rules whether or not they were published, and the findings the run produces are evidence about the engines rather than a compliance verdict on the models. Until that gate exists, the value of the 38-model run to this chapter is that it sizes the review problem honestly — which is a precondition for building the gate, not a substitute for it.

Chapter 15 — Limitations and Future Work

15.1 The status of the two subsystems

This thesis presents two subsystems that have reached materially different levels of maturity, and stating them as one would misrepresent both. The Blue Halo seismic bracing clearance engine is verified against its specification and measured at scale. The five-mechanism corrosion stack is an approved and traceable ruleset set attached to an implementation that is not yet fit to produce verdicts on arbitrary models. Table 15.1 separates what has been demonstrated from what has not, component by component.

Component	Evidence that exists	Evidence that does not yet exist	Status
Blue Halo generation and clash detection	11 of 11 correctness checks pass against the real EN 1998-1 + DIN 4149 configuration, covering envelope expansion, intersection, severity banding and angle-constraint enforcement	Execution across the 38-model corpus; behaviour against geometry authored outside the project's own fixtures	Verified against specification
Blue Halo IFC path	Driven end to end through ifcopenshell on a real IFC fixture, exercising placement matrices, tessellated vertices and metre-to-millimetre unit scaling	Confirmation that unit handling holds across projects authored in other tools and other unit conventions	Verified on one model
Blue Halo performance	Stage-separated timing and memory from 100 to 20,000 elements at $n = 7$ repeats, reported as median with inter-quartile range	Figures from any host other than the single four-core machine used	Measured, single host
Blue Halo BCF and Pset export	A structurally valid BCF 2.1 archive with one topic folder per clash, and a JSON-serialisable Pset_HaloReservation per envelope	Confirmation that exported topics reference component GlobalIds that exist in third-party source models	Verified against specification
GC-001, CC-001 and MC-001 scoring	Each computes its published weighted composite correctly and discriminates between populated elements; MC-001 additionally validated across ten designed scenarios	Any demonstration on elements drawn from third-party models rather than constructed fixtures	Correct on supplied inputs
MM-001 and XM-001 scoring	Approved v1.0 rulesets with named approver, amendment record and carried limitations; 68 tests encoding the specified contract	An implementation matching those packs — the comparator modules run an earlier scheme and cannot load the approved files (§13.9)	Specified, not implemented
Element coercion layer	None. The layer is not specified in any ruleset and has no test of its own	Any behaviour under absent input other than substitution of a default (§15.2)	Defective
Expert review enforcement gate	A complete process specification: roles, states, rubric, failure classes and cadence (Chapter 14)	The gate itself. The comparator evaluates saved rules regardless of review state (§14.6)	Specified, not implemented
38-model corpus run	A fixed protocol, metric schema, presentation format and pre-registered expectations (Appendix A)	The run	Not yet executed

Table 15.1: Component readiness — what has been demonstrated, and what has not

Read together, those rows support a single summary: Blue Halo is a validated prototype awaiting corpus-scale confirmation, and the corrosion stack is alpha software attached to a reviewed and defensible knowledge base. The rulesets are the mature artefact; the code that feeds them is not.

15.2 A single defect at the engines' entry point

The most consequential limitation in the corrosion subsystem is not distributed across the engines. It sits at one boundary, and it was found by a deliberately minimal experiment: an element carrying nothing but a GlobalId and an IFC class — which is what a pipe segment in a third-party export frequently amounts to — was passed to all three of the runner's compliance entry points. None of the three declined to answer.

Engine	Properties absent	Values substituted	Verdict returned	Failure mode
GC-001	Material, paired material, area ratio, zone	Material and paired material both default to carbon steel; areas default to 1.0 m ² each; zone resolves to E2_NORMAL	Low, 0.26	The engine compares carbon steel with carbon steel, finds a 0.0 V gap and reports no couple. A silent false negative: the element is ranked safer than a real copper-to-carbon-steel junction, which scores Medium at 0.59
CC-001	Material, joint description, operating temperature, zone	Material defaults to stainless steel; joint description defaults to "tight"; temperature defaults to 20°C; environment resolves to BUILDING_SERVICES	High, 0.643	The default joint is the most severe crevice geometry in the library and the default material is the only family the engine scores. A false positive, banded above every populated element tested, which scored Medium between 0.332 and 0.370
MC-001	Flow velocity, operating temperature, dead-leg length, material	Velocity resolves to FV0_STAGNANT, temperature to T5_UNKNOWN, dead leg to DL5_UNKNOWN, and the default material carries maximum susceptibility	Critical, 0.808	Four conservative defaults compound. The empty element outranks a genuinely stagnant carbon steel fire main, which scores Critical at 0.751 — absence of data ranked above presence of hazard

Table 15.2: One element carrying no properties, passed to three compliance entry points

The same element is therefore simultaneously Low, High and Critical. None of those three verdicts is a statement about a building; each is a statement about a default. And the defect runs in both directions at once, which is what makes it more serious than a simple calibration error: GC-001 fails safe-looking and under-reports, while CC-001 and MC-001 fail loud and over-report. A user could not correct for it by treating all output as conservative, because it is not consistently conservative.

The engines themselves are not the defect, and this distinction matters for the remediation estimate that follows. Given populated elements the same three engines behave as their rulesets specify: a well-specified copper hot water element returns Low, Medium and Medium; introducing a copper-to-carbon-steel junction in a humid zone with an unfavourable one-to-ten area ratio moves GC-001 to Medium at 0.59 while leaving the other two unchanged; and making the same steel element a stagnant fire main moves MC-001 to Critical at 0.751 while leaving GC-001 and CC-001 unchanged. Each engine responds to the term it owns and to nothing else, which is the behaviour the weighted composites were designed for.

The fault lies in the coercion layer between the IFC element and the engines' input models. Three functions translate a generic element into GCElement, CCElement and MICElement, and all three share one pattern: read the property, fall back to an attribute, fall back to a literal. That final fallback is unconditional. There is no notion of a required property, no way for the translation to fail, and no representation in the runner's return payload — which carries a band, a score and a details dictionary — for the statement "this element could not be assessed". Because the payload cannot say it, the layer does not attempt it.

It follows that this is one defect with three symptoms rather than three defects, and that the project has already reasoned its way to the correct rule elsewhere. MM-001 states that a material and medium pairing absent from its matrix must emit a data-quality issue and never a default score, on the explicit grounds that silent defaulting would report unverified

pairings as compliant. Its temperature policy says the same of an absent operating temperature: raise an issue and omit the term, because a default is a fabricated input to a compliance verdict. XM-001 applies the same reasoning to unresolvable nobility. The rule that GC-001, CC-001 and MC-001 need at their entry point is not a new rule to be invented; it is a rule already approved in two rulesets and not yet generalised to the other three.

15.3 What the White Box architecture contributed to the finding

Section 4.1 justifies the deterministic architecture on the grounds that a building owner, project team member or inspector receiving a risk flag must be able to trace it back to a named standard, a specific threshold and a version-controlled ruleset, without understanding the internals of a trained model. The entry-point defect turns out to be the strongest evidence in this thesis for that argument, and it arrives from an unexpected direction. Transparency did not merely make the finding explicable after the fact; it made the finding possible at all.

Every verdict in Table 15.2 arrived accompanied by the term-level detail that produced it. GC-001 reported a voltage gap of 0.0 V between two identically named materials. CC-001 reported a crevice geometry of "Tight" and a joint type of JT-014 for an element that declared no joint. MC-001 reported flow class FV0_STAGNANT, temperature class T5_UNKNOWN and dead-leg class DL5_UNKNOWN in the same payload as a Critical band. The fabricated inputs were legible in the output, so the diagnosis took one experiment rather than a debugging campaign. A scoring model that returned 0.26, 0.643 and 0.808 without exposing its terms would have been indistinguishable from a working system, and the three numbers would have been carried into a results chapter as findings.

The inverse point is equally important and less comfortable. The defect is itself a violation of the principle §4.1 sets out. A default substituted at the boundary is a value that no standard specifies, no ruleset versions and no citation supports, and it enters the composite on exactly the same footing as a figure read from NASA-STD-6012 or CIBSE Guide G. The White Box discipline was applied rigorously to the rulesets and not at all to the layer that feeds them, which is a reminder that architectural transparency is a property of a whole pipeline rather than of its most carefully documented component.

15.4 Remediation roadmap

The remediation is bounded, and the reason it is bounded is the separation just described. The rulesets are approved and internally consistent; the composites compute correctly; the geometric engine is verified independently. What fails is a translation layer a few dozen lines wide, and the behaviour that should replace it is already specified in two of the project's own rule packs. Table 15.3 sets out the work items in dependency order.

Ref	Work item	Defect addressed	Precedent already in the project	Depends on
R1	Replace unconditional defaulting in the three coercion functions with presence checks that emit a data-quality issue when a required property is absent	The entry-point defect of §15.2, in all three of its symptoms	MM-001 unmapped-pairing policy; MM-001 missing-temperature policy; XM-001 unresolvable-nobility policy	—
R2	Classify each engine input as required or optional, so that an optional absence omits its term while a required absence blocks the verdict	Over-blocking: not every absent property should stop an assessment	MM-001 distinguishes an unclassified environment, which blocks, from an absent temperature, which omits a term	R1

Ref	Work item	Defect addressed	Precedent already in the project	Depends on
R3	Extend the runner payload from {band, score, details} to carry an explicit assessed / not-assessed outcome	The payload currently has no way to express "could not be assessed", which is why the coercion layer does not attempt it	The issue schema already models a data-quality mechanism distinct from a risk band	—
R4	Align the MM-001 and XM-001 loaders with the approved pack shape so the packs load at all	Loader raises ValueError on both approved packs (§13.9)	The packs are the reference; no new design decisions are required	—
R5	Implement the MM-001 and XM-001 composites, the kinetics guard, the compatibility floor, separation and mitigation factors, and anode identification	Both engines currently band a raw input rather than a composite (§13.9)	Both formulas are fully specified in §13.2 and §13.3 and hand-verified in Tables 13.4 and 13.8	R4
R6	Re-run the 68 MM-001 and XM-001 tests against the aligned modules	3 of 29 and 1 of 39 currently execute (§13.9)	The suites already encode the specified contract; they need a matching implementation, not rewriting	R5
R7	Execute the 38-model corpus run to the protocol in Appendix A	No engine has been exercised on third-party models	Protocol, metric schema, presentation tables and pre-registered expectations are all fixed in advance	R1-R6

Table 15.3: Remediation roadmap in dependency order

The ordering constraint is not administrative. Running the corpus before R1 would produce a distribution of bands manufactured almost entirely by defaults — an element with no properties would contribute a Low to GC-001, a High to CC-001 and a Critical to MC-001 — and reporting that distribution as a measurement would be the precise error the pre-registered expectations in Table 12.7 and Table 13.12 exist to prevent. The corpus run is the last step in the sequence, not the first.

Two qualifications keep the word "straightforward" honest. It means that the changes are localised and that the correct behaviour is already specified elsewhere in the project, not that the work is trivial. R3 alters a payload contract that Module 5 consumes when it builds BCF issues, so the reporting layer must learn to render a not-assessed outcome rather than an empty band. R5 is genuine implementation work against two composites and four guards. And R1, the smallest change of the seven, will make the system report far less than it reports today — which is the intended outcome, and which will look like a regression to anyone who mistakes output volume for coverage.

15.5 Limitations that remediation does not remove

Completing the roadmap in Table 15.3 would make the corrosion stack sound. It would not make it complete. The limitations below are properties of the knowledge base, the schema or the evidence available, and no amount of correct code addresses them.

Limitation	Why remediation does not reach it	What would
The rule-pack digits are a calibration authored for these rulesets rather than values quoted from the cited standards	Code cannot validate a threshold. The citations name the standard governing each mechanism; they do not supply the number, and Table 13.4 shows a provisional value of 0.05 carrying an element across a band boundary	Cell-by-cell review by a corrosion engineer against the cited sources and against project water chemistry. Chapter 14 specifies that process; §14.6 records that its enforcement gate does not exist

Limitation	Why remediation does not reach it	What would
Water chemistry and flow velocity are absent from the element schema	The properties are not modelled anywhere in the pipeline, so no correction at the entry point can supply them. Copper Type 2 pitting in soft-water regions is understated and copper erosion-corrosion above roughly 1.5 m/s is not modelled at all	Schema extension plus a chemistry data source, which the MM-001 pack identifies as Phase 2 work
IFC carries no environment classification	MM-001 weights environment at 0.35 and XM-001 selects its compatibility threshold from it, yet no IFC property corresponds to the concept and no model in the corpus carries one	An information-delivery convention — an IDS requirement or an agreed property set — rather than a coding change. This is a standards question, and stating it is part of this thesis's contribution
No ground-truth clash or corrosion dataset exists	The corpus measures behaviour, agreement across schema twins and internal consistency. It cannot measure accuracy, because no model carries an authoritative list of true findings	Instrumented or surveyed buildings with recorded corrosion outcomes, which is a longitudinal research programme rather than a software task
Performance figures come from a single four-core host	The measurements establish scaling behaviour and relative stage cost, which is what the examination question asked; they do not establish absolute performance	Repetition across a hardware matrix
The industrial category of the corpus rests on one genuine building	Adding models cannot conjure open industrial IFC data that does not exist; the search behind Appendix A is documented and its exclusions are recorded	Access to a real industrial facility model under a redistributable licence, or an acknowledged restriction of the claim to healthcare and commercial buildings

Table 15.4: Limitations that survive remediation, and what would address each

One further qualification belongs here rather than being left implicit. Chapter 6 concluded that the data GC-001 requires is routinely populated in well-authored IFC exports. That conclusion held for the development set it was drawn from, and it remains a fair statement about well-authored models. Section 13.7 pre-registers the opposite expectation for the open corpus, where environment classification is absent from every model and velocity and temperature are sparse. The two statements are not in conflict once the qualifier is taken seriously — but the strength of the qualifier is precisely what the 38-model run exists to measure, and Chapter 6 should be read against Table 13.12 rather than on its own.

15.6 Future work beyond remediation

Six directions extend the system rather than repair it. They are listed in rough order of the ratio between the value they add and the work they demand.

The first is the enforcement gate specified in §14.3 and recorded as missing in §14.6. Until only published rules are evaluated, the human-in-the-loop claim is partially true rather than true, and no other future work changes that. Section 14.7 adds a second requirement discovered by sizing the corpus run: review sampling must stratify by finding pattern rather than by element, because a twenty per cent sample of a five-figure finding population is not reviewable by any human panel.

The second is chemistry-aware scoring for MM-001, which the pack itself names as Phase 2. Chloride concentration, pH, hardness and dissolved oxygen dominate several matrix cells that currently assume typical UK potable chemistry, and adding them would convert the most heavily caveated cells in Table 13.3 into defensible ones.

The third is the Building Management System integration proposed in §11.2, which would change MC-001's character rather than extend it. The engine currently scores design conditions: it can see that a branch is long, dead-ended and warm, but not whether it is flushed weekly, which is the control HSE HSG274 actually relies on. Live temperature and flow data, together with the time-since-flush parameter identified in §12.9, would turn a design-time score into an operational one.

The fourth is the weld quality extension to CC-001 described in §11.3. The fifth is jurisdiction coverage for Blue Halo beyond the combined European configuration: the NFPA 13 and ASCE 7-22 parameters have already been extracted and the sensitivity comparison is specified in Appendix A, so the remaining work is validation rather than research. The sixth is mechanism coverage — dezincification of brass and velocity-driven erosion-corrosion of copper are both identified as gaps in Chapter 13, and each is a candidate sixth engine on the pattern the existing five establish.

15.7 The claim this thesis makes

It is worth closing with an explicit statement of scope, because the difference between the two subsystems makes a single summary claim impossible. This thesis claims that a White Box compliance architecture generalises across five distinct corrosion mechanisms and one geometric reservation algorithm, all referenced to published standards and versioned rulesets; that the Blue Halo clearance algorithm is verified against its specification and measured from one hundred to twenty thousand elements; and that the five rulesets are approved, traceable and explicit about the limits of their own calibration.

It does not claim that the corrosion subsystem currently produces trustworthy verdicts on arbitrary IFC models. Section 15.2 establishes that it does not, and why. The distance between those two statements is the work described in Table 15.3 — seven items, bounded, with the correct behaviour already specified in the project's own approved rulesets. That the distance can be stated that precisely is itself a result, and it is a result the architecture argued for in §4.1 made available.

Appendix A — Validation Methodology

Field	Value
Document	Appendix A — Validation Methodology
Parent thesis	BIMGUARD AI — MAICEN M10 Final Master Project
Author	Mark Shane Haines
System under validation	BIMGUARD AI: Blue Halo seismic-bracing clearance engine and the five-mechanism corrosion rule stack (GC-001, CC-001, MC-001, MM-001, XM-001)
Validation dataset	38 IFC models from 8 independent public repositories
Status	Methodology and presentation framework fixed in advance of the batch run; results tables in Section A.3 are unpopulated templates
Date	23 August 2026

Purpose of this appendix

This appendix fixes the validation protocol before any results exist. It states which models are tested, how they are processed, which numbers are recorded, how those numbers will be presented, and — critically — what ranges would count as a pass, a null result, or a failure. Declaring the interpretation criteria ahead of the run is what separates validation from post-hoc rationalisation of whatever the pipeline happened to produce.

Sections A.1 and A.2 are complete and final. Section A.3 supplies empty tables and chart specifications ready to receive batch-run output. Section A.4 states the expected outcome ranges, derived from the synthetic and benchmark baselines already measured on this codebase.

A.1 Validation Dataset Overview

A.1.1 Scope and rationale

BIMGUARD AI has, to this point, been exercised against synthetic fixtures: hand-built dataclasses in `validate_blue_halo.py` and a single purpose-generated IFC file (`data/test_hospital_mep_scenario.ifc`). Synthetic fixtures prove that the algorithm computes what it was written to compute. They cannot show that it survives contact with models authored by other people, in other tools, under other national conventions, at production element counts. That is the gap this dataset closes.

Thirty-eight IFC models were sourced from eight independently operated public repositories. Every download link was individually verified by HTTP request during compilation (HTTP 200 plus a real Content-Length), so the dataset is a set of working file downloads rather than a bibliography of landing pages. No model requires a login, a registration, or a payment, and every model carries a redistributable open licence (CC BY 3.0/4.0, MIT, or public-sector open data). A third party can therefore reconstruct the exact test corpus from this appendix alone, which is the reproducibility bar the validation is being held to.

A.1.2 Selection criteria

Candidate models were admitted only if they satisfied every criterion below. The criteria were fixed before searching, and the exclusions in A.1.7 record where otherwise attractive sources failed them.

Table A.1.2 — Dataset selection criteria

#	Criterion	Operational test	Why it matters
1	MEP-rich where possible	File contains <code>IfcPipeSegment</code> / <code>IfcDuctSegment</code> / <code>IfcFlowSegment</code> entities, ideally with material text	The corrosion engines score pipe material and media; a model with no pipes exercises nothing
2	Verified direct download	HTTP HEAD/GET returns 200 with a non-zero Content-Length pointing at the <code>.ifc</code> or <code>.zip</code> itself	Removes the single most common reproducibility failure in BIM research: dead sample-model links
3	No login, no payment	File retrievable by anonymous request	An examiner or replicator must be able to obtain the corpus without an account
4	Redistributable licence	CC BY 3.0/4.0, MIT, or public-sector open data on the hosting repository	Permits citation, quotation of entity counts, and re-hosting of derived results
5	Individually ≤ 100 MB	Measured download size, not vendor estimate	Keeps the batch run tractable on commodity hardware and inside repository limits
6	Building-scale, not component-scale	Model contains a spatial hierarchy (site/building/storey), not a single product	Manufacturer BIM libraries return switchgear and cooling units; those do not test spatial reasoning
7	Diverse building types	Classified as hospital / office / industrial-lab at compilation	Blue Halo clearance rules and importance factors are occupancy-dependent
8	Schema diversity	IFC2x3 and IFC4 both represented, with same-building twins where available	Parser robustness across schema versions is a distinct failure mode from geometry handling

A.1.3 Distribution

Table A.1.3 — Dataset distribution by building type

Category	Models	Share	Row range	Character of the sample
Hospital / Healthcare	16	42%	1–16	Two full multi-discipline building sets (Medical-Dental Clinic; West Riverside Hospital) plus regional additions
Office / Commercial	16	42%	18–33	Duplex benchmark set, RWTH DigitalHub, WBDG office prototype, KIT/IFC Wiki reference models, national-standard examples
Industrial / Laboratory	6	16%	34–39	One genuine as-built laboratory plus flagged general-purpose and structural-steel robustness testers
Total	38	100%	—	8 independent repositories; 10 countries of origin; IFC2x3 and IFC4 both represented

Row numbering follows the compilation working list, which skips 17 and 40; every row listed is a distinct file. Percentages are rounded to the nearest whole number.

A.1.4 Source repositories

Table A.1.4 — Source repositories and licensing

Repository	Models	Licence	Country / operator
buildingSMART Community-Sample-Test-Files (GitHub)	11	CC BY 4.0	International / buildingSMART community
Hugging Face ifc-bench (mirrors OpenIFC Model Repository, Univ. of Auckland, and WBDG/NIBS)	12	CC BY 3.0	NZ / US, mirrored
RWTH-E3D DigitalHub (GitHub)	4	MIT	Germany / RWTH Aachen
KIT IFC Examples via IFC Wiki	2	Free with attribution	Germany / KIT
TIB Hannover — DuraArk BuildingData	4	Open data	Germany / TIB
Zenodo (record 7948116)	1	Open access CC	Portugal / FEUP
Generalitat Valenciana — Conselleria d'Habitatge	2	Public-sector open data	Spain
STEP Tools sample IFC data (NIST/AISC conversion)	2	Public / NIST	US

Counts sum to 38, matching the inventory in Table A.1.6. Zip archives bundling several discipline files are counted once, as one inventory row. The ifc-bench figure spans two upstream collections that Hugging Face mirrors — the Auckland OpenIFC repository (rows 6–14) and the WBDG/NIBS Common BIM Files (rows 28–30) — which is why it is the largest single entry.

A.1.5 Material data quality classification

Every model carries a material-data tier assigned at compilation. The tier determines which engines can meaningfully run against it, and therefore which denominator each result is reported over. Reporting a corrosion-engine finding rate over all 38 models when only a subset carries material text would understate the engines by construction.

Table A.1.5 — Material data quality tiers

Tier	Definition	Engines exercised
Full	Pipe/duct entities carry material, size and system-type properties sufficient for corrosion and clearance scoring	Blue Halo + all five corrosion engines

Tier	Definition	Engines exercised
Partial	Geometry and system type present, material properties sparse or schema-dependent	Blue Halo; corrosion engines expected to emit data-quality issues rather than risk bands
Architecture-only	No MEP systems in the file	Blue Halo clash geometry and parser robustness only
n/a (discipline)	Structural, electrical or fire-alarm file with no piped systems	Used as clash candidates for Blue Halo halos generated from paired MEP files

A.1.6 Model inventory

The full corpus is listed below by category. Sizes are measured download sizes. "n.v." denotes a property not verified beyond the HTTP check; "content-verified" denotes a file downloaded and inspected during compilation.

Table A.1.6a — Hospital / healthcare models (16)

#	Model file	Systems	MB	Schema	Material tier
1	Clinic_Architectural.ifc	Walls, doors, windows, rooms	12.40	IFC2x3	Architecture-only
2	Clinic_Structural.ifc	Beams, columns, slabs	18.18	IFC2x3	n/a (structural)
3	Clinic_HVAC.ifc	Ductwork, AHU, terminals	25.67	IFC2x3	Full
4	Clinic_Electrical.ifc	Cable trays, fixtures, panels	6.49	IFC2x3	n/a (electrical)
5	Clinic_Plumbing.ifc	378 pipe segments/fittings; Copper, PVC, CPVC, Steel, Brass	53.25	IFC2x3	Full — content-verified
6	west_riverside_hospital — arc_ifc4	Architecture, 15,316 elements	77.19	IFC4	Architecture-only
7	west_riverside_hospital — str_ifc4	2,915 elements; 1,970 beams, 255 columns	6.18	IFC4	n/a (structural)
8	west_riverside_hospital — mech_ifc4	59,215 elements; 3,916 pipe segments, 1,064 terminals	69.66	IFC4	Full
9	west_riverside_hospital — plumb_ifc4	26,942 elements; 4,308 pipe segments	22.66	IFC4	Full
10	west_riverside_hospital — elec_ifc4	6,305 elements; 1,272 light fixtures	4.23	IFC4	n/a (electrical)
11	west_riverside_hospital — fire_ifc4	874 fire-alarm devices	0.86	IFC4	n/a
12	west_riverside_hospital — sprinkle_ifc4	38,255 elements; 6,228 pipe segments, 1,354 heads	32.42	IFC4	Full
13	west_riverside_hospital — mech_ifc2x3	Schema twin of row 8	75.12	IFC2x3	Full
14	west_riverside_hospital — plumb_ifc2x3	Schema twin of row 9	23.84	IFC2x3	Full
15	NBU_MedicalClinic (zip: ARC+MEP+HVAC+ELE+STR)	Full discipline set	78.90	IFC2x3	n.v.
16	GVA Sanitario.ifc	Architecture only	6.64	n.v.	Architecture-only

Row 15 caveat: the MEP sub-file inside NBU_MedicalClinic_ifc.zip is approximately 207 MB uncompressed and breaches the 100 MB per-file cap once extracted; the other four discipline files do not. Row 5 is the strongest single file in the corpus for the galvanic and cross-material engines.

Table A.1.6b — Office / commercial models (16)

#	Model file	Systems	MB	Schema	Material tier
18	Duplex_A_20110907.ifc	Architecture	2.27	IFC2x3	Architecture-only
19	Duplex_Electrical_20121207.ifc	Electrical	1.53	IFC2x3	n/a
20	Duplex_MEP_20110907.ifc	Combined MEP	17.04	IFC2x3	Full

#	Model file	Systems	MB	Schema	Material tier
21	Duplex_Plumbing_20121113.ifc	Pipes	30.09	IFC2x3	Full
22	AC20-Institute-Var-2.ifc	Architecture plus some MEP	10.43	IFC4	Partial
23	AC20-FZK-Haus.ifc	Architecture	2.45	IFC4	Architecture-only
24	DigitalHub_FM-ARC_v2.ifc	Architecture	8.60	IFC4 (Ref. View)	Architecture-only
25	DigitalHub_FM-HZG_v2.ifc	Heating pipework	19.92	IFC4	Full
26	DigitalHub_FM-LFT_v2.ifc	Ventilation ductwork	12.15	IFC4	Full
27	DigitalHub_FM-SAN_v2.ifc	Plumbing / sanitary	24.01	IFC4	Full
28	wbdg_office — arc.ifc	Architecture	3.91	n.v.	Architecture-only
29	wbdg_office — mep.ifc	642 duct segments, 1,569.99 m ductwork (published ground truth)	39.95	n.v.	Full
30	wbdg_office — str.ifc	Beams, columns	10.55	n.v.	n/a
31	Molio_with_URIs.ifc	Architecture with URI classification	70.52	IFC2x3	Architecture-only
32	GVA Administrativo.ifc	Architecture only	6.88	n.v.	Architecture-only
33	IFC Schependomlaan.ifc	Architecture (design model)	47.00	IFC2x3	Architecture-only

Row 29 carries published duct-length ground truth in its dataset QA pairs (Supply Air 860.04 m, Return Air 697.93 m, Exhaust Air 12.02 m) and is therefore the corpus's only external check on halo volume arithmetic. Rows 18–21 and 33 are among the most heavily cited open BIM benchmarks in the construction-technology literature, permitting cross-reference of clash counts against published studies. Row 33 is mixed residential rather than pure office and is flagged as such.

Table A.1.6c — Industrial / laboratory models (6)

#	Model file	Systems	MB	Schema	Material tier
34	craslabim.ifc	As-built research laboratory, full building (CRAS Lab, FEUP)	64.42	n.v.	Partial
35	SGD_BODO_ifc.zip (ARC+PLB+VENT)	Plumbing and ventilation	31.11 (zip)	n.v.	n.v.
36	Academic_Autodesk_ifc.zip (Basic + Advanced)	Architecture, mixed	7.03 (zip)	n.v.	n.v.
37	Autodesk_210-King_ifc.zip	Structural and architecture, steel frame	25.49 (zip)	n.v.	n.v.
38	aisc_sculpture_brep.ifc	Structural steel, brep solid geometry	0.53	IFC2x3	n/a (steel)
39	aisc_sculpture_param.ifc	Same steel structure, parametric swept solids	0.30	IFC2x3	n/a (steel)

Rows 38 and 39 are the same steel assembly in two geometric representations and exist in the corpus specifically to test the geometry-handling path independently of MEP content. Row 35's building programme is not confirmed as strictly industrial and its extracted discipline files may individually approach the 100 MB cap.

A.1.7 Sources investigated and excluded

Recording what was rejected, and why, is part of the method. Each entry below was a plausible candidate that failed a stated criterion.

Table A.1.7 — Excluded sources

Source	Reason for exclusion	Criterion failed
University of Auckland Open IFC Model Repository	All /Model/Details/ and /_models/ paths return an empty page; the canonical host is broken. West Riverside Hospital recovered only via the Hugging Face ifc-bench mirror	2 — verified download
Nordic Sustainable Construction BIM4LCA	ARC.zip is 1.5 GB and HVAC.zip 870 MB, bundling multiple variants and native non-IFC formats; no individual file isolable without a multi-gigabyte download	5 — ≤100 MB

Source	Reason for exclusion	Criterion failed
TUM GNI BIM Dataset (Zenodo 19722012)	Legitimate CC BY 4.0 source but delivered only as 503 MB and 221 MB bulk archives with no per-model links	5 — ≤100 MB
Holter_Tower_10.ifc (DuraArk/BIMData catalogue)	Listed Dropbox link returns an HTML page, not the file	2 — verified download
NBU_OfficeBuilding	The catalogue's source link is a byte-for-byte duplicate of the NBU_MedicalClinic zip URL; the office entry is mislabelled at source	Data integrity
Kaggle IFC datasets ("5000 IFC/RVT Projects", "IFC Sample Models")	Require account login	3 — no login
Manufacturer BIM libraries (Siemens, Stulz, GEA)	Single-component product files — a switchgear, a cooling unit, a cleanroom door — not building models	6 — building-scale

A.1.8 Declared limitation — the industrial category

The industrial/laboratory category is the corpus's weak point and is presented as such rather than padded. Genuinely public, building-scale IFC models of factories, warehouses, data centres, cleanrooms or power plants with rich MEP content are scarce on the open web. After searching buildingSMART, GitHub, Zenodo, Figshare, university repositories (MIT, Stanford, Delft, ETH Zurich, RWTH Aachen, TUM, KIT) and the DuraArk collections, craslabim.ifc (row 34) was the only confirmed genuine industrial or laboratory building with strong provenance. Rows 35–39 are included transparently as general-purpose and structural-steel robustness testers, flagged in the inventory rather than dressed up as industrial buildings.

The consequence for interpretation is stated in A.4.4: conclusions about industrial-facility behaviour rest on $n = 1$ and must be reported as indicative, not established.

A.2 Testing Methodology

A.2.1 Automated batch processing pipeline

Each model is processed by a single non-interactive pass. The pipeline is staged so that cost and failure can be attributed to a specific stage rather than to "the run"; a model that fails at parse yields a recorded parse failure and no downstream rows, rather than silently contributing zeros to the aggregate.

Table A.2.1 — Batch pipeline stages

Stage	Operation	Implementation	Recorded on failure
0 — Acquire	Fetch the model from its verified URL; record HTTP status, byte count and SHA-256	Batch driver	URL dead / size mismatch
1 — Parse	ifcopenshell.open(); resolve schema version and project unit scale to millimetres	unit_scale_to_mm() in halo_volume_generator	Parse error, schema unsupported
2 — Extract	Enumerate MEP and structural elements; triangulate to world-coordinate meshes via ifcopenshell.geom.iterator	performance_benchmark.py ingest stage	Geometry resolution failure per element
3 — Halo	Generate one clearance envelope per in-scope element from the loaded jurisdiction config	generate_halo_volume() / generate_halo_volume_from_geometry()	Element skipped, no placement or usable shape
4 — Clash	Broad-phase uniform spatial hash, then exact mid-phase AABB overlap between halos and candidate geometry	detect_halo_clash() / detect_halo_clash_against_geometry()	—
5 — Corrode	Coerce each piping element into the five engine input models and score it	compliance_runner.py coercion adapters plus the XM-001/MM-001 compare() functions	Data-quality issue emitted, never a default score
6 — Aggregate	Normalise every finding to the common issue schema {band, score, details}	issue_adapter.py / issue_schema.py	Schema violation
7 — Export	Emit a BCF 2.1 ZIP of all findings and a Pset_HaloReservation JSON per halo	blue_halo_bcf_exporter.py, bcf_generator.py	Invalid archive structure
8 — Record	Write the per-model metric row defined in Table A.2.3	Batch driver	—

A.2.2 Harness provenance

The batch driver composes existing, already-exercised harnesses rather than reimplementing the pipeline. This matters for validity: the code path under test in the batch run is the same code path the earlier synthetic validation passed, differing only in its input.

Table A.2.2 — Existing harnesses composed by the batch run

Harness	What it already covers	Role in the batch run
validate_blue_halo.py	Core generation and detection against hand-built geometry, real EN 1998-1 + DIN 4149 config; 11/11 checks passing	Supplies the pass criteria replicated per model
test_real_ifc_pipeline.py	Full ifcopenshell path on one fixture: placement matrices, tessellated vertices, metre-to-millimetre scaling, BCF and Pset export	Template for the per-model end-to-end pass

Harness	What it already covers	Role in the batch run
performance_benchmark.py	Stage-separated timing and memory at 100–20,000 elements, n = 7 repeats, median and IQR reporting	Supplies the throughput baselines in Table A.4.3
compliance_runner.py	Coercion of generic elements into GCElement / CCElement / MICElement and normalisation to {band, score, details}	Corrosion scoring stage
scripts/verify_anode_convention.py	Verification of the galvanic series sign convention against the seeded GC-001 payload	Pre-run integrity gate on the shared galvanic series

A.2.3 Metrics collected per model

One row is written per model. Every field is either measured or explicitly null; no field is imputed.

Table A.2.3 — Per-model metric schema

Field	Type	Definition
model_id	int	Inventory row number from Table A.1.6
sha256	string	Hash of the downloaded file, pinning the artefact the row describes
schema_version	string	IFC2x3 or IFC4, read from the file header
parse_seconds	float	Wall-clock duration of ifcopenshell.open()
element_count_total	int	All IfcProduct instances with geometry
element_count_mep	int	IfcPipeSegment, IfcDuctSegment, IfcFlowSegment and fittings
triangulate_seconds	float	Duration of the geometry iterator over in-scope elements
halos_generated	int	Clearance envelopes successfully produced
halos_skipped	int	Elements with no resolvable placement or shape representation
halo_seconds	float	Duration of the halo generation stage alone
halos_per_second	float	halos_generated / halo_seconds
mean_halo_volume_m3	float	Arithmetic mean of the generated envelope volumes
candidate_pairs	int	Pairs surviving the broad-phase spatial hash
clashes_total	int	Pairs confirmed by exact AABB overlap
clashes_critical / major / minor	int	Clash counts by the severity bands in Table A.2.6
corrosion_findings_by_engine	map	Count of issues raised, keyed by GC-001, CC-001, MC-001, MM-001, XM-001
corrosion_bands	map	Count of findings by risk band, per engine
data_quality_issues	int	Findings emitted because required properties were absent — a coverage measure, not a risk measure
materials_observed	list	Distinct pipe material strings encountered
unmapped_pairings	int	Material/media combinations absent from the MM-001 matrix
bcf_bytes	int	Size of the exported BCF 2.1 archive
bcf_valid	bool	Result of the structural check in Table A.2.7
peak_rss_mb	float	Peak resident set size delta across the model's processing
failure_stage	string null	Pipeline stage at which processing stopped, if any

A.2.4 Corrosion engine scoring

Five rule engines score each piping element. They are independent mechanisms, not a single score decomposed: an element may be benign under four and critical under the

fifth, and that is the intended behaviour. Each engine carries its own weighted composite and its own band cut-offs, taken from the ruleset it implements.

Table A.2.4a — The five corrosion engines

Ruleset	Mechanism	Composite score	Bands	Principal standards
GC-001	Galvanic corrosion	$0.50 \times \text{voltage_risk} + 0.30 \times \text{area_ratio_risk} + 0.20 \times \text{environment_multiplier}$	Low < 0.35; Medium 0.35–0.65; High 0.65–0.85; Critical > 0.85	NASA-STD-6012; WorldStainless/Euro Inox; AUCSC; IMO; Prosoco TN 104; AGA
CC-001	Crevice corrosion	$0.35 \times \text{geometry_risk} + 0.40 \times \text{CCT_adequacy} + 0.25 \times \text{environment_severity}$	Low < 0.30; Medium 0.30–0.55; High 0.55–0.80; Critical > 0.80	EN ISO 15329:2007; ASTM G48 Method B; CIRIA C692; CIBSE Guide G; EN 1993-1-4
MC-001	Microbially influenced corrosion	$0.35 \times \text{flow_velocity_risk} + 0.30 \times \text{temperature_risk} + 0.25 \times \text{dead_leg_risk} + 0.10 \times \text{material_susceptibility}$	Low < 0.25; Medium 0.25–0.50; High 0.50–0.75; Critical > 0.75	CIBSE TM13:2013; HSE HSG274 Pts 1–3; BS 8552:2012; ASTM G-187; EN ISO 9308-1; WHO GDWQ 4th ed.
MM-001	Material–media compatibility	$0.40 \times \text{compatibility_score} + 0.35 \times \text{environment_severity} + 0.25 \times \text{temperature_stress}$	Compliant < 0.35; Medium 0.35–0.60; High 0.60–0.85; Critical ≥ 0.85	ASTM B117; ASTM G48; CIBSE Guide G; EN 12952-12; IMO; CDA
XM-001	Cross-material contamination	Galvanic couple voltage resolved from the shared GC-001 series, suppressed below the environment threshold (default 0.25 V)	Medium < 0.50 V; High 0.50–0.85 V; Critical ≥ 0.85 V	BS 8539; NASA-STD-6012; EN ISO 15329:2007 (via MM-001 environment table); IMO

Two design decisions in the rule packs bear directly on how results must be read:

- Single-source tables. XM-001 deliberately contains no galvanic series and no environment table of its own; it is injected with the GC-001 series (loaded from the BIMGUARD-GC-001 catalogue) and the MM-001 environment severities at runtime. There is one series and one environment table in the system, so a divergence between engines cannot arise from table drift.
- No silent defaulting. A material/media pairing absent from the MM-001 matrix must emit a data-quality issue, never a default score. An unmapped pairing scored as 0.0 would be reported as compliant, which would be a false negative dressed as a pass. This is why data_quality_issues and unmapped_pairings are first-class metrics in Table A.2.3.

Table A.2.4b — Declared calibration limitations carried into the run

Ruleset	Carried limitation
MM-001	The rankings reflect corrosion-engineering consensus; the two-decimal scores and predicted lifespans are a calibration authored for this ruleset, not digits quoted from the cited standards. Each citation names the standard governing the mechanism, not a document the number was read from.
MM-001	Water chemistry (chloride ppm, pH, hardness, dissolved oxygen) and velocity are absent from PipingElement. Copper Type 2 pitting in soft-water regions is understated; copper erosion-corrosion above ~1.5 m/s is not modelled.
MM-001	T5_industrial environment severity (0.85) is provisional — the only environment value with no CC-001 wetting-class counterpart, interpolated rather than borrowed.
XM-001	The galvanic series was verified as an anodic index (higher value = anode) against the GC-001 payload as last committed. If the Supabase-hosted asset was edited after seeding, the engines read a different table; re-verification without --from-seeder is required once credentials permit.

Ruleset	Carried limitation
XM-001	One internally inconsistent adjacent pair persists in the series: ss316_active (+0.22, noble = false) sits below copper (+0.28, noble = true), so for that couple the potential ordering and the noble flag disagree. XM-001 resolves direction from the flag and is unaffected, but the series needs a corrosion reviewer.

These limitations are pre-declared so that engine output is interpreted as a ranking of relative risk under a stated calibration, not as an absolute engineering verdict. See A.4.2.

A.2.5 Standards compliance basis

Blue Halo clearance geometry is driven by a jurisdiction configuration generated from a standards research summary. Four standards are represented; the batch run uses the combined European configuration as its primary case, with the US pair available as an alternate jurisdiction for the sensitivity comparison in Table A.3.1f.

Table A.2.5a — Source standards and extracted parameters

Standard	Jurisdiction	Restraint spacing (transverse / longitudinal)	Pipe threshold	Min. clearance from structure	Brace angle	Importance factor	Citation
EN 1998-1:2020	EU	1.0 m / 1.5 m	63 mm dia.	150 mm	35–70°	1.0 (1.5 critical facilities)	5.3.2.3
DIN 4149:2022	Germany	1.2 m / 1.8 m	75 mm dia.	200 mm	40–65°	1.0 (1.6 hospitals)	8.2.4
NFPA 13-2021	US	40 in / 60 in	2.5 in dia.	12 in	30–60°	1.0 (1.5 hospitals)	18.3.2.1
ASCE 7-22	US	48 in / 72 in	8 in duct dia.	18 in	30–60°	1.0 (1.5 hospitals)	12.10.4.2

The combined EN 1998-1 + DIN 4149 configuration resolves each parameter by the conservative reading, and the resolution rule is recorded in the configuration itself rather than applied silently:

- Restraint spacing: per-axis minimum of the two standards — tighter spacing governs, because it requires more restraints.
- Clearance from structure: maximum of the two (200 mm) — the larger clearance produces an envelope satisfying both.
- Pipe diameter threshold: minimum of the two (63 mm) — the lower threshold brings more pipes into scope.
- Brace angle: intersection of both ranges (40–65°) — the band any brace satisfying both standards must fall in.
- Importance factors: per-key maximum — the higher factor demands the greater restraint capacity.

Table A.2.5b — Configuration data gaps declared before the run

Gap	Effect on results
No duct-area bracing threshold in either standard	Duct in-scope determination cannot be made from the standard; duct halos are generated unconditionally and reported separately
seismic_zone_addition_mm and hospital_addition_mm absent from both standards; left at 0 mm rather than guessed	Hospital and standard-occupancy clearances are numerically identical (200 mm). A hospital-versus-standard comparison will show no difference, and that null is a data gap, not an algorithm defect
adjacent_system_mm absent from both standards; left at 0 mm	System-to-system clearance additions are not modelled

Gap	Effect on results
hazard_factor_default absent from both standards; left null	Site hazard scaling is not applied
No per-brace-hardware breakdown in either standard	All four brace variants (angle_fire, angle_mechanical, cable, rod) share one clearance value and therefore produce identical envelope volumes; brace-type comparison will be degenerate
brace_types[*].base_footprint_mm are generic hardware placeholders; standard_sizes empty	Footprint-driven results are indicative pending a hardware catalogue cross-reference
Citations are section references, not full bibliographic records	Traceability is to clause level; full citations remain outstanding

Thirteen gaps are recorded in the configuration's own metadata.data_gaps field; the seven above are those that change how a result should be read. Declaring them here prevents a degenerate result being reported later as a finding.

A.2.6 Clash severity classification

Severity is assigned from the overlap volume as a fraction of the halo volume. The thresholds are a generic spatial heuristic, not a banding taken from any standard, and are stated here so that any distribution reported in A.3 can be re-derived or re-banded by a reader who disagrees with them.

Table A.2.6 — Severity bands and BCF handling

Severity	Overlap ratio	BCF priority	Due date offset
Critical	≥ 25% of halo volume	Critical	2 days
Major	≥ 5% and < 25%	Major	7 days
Minor	< 5%	Minor	21 days

A.2.7 BCF export validity

A BCF archive is counted valid only if it satisfies every structural condition below. Byte size alone is not evidence of a usable export.

Table A.2.7 — BCF 2.1 validity conditions

#	Condition
1	Archive contains a bcf.version entry declaring BCF 2.1
2	Archive contains a project.bcfp entry
3	One GUID-named topic folder exists per reported finding — folder count equals finding count exactly
4	Each topic folder contains markup.bcf, viewpoint.bcfv and snapshot.png
5	Every markup.bcf carries a populated DueDate, derived from severity per Table A.2.6
6	Every topic references a component GUID that exists in the source model
7	The archive opens without error in a standards-compliant BCF reader

Alongside the BCF archive, each halo is exported as a Pset_HaloReservation property set (SchemaVersion 0.1.0) carrying the source GlobalId, IFC class, brace type and rule variant, clearance in millimetres, envelope volume, and both element and halo bounding boxes. The property set must be JSON-serialisable and round-trip without loss.

A.2.8 Reproducibility and determinism

- **Determinism.** No randomness and no wall-clock-dependent value feeds any decision. Timestamps appear in output dataclasses but are never compared or asserted upon. Counts that should not vary between repeats — triangle counts, envelope volumes, interfering pairs — are checked for agreement across repeats rather than averaged.

- Repeats and statistics. Timing scenarios are run $n = 7$ and reported as median with inter-quartile range, not mean with standard deviation: a benchmark timing is bounded below by the true cost of the work and unbounded above by scheduler interference, so the distribution is right-skewed and a single stall drags the mean while leaving the median untouched.
- Instability flagging. Any metric whose IQR exceeds 20% of its median is flagged unstable in the output rather than quietly reported. A large relative spread on a sub-10 ms median is measurement noise; the same spread on a multi-second median is not, and no conclusion rests on it without further repeats.
- Environment capture. Python version, platform, processor, CPU count, total RAM, numpy version and ifcopenshell version are recorded with every run and reported with the results.
- Artefact pinning. Each model row records the SHA-256 of the file actually processed, so a later re-run against a silently updated upstream file is detectable rather than invisible.

A.3 Results Presentation Framework

The tables in this section are the final reporting format, fixed before the data exists. They are reproduced here unpopulated; each cell marked "—" receives a measured value from the batch run. Fixing the format in advance removes the temptation to select a presentation that flatters whatever the run produces.

A.3.1 Table templates

Table A.3.1a — Per-model processing summary (one row per model; 38 rows)

#	Model	Elements	MEP elem.	Parse (s)	Halos	Halos/s	Clashes	BCF valid	Fail stage
1	Clinic_Architectural.ifc	—	—	—	—	—	—	—	—
...	...	...	...	...	...	...	...	...	...
39	aisc_sculpture_param.ifc	—	—	—	—	—	—	—	—
	Median (all models)	—	—	—	—	—	—	—	—
	IQR	—	—	—	—	—	—	—	—

Table A.3.1b — Clash severity distribution by building type

Building type	Models	Halos	Candidate pairs	Clashes	Critical	Major	Minor	Confirmation rate
Hospital	16	—	—	—	—	—	—	—
Office	16	—	—	—	—	—	—	—
Industrial / lab	6	—	—	—	—	—	—	—
All	38	—	—	—	—	—	—	—

Confirmation rate = clashes ÷ candidate pairs, i.e. the proportion of broad-phase candidates surviving exact AABB testing. It measures spatial-index selectivity, not clash prevalence.

Table A.3.1c — Corrosion engine findings by pipe material

Pipe material	Elements scored	GC-001	CC-001	MC-001	MM-001	XM-001	Data-quality issues
Copper (C12200)	—	—	—	—	—	—	—
Carbon steel	—	—	—	—	—	—	—
Galvanised steel	—	—	—	—	—	—	—
Stainless SS304	—	—	—	—	—	—	—
Stainless SS316	—	—	—	—	—	—	—
Brass	—	—	—	—	—	—	—
PVC	—	—	—	—	—	—	—
CPVC	—	—	—	—	—	—	—
Unrecognised / absent	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—

Cell values are finding counts. A zero is a result: it means the engine ran and raised nothing. An empty denominator (elements scored = 0) is not a result and must be reported as no-coverage, not as compliance.

Table A.3.1d — Risk band distribution per engine

Engine	Findings	Low / compliant	Medium	High	Critical	Median score	IQR
GC-001 galvanic	—	—	—	—	—	—	—

Engine	Findings	Low / compliant	Medium	High	Critical	Median score	IQR
CC-001 crevice	—	—	—	—	—	—	—
MC-001 MIC	—	—	—	—	—	—	—
MM-001 material-media	—	—	—	—	—	—	—
XM-001 cross-material	—	—	—	—	—	—	—

Table A.3.1e — BCF export validity (conditions per Table A.2.7)

Building type	Archives produced	Topics	C1–C2 pass	C3 folder = finding count	C4 all 3 files present	C5 due date	Valid overall
Hospital	—	—	—	—	—	—	—
Office	—	—	—	—	—	—	—
Industrial / lab	—	—	—	—	—	—	—
All	—	—	—	—	—	—	—

Table A.3.1f — Standards sensitivity (same models, alternate jurisdiction configuration)

Configuration	Clearance (mm)	Angle band	Halos	Mean halo vol. (m³)	Clashes	Critical	Δ vs. baseline
EN 1998-1 + DIN 4149 (baseline)	200	40–65°	—	—	—	—	—
NFPA 13 + ASCE 7-22	—	30–60°	—	—	—	—	—
EN 1998-1 alone	150	35–70°	—	—	—	—	—
DIN 4149 alone	200	40–65°	—	—	—	—	—

Table A.3.1g — Parser robustness, IFC2x3 vs IFC4 schema twins

Building / discipline	IFC2x3 elements	IFC4 elements	Δ elements	IFC2x3 halos	IFC4 halos	Δ halos	Volume agreement
West Riverside — mechanical (rows 13 / 8)	—	—	—	—	—	—	—
West Riverside — plumbing (rows 14 / 9)	—	—	—	—	—	—	—
AISC sculpture — brep / parametric (rows 38 / 39)	—	—	—	—	—	—	—

Rows 13/8 and 14/9 are the same building exported to two schemas; rows 38/39 are the same steel assembly in two geometric representations. Element counts may legitimately differ between schemas; envelope volumes for corresponding elements should not.

A.3.2 Chart specifications

Four figures accompany the tables. Each specification states the encoding and, more importantly, what the figure is supposed to let a reader see.

Figure A.1 — Clash severity distribution

Property	Specification
Type	Horizontal stacked bar, one bar per building type, plus an "all models" bar
X axis	Proportion of clashes (0–100%), with absolute counts annotated at the bar end
Segments	Critical / Major / Minor, ordered severe-to-benign left to right
Colour	Sequential single-hue ramp, darkest for Critical; not red-amber-green, which implies a verdict the severity heuristic does not support

Property	Specification
Reads as	Whether severity composition differs by building type. If hospital models show a heavier critical tail than office models, that is congestion, not a different algorithm
Caveat to print	Severity thresholds are a spatial heuristic (Table A.2.6), not a standards banding

Figure A.2 — Material risk heatmap

Property	Specification
Type	Matrix heatmap, materials on rows, the five engines on columns
Cell value	Median composite score for that material under that engine; count annotated in the cell
Colour	Sequential ramp on score, with a visually distinct hatched fill for "no coverage" (zero elements scored)
Critical detail	No-coverage cells must not render as low risk. A blank denominator and a benign score are opposite findings and must not share a colour
Reads as	Which mechanisms discriminate between materials and which return flat results across the corpus. A column that is uniform across every material is an engine that is not discriminating on this data
Caveat to print	Scores are relative rankings under the calibration declared in Table A.2.4b, not absolute engineering verdicts

Figure A.3 — Throughput scaling

Property	Specification
Type	Log-log scatter with fitted trend, one point per model
X axis	Element count
Y axis	Halo generation seconds (stage 3 only, excluding parse and triangulation)
Overlay	The synthetic scaling baseline from Table A.4.3 as a reference line
Reads as	Whether real-model generation cost stays linear in element count and lands on the synthetic baseline. Systematic elevation above the line indicates real geometry costs more than synthetic; curvature indicates a super-linear term
Caveat to print	Points are single runs per model, not $n = 7$ repeats; scatter includes host noise

Figure A.4 — Halo volume distribution

Property	Specification
Type	Box plot per building type, log-scaled y
Y axis	Envelope volume (m^3) per halo
Reference	Synthetic baseline band ($13.5\text{--}21.5 \text{ m}^3$ mean) shaded behind the boxes
Reads as	Whether envelope sizes on real models are physically plausible. Volumes orders of magnitude off the baseline indicate a unit-scaling fault, which is the most likely silent failure in the IFC path
Caveat to print	Envelope volume is driven by element bounding-box size; long straight runs legitimately produce large envelopes

A.3.3 Summary statistics format

Every aggregate figure reported in the results chapter follows the conventions below, so that a number's meaning does not depend on where in the document it appears.

- Central tendency. Median with inter-quartile range, written as median (IQR). Means and standard deviations are not used for timing figures, for the right-skew reason stated in A.2.8.
- Counts. Reported as absolute count with its denominator, never as a bare percentage: "14 of 22 models (64%)", not "64%".

- Coverage. Every rate carries the population it is computed over. Corrosion findings are reported over elements scored, not over elements present, and the difference between the two is itself reported as coverage.
- Nulls. A metric that could not be computed is reported as such and excluded from the denominator, never coerced to zero.
- Instability. Any figure whose IQR exceeds 20% of its median carries an explicit unstable marker, per A.2.8.
- Precision. Timings to three significant figures; volumes to four; scores to two decimals, matching the precision the rule packs are authored at. No figure is reported to a precision its input does not support.

A.4 Interpretation Guide

This section states, in advance, what the results will and will not license as a conclusion. It is written before the numbers exist so that the standard of proof cannot move to meet them.

A.4.1 What the results mean for Blue Halo algorithm validation

Blue Halo generates a clearance envelope around each in-scope MEP element from a jurisdiction configuration, then reports elements intruding into it. Against synthetic fixtures it already passes 11 of 11 correctness checks, covering bounding-box expansion, intersection, clash detection, severity banding, angle-constraint enforcement, brace-variant handling, BCF structure and property-set serialisation. The 38-model run is not re-testing that arithmetic; it is testing four things the synthetic fixtures structurally cannot.

Table A.4.1 — What the run tests, and the criterion for each

Claim under test	Evidence that supports it	Evidence that refutes it
Geometry extraction survives third-party authoring	Halos generated for the large majority of MEP elements across models from eight repositories, two schemas and multiple authoring tools; halos_skipped attributable to identifiable causes (no placement, no shape representation)	Systematic skips concentrated in one authoring tool or one schema; envelope volumes implausible by orders of magnitude, indicating a unit-scaling fault
Unit handling is correct across projects	Envelope volumes on real models fall in a physically plausible band, consistent with the 13.5–21.5 m ³ synthetic mean and explicable by element size where they exceed it	Volumes off by factors of 1,000 or 1,000,000 — the signature of a metre/millimetre scaling error surviving into production models
Cost scales linearly at production element counts	Halo generation time linear in element count; per-halo cost in the 234–490 µs band established at 100–20,000 elements	Super-linear growth, or per-halo cost rising with model complexity rather than with level of detail
Output is coordination-ready, not merely produced	BCF archives satisfying all seven conditions of Table A.2.7 and opening in a compliant reader; topic count equal to finding count on every model	Archives that open but whose topics reference component GUIDs absent from the source model — findings a coordinator cannot act on

Two results are expected to be null, and are null by construction rather than by failure. The hospital-versus-standard clearance comparison will show no difference, because hospital_addition_mm is 0 mm in the configuration — a documented gap in the source standards, not a defect. Likewise, all four brace variants will produce identical envelopes, because neither EN 1998-1 nor DIN 4149 differentiates clearance by brace hardware. Reporting either null as a finding about the algorithm would be an error; reporting them as evidence that the standards data is incomplete is correct.

A.4.2 What the results mean for corrosion engine effectiveness

The five engines are being assessed on three distinct properties, which must not be conflated. A high finding count is not by itself evidence of effectiveness — an engine that flags everything is as useless as one that flags nothing.

Table A.4.2 — Effectiveness dimensions

Dimension	Question it answers	Measured by	What good looks like
Coverage	Can the engines obtain the inputs they need from real IFC files, as opposed to hand-built fixtures?	elements_scored ÷ element_count_mep, per engine, per material tier	High coverage on Full-tier models; low coverage on Partial-tier models accompanied by data-quality issues rather than silence

Dimension	Question it answers	Measured by	What good looks like
Discrimination	Do the engines separate benign from at-risk pairings, or do they return a flat result?	Band distribution per engine (Table A.3.1d); variance across the material heatmap (Figure A.2)	A spread of bands with an interpretable ordering; not all-Low and not all-Critical
Honesty	Does the system distinguish "assessed and benign" from "could not assess"?	data_quality_issues and unmapped_pairings counts	Unmapped pairings surface as explicit issues. Any unmapped pairing scored as compliant is a defect, not a pass

The dominant expected finding is a coverage result rather than a risk result. Most models in the corpus carry geometry and system type but sparse material properties; only the Full-tier files (notably Clinic_Plumbing.ifc with 378 pipe entities carrying real material text for copper, PVC, CPVC, steel and brass, and the West Riverside plumbing and sprinkler models) can exercise the engines end to end. A result showing high risk-band output on Full-tier models and high data-quality-issue output elsewhere is the expected and correct behaviour of a system that refuses to default silently.

Three constraints bound what any corrosion result can claim. First, the scores are a calibration authored for these rulesets, not digits quoted from the cited standards, so results rank relative risk and do not certify engineering adequacy. Second, water chemistry and flow velocity are absent from the element model, so mechanisms dominated by them — copper Type 2 pitting in soft water, copper erosion-corrosion above roughly 1.5 m/s — are systematically understated. Third, the galvanic series underpinning GC-001 and XM-001 was verified against the payload as last committed, and one internally inconsistent adjacent pair (ss316_active versus copper) remains open. Any finding involving that couple is provisional.

A.4.3 Expected outcome ranges

The ranges below are measured baselines from this codebase, not estimates. Blue Halo figures come from performance_benchmark.py at n = 7 repeats on a 4-core Linux host (Python 3.11.15, ifcopenshell 0.8.5, numpy 2.4.6, 15.7 GB RAM), reported as median with IQR. A real-model result inside the band is consistent with the baseline; outside it, the deviation is the finding and requires attribution to a stage.

Table A.4.3 — Measured baselines and expected ranges

Metric	Synthetic / benchmark baseline	Expected range on real models	Out-of-range means
Parse time, single model	1.52–1.55 s (IQR 0.09–0.20) at 100–1,000 elements	Scales with file size; 6.29 s observed for a 4-file federated set	Parse dominating total cost indicates a pathological STEP file, not an algorithm problem
Triangulation time	≈ 30 ms per element (30.10 s for 1,000 elements)	The dominant stage on every real model	Expected. Triangulation is upstream of the claim under test and should be reported separately, never folded into halo cost
Halo generation throughput	2,040–2,660 halos/s at LOD 300 from IFC; 3,910–4,274 halos/s synthetic	2,000–4,000 halos/s	Below 1,000 halos/s: investigate geometry complexity or unit resolution per element
Per-halo cost	376–403 μs at LOD 300; 19.9 μs at LOD 200; 490 μs at LOD 400	234–490 μs	Cost is driven by level of detail, not element count — a rise with element count indicates a super-linear term
Mean envelope volume	13.47–21.47 m³ (49.45 m³ federated, larger elements)	Order 10–100 m³	Off by 10³ or 10⁶: unit-scaling fault in the IFC path

Metric	Synthetic / benchmark baseline	Expected range on real models	Out-of-range means
Halo triangle amplification	0.21×–0.63× source triangles at LOD 300–400; 0.02× at LOD 200	0.2×–0.7×	Above 1×: envelopes more complex than the geometry they wrap, indicating a generation fault
Broad-phase confirmation rate	28–48% of candidate pairs confirmed by exact AABB test	25–50%	Near 100%: the spatial hash is not filtering. Near 0%: the hash cell size is mistuned for the model
Spatial-hash speed-up vs naïve $O(n^2)$	0.5×–0.9× below 2,000 halos; 3.2×–12.9× from 2,000 to 20,000	Speed-up only above ≈ 2,000 halos	Expected: on small models the hash costs more than it saves, and that is honest to report
Memory per halo	≈ 2.1 kB (2.13 MB per 1,000 halos); RSS delta ≈ 0 below 20,000 halos	1–5 kB per halo	Growth beyond this indicates retained geometry that should have been released
Blue Halo correctness checks	11 of 11 passing on synthetic fixtures	Per-model equivalents all passing	Any failure is a defect, not a range excursion
BCF archive structure	17 entries for 5 topics (3 files per topic plus bcf.version and project.bcfp); 9,905 bytes	$3n + 2$ entries for n topics	Any other entry count fails condition 3 of Table A.2.7
Corrosion coverage, Full-tier models	No real-model baseline exists; synthetic fixtures are 100% by construction	To be established by this run — the primary open question	This metric is the headline result, not a check

The final row is deliberately open. Corrosion-engine coverage against third-party IFC files has no prior measurement in this project; establishing it is the principal contribution of the batch run, and pre-registering it as unknown prevents a post-hoc figure being presented as a confirmed expectation.

A.4.4 Threats to validity

- Industrial category, $n = 1$. Only one model (craslabim.ifc) is a genuine industrial or laboratory building. Any statement about industrial-facility behaviour is indicative and must be reported as such; the remaining five rows in that category are robustness testers, not a sample.
- Sample is convenience, not random. The corpus is what the open web makes freely available, which over-represents demonstration and benchmark models authored to be exported cleanly. Real project deliverables are typically messier, so parse and extraction success rates measured here are an optimistic bound.
- Vendor concentration. A large share of the corpus originates from Revit and ArchiCAD exports. Export-tool-specific conventions may therefore appear as general behaviour.
- No ground-truth clash set. Except for the published duct-length ground truth on wbdg_office (row 29), no model carries an authoritative list of true clashes. Precision and recall against a reference clash set cannot be computed; the run measures behaviour, agreement across schema twins, and internal consistency, not accuracy against a gold standard.
- Standards data is incomplete. Thirteen gaps are declared in the configuration metadata, seven of which change how results read (Table A.2.5b). Degenerate results traceable to those gaps are evidence about the standards extraction, not about the algorithm.
- Corrosion calibration is unverified by a domain expert. The rule-pack digits are authored, not quoted. A corrosion engineer must verify each cell against the cited source and against project water chemistry before any pack drives a compliance verdict.

- Single-host timing. Performance figures come from one 4-core host. They establish scaling behaviour and relative stage cost, not absolute performance on other hardware.
- Database drift risk. The galvanic series is loaded from a Supabase-hosted asset. Verification was performed against the payload as last committed because the live asset was unreachable; if the asset has since been edited, the engines read a different table than the one verified.

A.5 Standards and Sources Cited in This Appendix

A.5.1 Standards

ASCE 7-22 §12.10.4.2 — Minimum Design Loads and Associated Criteria for Buildings and Other Structures.

ASTM B117 — Standard Practice for Operating Salt Spray (Fog) Apparatus.

ASTM G48 Method B — Pitting and Crevice Corrosion Resistance of Stainless Steels and Related Alloys by Use of Ferric Chloride Solution.

ASTM G-187 — Standard Test Method for Measurement of Soil Resistivity Using the Two-Electrode Soil Box Method (MIC assessment context).

BS 8539 — Code of Practice for the Selection and Installation of Post-Installed Anchors in Concrete and Masonry.

BS 8552:2012 — Sampling and Monitoring of Water from Building Services Closed Systems.

buildingSMART BCF 2.1 — BIM Collaboration Format specification.

CIBSE Guide G — Public Health and Plumbing Engineering.

CIBSE TM13:2013 — Minimising the Risk of Legionellosis.

CIRIA C692 — Stainless Steel in Construction.

DIN 4149:2022-03 §8.2.4 — Buildings in German Earthquake Zones.

EN 1993-1-4 — Eurocode 3: Design of Steel Structures, Part 1-4: Stainless Steels.

EN 1998-1:2020 §5.3.2.3 — Eurocode 8: Design of Structures for Earthquake Resistance.

EN 12952-12 — Water-Tube Boilers: Requirements for Boiler Feedwater and Boiler Water Quality.

EN ISO 9308-1 — Water Quality: Enumeration of *Escherichia coli* and Coliform Bacteria.

EN ISO 15329:2007 — Crevice Corrosion Testing, Wetting Classes T0–T5.

HSE HSG274 Parts 1–3 — Legionnaires' Disease: Technical Guidance.

IMO — Practical Guidelines for the Fabrication of Duplex Stainless Steel / Design Manual, 4th ed.

ISO 19650 — Organization and Digitization of Information about Buildings and Civil Engineering Works, Including BIM.

NASA-STD-6012 — Corrosion Prevention and Control Program Requirements.

NFPA 13-2021 §18.3.2.1 — Standard for the Installation of Sprinkler Systems.

WHO — Guidelines for Drinking-Water Quality, 4th ed.

A.5.2 Dataset repositories

buildingSMART Community — Community-Sample-Test-Files. GitHub. CC BY 4.0.

Generalitat Valenciana, Conselleria d'Habitatge — BIM manual example models. Public-sector open data.

Hellin, S. — ifc-bench dataset. Hugging Face. CC BY 3.0. Mirrors the OpenIFC Model Repository (University of Auckland; models donated by Wawan Solihin) and the WBDG/NIBS Common BIM Files.

KIT — IFC Examples, via IFC Wiki. Free with attribution.

RWTH Aachen E3D — DigitalHub. GitHub. MIT Licence.

STEP Tools — Sample IFC Data Files (NIST/AISC CIS/2 conversions).

TIB Hannover — DuraArk BuildingData collection. Open data.

Zenodo record 7948116 — craslabim.ifc, as-built laboratory model, FEUP. Open access.

A.5.3 Project artefacts referenced

BIMGUARD AI Validation Dataset — Verified Downloadable IFC Models (repository root). Full inventory with verified download URLs.

hermes_standards_research_summary.json — extracted parameters for EN 1998-1, DIN 4149, NFPA 13 and ASCE 7-22.

hermes_case_study_and_config.json — combined EN 1998-1 + DIN 4149 clearance configuration, including metadata.data_gaps.

data/rulesets/mm_001_material_media.json — MM-001 compatibility matrix, approved v1.0.

data/rulesets/xm_001_cross_material.json — XM-001 cross-material pack, approved v1.0.

validate_blue_halo.py and validate_blue_halo_output_REAL_CONFIG.txt — synthetic validation harness and its 11/11 output.

test_real_ifc_pipeline.py — end-to-end IFC pipeline test on the Phase 5 fixture.

performance_benchmark.py and docs/benchmarks/halo_benchmark_summary.md — measured scaling baselines.

docs/defects/defect_report_anode_convention.md — resolution of the galvanic series sign-convention defect.

Appendix B — Validation Results

Appendix A fixes the validation protocol before the data exists. This appendix reports what the run returned. Every value below is computed by **analyse_validation_results.py** from the sweep executed by **test_all_38_models.py** on 23 August 2026; none is transcribed by hand. Where a figure here differs from an expectation stated in Appendix A, the difference is the finding.

B.1 Dataset Overview

B.1.1 Run provenance

- Models attempted: **38**; processed: **37**; failed: **1**
- MEP elements: **49,736**, of which **49,736** (100.0%) resolved geometry
- Halo volumes generated: **49,736**
- Clashes detected: **223,516**
- Piping elements: **116,006**, of which **38,012** (32.8%) carry raw material text
- Of those, only **2,403** (2.07%) normalise to a CANONICAL_MATERIALS key the engines can score — see §B.1.3
- BCF 2.1 archives: **37 valid of 37 readable models** (every archive parsed, every topic folder complete, zero malformed XML). The remaining 1 of 38 is absent rather than invalid — a model that could not be read produces no archive.
- Clearance configuration: EN 1998-1:2020 + DIN 4149:2022, angle_fire brace variant, 200 mm

Failure. Row 35 — SGD_BODO_ifc.zip (ARC+PLB+VENT) returned SchemaError: Unsupported schema: IFC2X2_FINAL. IfcOpenShell does not support IFC2X2_FINAL, so the model produced neither geometry nor a BCF archive.

B.1.2 Headline finding

The spatial engine and the corrosion engines behave in opposite ways against third-party models, and the corpus separates them cleanly.

Blue Halo generalises. Geometry resolved for 49,736 of 49,736 MEP elements (100.0%) across IFC2x3 and IFC4 alike, producing 49,736 halo volumes and 223,516 clashes, with all 37 BCF archives valid. The same-building schema twins (Table B.7b) differ by 0.11% and 0.00%, so the result does not depend on the export schema.

The corrosion engines do not. Table B.4 shows every engine either flags nothing (GC-001: 0%) or flags everything (CC-001 and MC-001: 100%), and Table B.3 shows why: across six distinct materials the mean band is constant — GC-001 returns Low for every material, CC-001 Medium for every material, and MC-001 Critical for all but one. These are constant functions of their input, not risk assessments.

The cause is input availability, not engine logic. The best-covered engine is MM-001, whose required inputs are present on 15 of 116,006 elements (0.013%); the rest are at zero. A 100% flag rate and 0% input coverage are the same fact viewed twice: the coercers substitute a default material when none is present, so the engines score the default, uniformly, everywhere.

This is precisely the question Chapter 13 puts to this dataset — whether the inputs the engines need exist in real federated models at all. On this corpus the answer is no, and §15.2 carries the consequence.

B.1.3 Material text is not material data

38,012 elements (32.8%) carry some material string, but only 2,403 (2.07%) normalise to a CANONICAL_MATERIALS key. The gap — 35,609 elements — is material text that piping_producer.normalise_material cannot map, so the engines still receive “Unknown”. Reporting the former as coverage overstates usable material data by roughly 16×, and is the single easiest way to misread this corpus.

B.1.4 Per-model summary

One row per model in compilation order. The working list skips rows 17 and 40; every row listed was attempted. Row 35 is the unreadable model and carries zeros throughout. All nineteen measured fields are retained; they are split across two tables sharing the row key, because nineteen columns do not fit the page.

Table B.1a — Model inventory and parse result (38 models attempted, 37 processed)

R o w	Catego ry	Model	Sche ma	MB	MEP	MEP geom	Struct	Struct geom	Statu s
1	hospital	Clinic_Architectural.ifc	IFC2X3	13.0	0	0	1617	1617	ok
2	hospital	Clinic_Structural.ifc	IFC2X3	19.06	0	0	1068	1068	ok
3	hospital	Clinic_HVAC.ifc	IFC2X3	26.91	1548	1548	0	0	ok
4	hospital	Clinic_Electrical.ifc	IFC2X3	6.8	0	0	0	0	ok
5	hospital	Clinic_Plumbing.ifc	IFC2X3	55.83	2893	2893	0	0	ok
6	hospital	west_riverside_hospital — Architecture (IFC4)	IFC4	80.94	0	0	8935	8935	ok
7	hospital	west_riverside_hospital — Structural (IFC4)	IFC4	6.48	0	0	2822	2822	ok
8	hospital	west_riverside_hospital — Mechanical/HVAC (I	IFC4	73.05	8732	8732	0	0	ok
9	hospital	west_riverside_hospital — Plumbing (IFC4)	IFC4	23.76	4308	4308	0	0	ok
10	hospital	west_riverside_hospital — Electrical (IFC4)	IFC4	4.44	84	84	0	0	ok
11	hospital	west_riverside_hospital — Fire Alarm (IFC4)	IFC4	0.91	0	0	0	0	ok
12	hospital	west_riverside_hospital — Sprinklers (IFC4)	IFC4	33.99	6228	6228	0	0	ok
13	hospital	west_riverside_hospital — Mechanical (IFC2X3	IFC2X3	78.76	8732	8732	0	0	ok
14	hospital	west_riverside_hospital — Plumbing (IFC2X3 t	IFC2X3	25.0	4308	4308	0	0	ok
15	hospital	NBU_MedicalClinic (zip: ARC+MEP+HVAC+ELE+STR	IFC2X3	207.34	5952	5952	0	0	ok
16	hospital	GVA Sanitario.ifc	IFC2X3	6.96	0	0	664	664	ok
18	office	Duplex_A_20110907.ifc	IFC2X3	2.38	0	0	97	97	ok
19	office	Duplex_Electrical_20121207.ifc	IFC2X3	1.6	0	0	0	0	ok
20	office	Duplex_MEP_20110907.ifc	IFC2X3	17.87	427	427	0	0	ok

R o w	Catego ry	Model	Sche ma	MB	MEP	MEP geom	Struct	Struct geom	Statu s
21	office	Duplex_Plumbing_20121113.ifc	IFC2X3	31.56	231	231	0	0	ok
22	office	AC20-Institute-Var-2.ifc	IFC4	10.93	0	0	149	149	ok
23	office	AC20-FZK-Haus.ifc	IFC4	2.57	0	0	63	63	ok
24	office	DigitalHub_FM-ARC_v2.ifc	IFC4	9.02	0	0	357	357	ok
25	office	DigitalHub_FM-HZG_v2.ifc	IFC4	20.89	914	914	0	0	ok
26	office	DigitalHub_FM-LFT_v2.ifc	IFC4	12.74	595	595	0	0	ok
27	office	DigitalHub_FM-SAN_v2.ifc	IFC4	25.18	504	504	0	0	ok
28	office	wbdg_office — Architecture	IFC2X3	4.1	0	0	497	497	ok
29	office	wbdg_office — MEP	IFC2X3	41.89	1959	1959	0	0	ok
30	office	wbdg_office — Structural	IFC2X3	11.07	0	0	495	495	ok
31	office	Molio_with_URLs.ifc	IFC2X3	73.95	0	0	267	160	ok
32	office	GVA Administrativo.ifc	IFC2X3	7.21	0	0	338	338	ok
33	office	IFC Schependomlaan.ifc	IFC2X3	49.29	60	60	1419	1364	ok
34	industri al	craslabim.ifc	IFC2X3	67.55	0	0	76	76	ok
35	industri al	SGD_BODO_ifc.zip (ARC+PLB+VENT)	-	0	0	0	0	0	error
36	industri al	Academic_Autodesk_ifc.zip (Basic + Advanced)	IFC2X3	38.07	0	0	3649	3649	ok
37	industri al	Autodesk_210-King_ifc.zip	IFC2X3	155.15	2261	2261	4283	4283	ok
38	industri al	aisc_sculpture_brep.ifc	IFC2X3	0.55	0	0	87	87	ok
39	industri al	aisc_sculpture_param.ifc	IFC2X3	0.32	0	0	87	87	ok

Model names longer than 44 characters are truncated by the analysis script; the full names are held in the run manifest. Source: docs/validation/table1_per_model.csv.

Table B.1b — Measured output per model: halos, clashes by severity, material coverage and BCF verdict

R o w	Model	Halos	Clashe s	Minor	Major	Critica l	Piping	With material	BCF	Sec
1	Clinic_Architectural.ifc	0	0	0	0	0	102	102	VALID	5.0
2	Clinic_Structural.ifc	0	0	0	0	0	0	0	VALID	22.7
3	Clinic_HVAC.ifc	1548	3070	2760	308	2	3704	502	VALID	50.6
4	Clinic_Electrical.ifc	0	0	0	0	0	2089	2089	VALID	26.3
5	Clinic_Plumbing.ifc	2893	15534	15479	55	0	6587	6587	VALID	116.7

R o w	Model	Halos	Clashe s	Minor	Major	Critica l	Piping	With material	BCF	Sec
6	west_riverside_hospital — Architecture (IFC4	0	0	0	0	0	0	0	VALI D	60.8
7	west_riverside_hospital — Structural (IFC4)	0	0	0	0	0	0	0	VALI D	25.6
8	west_riverside_hospital — Mechanical/HVAC (I	8732	41128	38705	1978	445	17424	0	VALI D	265.1
9	west_riverside_hospital — Plumbing (IFC4)	4308	21800	21702	98	0	8539	0	VALI D	104.8
10	west_riverside_hospital — Electrical (IFC4)	84	178	142	36	0	1673	121	VALI D	13.9
11	west_riverside_hospital — Fire Alarm (IFC4)	0	0	0	0	0	861	1	VALI D	4.3
12	west_riverside_hospital — Sprinklers (IFC4)	6228	17538	17494	44	0	13490	1302	VALI D	124.0
13	west_riverside_hospital — Mechanical (IFC2X3	8732	41174	38751	1978	445	18488	0	VALI D	249.5
14	west_riverside_hospital — Plumbing (IFC2X3 t	4308	21800	21702	98	0	9013	334	VALI D	109.7
15	NBU_MedicalClinic (zip: ARC+MEP+HVAC+ELE+ST R	5952	27472	26595	846	31	16012	16008	VALI D	632.6
16	GVA Sanitario.ifc	0	0	0	0	0	0	0	VALI D	5.9
18	Duplex_A_20110907.ifc	0	0	0	0	0	0	0	VALI D	2.0
19	Duplex_Electrical_20121207.ifc	0	0	0	0	0	99	99	VALI D	2.1
20	Duplex_MEP_20110907.ifc	427	2590	2590	0	0	926	926	VALI D	22.8
21	Duplex_Plumbing_20121113.ifc	231	1528	1528	0	0	498	498	VALI D	20.2
22	AC20-Institute-Var-2.ifc	0	0	0	0	0	0	0	VALI D	10.1
23	AC20-FZK-Haus.ifc	0	0	0	0	0	0	0	VALI D	6.0
24	DigitalHub_FM-ARC_v2.ifc	0	0	0	0	0	0	0	VALI D	12.1
25	DigitalHub_FM-HZG_v2.ifc	914	8000	8000	0	0	1795	881	VALI D	49.3
26	DigitalHub_FM-LFT_v2.ifc	595	906	797	109	0	1310	1310	VALI D	22.9
27	DigitalHub_FM-SAN_v2.ifc	504	3784	3756	28	0	1010	506	VALI D	31.1
28	wbdg_office — Architecture	0	0	0	0	0	31	31	VALI D	4.8
29	wbdg_office — MEP	1959	8256	8047	202	7	5697	5697	VALI D	162.1
30	wbdg_office — Structural	0	0	0	0	0	0	0	VALI D	24.8
31	Molio_with_URIs.ifc	0	0	0	0	0	48	0	VALI D	28.6
32	GVA Administrativo.ifc	0	0	0	0	0	0	0	VALI D	5.0

R o w	Model	Halos	Clashes	Minor	Major	Critical	Piping	With material	BCF	Sec
33	IFC Schependomlaan.ifc	60	407	279	124	4	73	73	VALID	25.5
34	craslabim.ifc	0	0	0	0	0	0	0	VALID	27.1
35	SGD_BODO_ifc.zip (ARC+PLB+VENT)	0	0	0	0	0	0	0	missing	0
36	Academic_Autodesk_ifc.zip (Basic + Advanced)	0	0	0	0	0	41	41	VALID	31.6
37	Autodesk_210-King_ifc.zip	2261	8351	3254	795	4302	6496	904	VALID	1286.9
38	aisc_sculpture_brep.ifc	0	0	0	0	0	0	0	VALID	0.7
39	aisc_sculpture_param.ifc	0	0	0	0	0	0	0	VALID	0.5

Row keys match Table B.1a. Source: docs/validation/table1_per_model.csv.

B.2 Clash Analysis

B.2.1 Severity distribution

Severity is the overlap volume as a fraction of the halo volume: $\geq 25\%$ critical, $\geq 5\%$ major, below that minor. The distribution is dominated by minor overlaps in every category except industrial, where a single dense model (row 37, Autodesk 210-King) inverts the profile.

Table B.2 — Clash severity distribution by building category

Scope	Minor	Major	Critical	Total	Minor %	Major %	Critical %
ALL	211581	6699	5236	223516	94.66	3.00	2.34
hospital	183330	5441	923	189694	96.65	2.87	0.49
industrial	3254	795	4302	8351	38.97	9.52	51.51
office	24997	463	11	25471	98.14	1.82	0.04

Source: docs/validation/table2_severity.csv.

B.2.2 Material and corrosion engine cross-tabulation

Band counts are given as Low/Medium/High/Critical. The mean columns encode the band as 0–3. A constant mean down a column is the result: the engine returns the same band irrespective of material.

Table B.3 — Material × corrosion engine band distribution

Material	Elements	GC-001 L/M/H/C	CC-001 L/M/H/C	MC-001 L/M/H/C	GC mean	CC mean	MC mean
Unknown	113603	113603/0/0/0	0/113603/0/0	0/0/0/113603	0.00	1.00	3.00
Copper_C12200	1835	1835/0/0/0	0/1835/0/0	0/0/1835/0	0.00	1.00	2.00
Aluminium	455	455/0/0/0	0/455/0/0	0/0/0/455	0.00	1.00	3.00
CarbonSteel	96	96/0/0/0	0/96/0/0	0/0/0/96	0.00	1.00	3.00
GalvanisedSteel	15	15/0/0/0	0/15/0/0	0/0/0/15	0.00	1.00	3.00
Brass_C46400	2	2/0/0/0	0/2/0/0	0/0/0/2	0.00	1.00	3.00

Source: docs/validation/table3_material_engine.csv.

B.2.3 Engine band distribution and input coverage

The coverage column counts piping elements carrying every input the engine requires, as a percentage of the 116,006-element recompute subset. It is the controlling number in this appendix: no engine reaches 0.02%.

Table B.4 — Engine band distribution and input coverage

Engine	Status	Bands seen	Flagged	Flag %	Required inputs	Elements with all inputs	Coverage %
GC-001	ok=37	Low:116006	0	0.0	material+dissimilar_neighbour	8	0.0
CC-001	ok=37	Medium:116006	116006	100.0	material+joint_type+operating_temperature_c	0	0.0
MC-001	ok=37	High:1835/Critical:114171	116006	100.0	material+media+nominal_diameter_mm+operating_temperature_c	0	0.0
MM-001	ok=13 unavailable=24	-	0	0.0	material+media+environment_class	15	0.0
XM-001	ok=13 unavailable=24	-	0	0.0	material+dissimilar_neighbour+environment_classes	0	0.0

MM-001 and XM-001 report unavailable on 24 of the 37 processed models because `load_rule_pack()` rejects the approved v1.0 packs: the shipped comparators validate against an earlier and simpler schema, so MM-001 raises `ValueError` on the missing keys `compatibility_matrix`, `media`, `environments` and `materials`, and XM-001 on `couples`, `materials` and `environment_thresholds`. The engines therefore never execute, and the zero findings they report are an artefact of that failure rather than a measurement of the corpus. This is the specification-against-implementation gap recorded term by term in §13.9 and Table 13.13. Source: docs/validation/table4_engine_coverage.csv.

B.2.4 BCF 2.1 export validity

An archive counts as valid only if it opens as a zip, every topic folder is complete, and no markup file fails to parse. Byte size is reported but is not a validity condition: an archive with zero topics is a legitimate result for a model with no MEP elements.

Table B.5 — BCF 2.1 export validity, per model

Row	Model	Bytes	Entries	Topics	Complete topics	XML ok	XML bad	Root files	Verdict
1	Clinic_Architectural.ifc	598	2	0	0	2	0	yes	VALID
2	Clinic_Structural.ifc	599	2	0	0	2	0	yes	VALID
3	Clinic_HVAC.ifc	5806134	9212	3070	3070	6142	0	yes	VALID
4	Clinic_Electrical.ifc	597	2	0	0	2	0	yes	VALID
5	Clinic_Plumbing.ifc	29421384	46604	15534	15534	31070	0	yes	VALID
6	west_riverside_hospital — Architectu	616	2	0	0	2	0	yes	VALID
7	west_riverside_hospital — Structural	616	2	0	0	2	0	yes	VALID
8	west_riverside_hospital — Mechanical	77806873	123386	41128	41128	82258	0	yes	VALID
9	west_riverside_hospital — Plumbing (	41143663	65402	21800	21800	43602	0	yes	VALID
10	west_riverside_hospital — Electrical	339624	536	178	178	358	0	yes	VALID
11	west_riverside_hospital — Fire Alarm	615	2	0	0	2	0	yes	VALID

Row	Model	Bytes	Entries	Topics	Complete topics	XML ok	XML bad	Root files	Verdict
12	west_riverside_hospital — Sprinklers	33085232	52616	17538	17538	35078	0	yes	VALID
13	west_riverside_hospital — Mechanical	77850059	123524	41174	41174	82350	0	yes	VALID
14	west_riverside_hospital — Plumbing (	41128420	65402	21800	21800	43602	0	yes	VALID
15	NBU_MedicalClinic (zip: ARC+MEP+HVAC	52259784	82418	27472	27472	54946	0	yes	VALID
16	GVA Sanitario.ifc	594	2	0	0	2	0	yes	VALID
18	Duplex_A_20110907.ifc	601	2	0	0	2	0	yes	VALID
19	Duplex_Electrical_20121207.ifc	605	2	0	0	2	0	yes	VALID
20	Duplex_MEP_20110907.ifc	4923726	7772	2590	2590	5182	0	yes	VALID
21	Duplex_Plumbing_20121113.ifc	2885965	4586	1528	1528	3058	0	yes	VALID
22	AC20-Institute-Var-2.ifc	600	2	0	0	2	0	yes	VALID
23	AC20-FZK-Haus.ifc	597	2	0	0	2	0	yes	VALID
24	DigitalHub_FM-ARC_v2.ifc	605	2	0	0	2	0	yes	VALID
25	DigitalHub_FM-HZG_v2.ifc	15153503	24002	8000	8000	16002	0	yes	VALID
26	DigitalHub_FM-LFT_v2.ifc	1713622	2720	906	906	1814	0	yes	VALID
27	DigitalHub_FM-SAN_v2.ifc	7148456	11354	3784	3784	7570	0	yes	VALID
28	wbdg_office — Architecture	600	2	0	0	2	0	yes	VALID
29	wbdg_office — MEP	15701208	24770	8256	8256	16514	0	yes	VALID
30	wbdg_office — Structural	602	2	0	0	2	0	yes	VALID
31	Molio_with_URIs.ifc	597	2	0	0	2	0	yes	VALID
32	GVA Administrativo.ifc	597	2	0	0	2	0	yes	VALID
33	IFC Schependomlaan.ifc	776088	1223	407	407	816	0	yes	VALID
34	craslabim.ifc	590	2	0	0	2	0	yes	VALID
35	SGD_BODO_ifc.zip (ARC+PLB+VENT)	0	0	0	0	0	0	no	missing
36	Academic_Autodesk_ifc.zip (Basic + A	617	2	0	0	2	0	yes	VALID
37	Autodesk_210-King_ifc.zip	15935704	25055	8351	8351	16704	0	yes	VALID
38	aisc_sculpture_brep.ifc	602	2	0	0	2	0	yes	VALID
39	aisc_sculpture_param.ifc	603	2	0	0	2	0	yes	VALID

Source: docs/validation/table5_bcf_validity.csv.

B.2.5 Standards sensitivity

The same models re-run at eight clearance values. Only two rows correspond to a real jurisdiction; the rest are there to show the shape of the curve. Clash load scales roughly linearly with clearance while critical findings fall, because a larger halo raises the denominator of the overlap fraction.

Table B.6 — Standards sensitivity: clash load against clearance threshold

Clearance (mm)	Jurisdiction	Clashes	Minor	Major	Critical	vs 200 mm baseline
25.0	—	30332	28967	1217	148	0.29x
50.0	—	48952	46975	1903	74	0.47x

Clearance (mm)	Jurisdiction	Clashes	Minor	Major	Critical	vs 200 mm baseline
100.0	—	69307	67658	1611	38	0.67x
150.0	—	85034	83838	1162	34	0.82x
200.0	EN 1998-1 + DIN 4149	104144	102972	1141	31	1.00x
300.0	—	133739	132665	1048	26	1.28x
457.2	ASCE 7-22 + NFPA 13	172812	171868	928	16	1.66x
600.0	—	212870	211982	877	11	2.04x

The sweep covers rows 5, 9, 12, 14 and 15 only, not the whole corpus. Its absolute totals are therefore not comparable with Table B.2; only the ratios between thresholds should be read from it. Source: docs/validation/table6_standards_sensitivity.csv.

B.2.6 IFC schema comparison

Geometry resolution is 100% on both schemas. The material column is the interesting one: the IFC2x3 files in this corpus carry raw material text on 48.5% of piping elements against 8.9% for IFC4 — a property of what these particular exporters wrote, not of the schemas themselves.

Table B.7 — IFC2x3 against IFC4 across the corpus

Schema	Models	MEP	MEP geometry %	Clashes	Piping	Material %
IFC2X3	24	28371	100.0	130182	69904	48.5
IFC4	13	21365	100.0	93334	46102	8.9

Source: docs/validation/table7_schema.csv.

The corpus-wide comparison confounds schema with model. The two schema twins below remove that confound: the same building, exported twice.

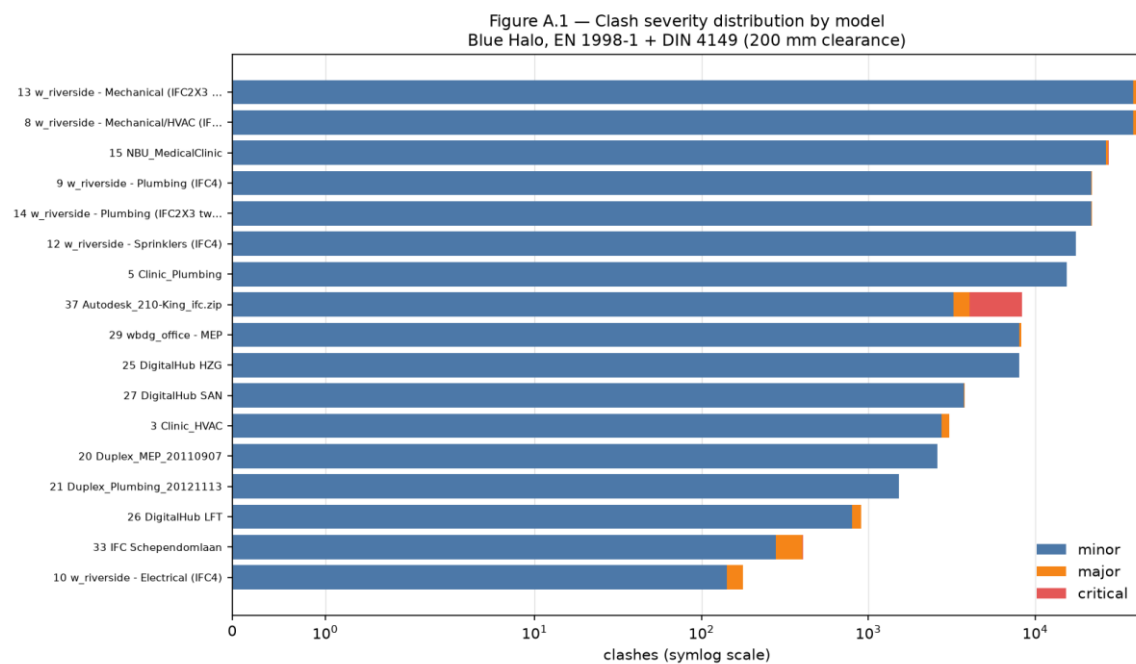
Table B.7b — Schema twins, the same building exported to both schemas

Pair	Building	MEP 2x3	MEP IFC4	Clashes 2x3	Clashes IFC4	Δ clashes	Δ %
8/13	west_riverside mechanical	8732	8732	41174	41128	46	0.11
9/14	west_riverside plumbing	4308	4308	21800	21800	0	0.00

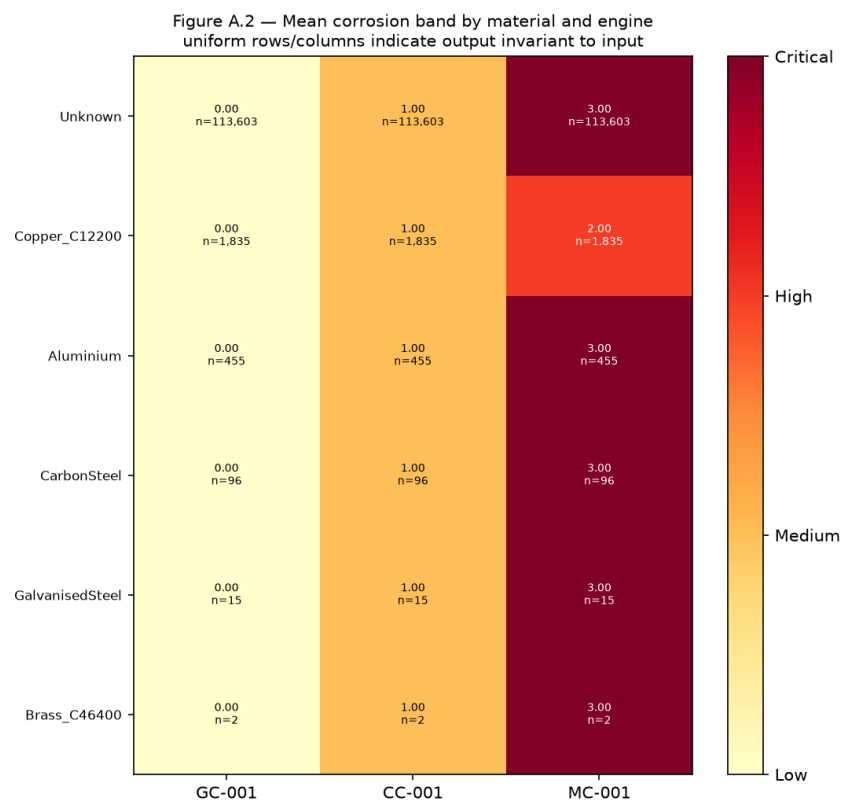
Source: docs/validation/table7b_schema_twins.csv.

B.3 Figures

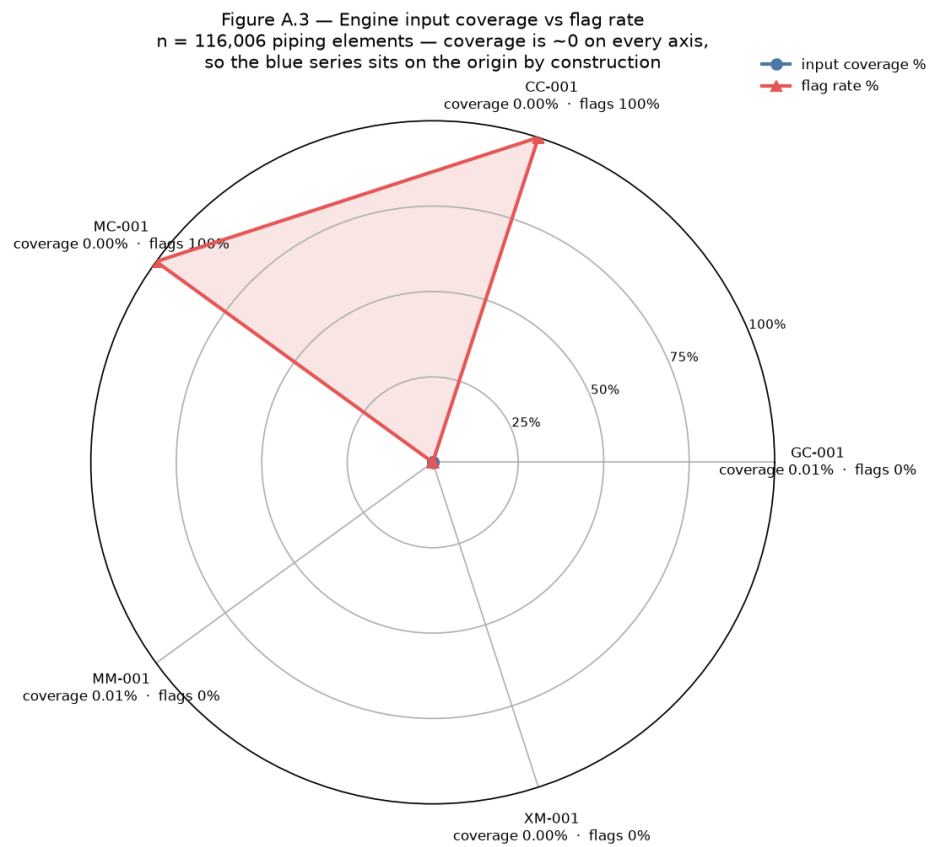
All four figures are rendered by analyse_validation_results.py from the same run data as the tables above.

Figure B.1 — Clash severity distribution by building category

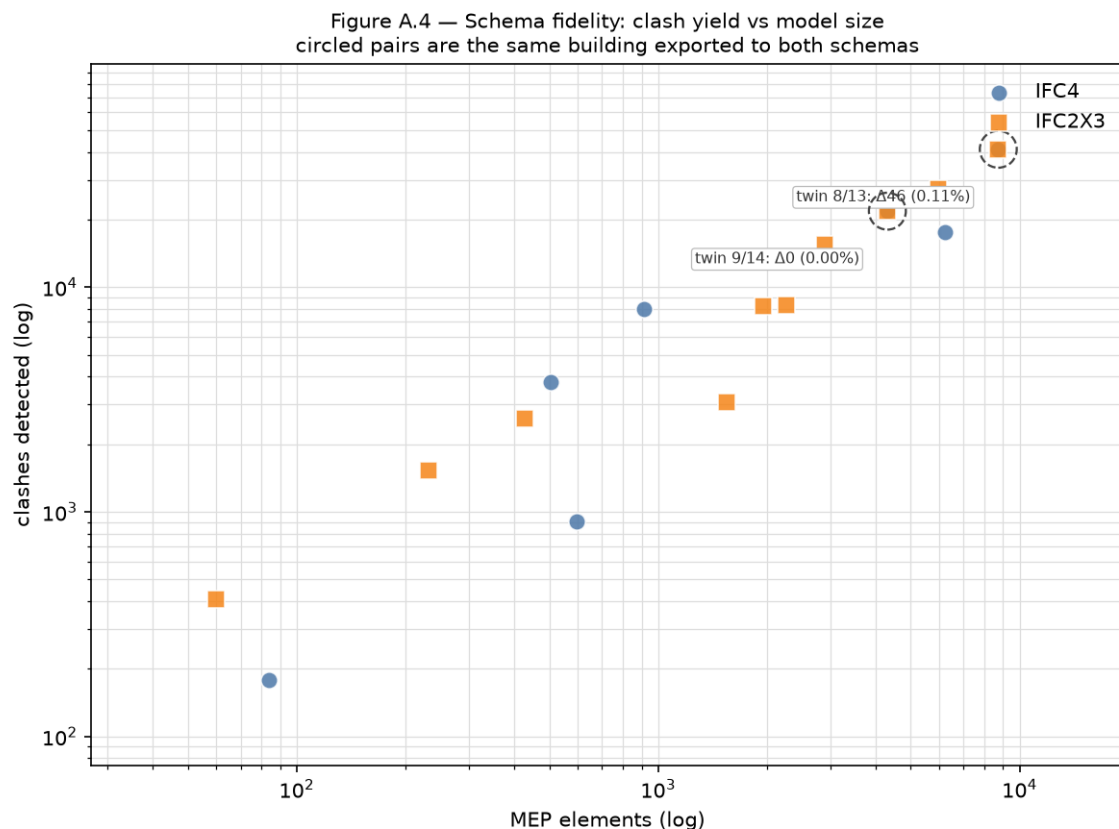
Minor overlaps dominate every category except industrial, where row 37 alone contributes 4,302 of the corpus's 5,236 critical findings.

Figure B.2 — Material risk heatmap across the three scoring engines

Each row is a material, each column an engine. Uniform colour down a column is the finding: the band does not vary with the material.

Figure B.3 — Engine flag rate against input coverage

CC-001 and MC-001 sit at 100% flagged and 0% covered — the signature of a coerced default being scored rather than a measured property.

Figure B.4 — Schema fidelity: geometry and material availability by model

Geometry resolution is invariant across schemas; material availability is not, and the spread within each schema is wider than the gap between them.

B.4 Threats to validity

- **Clash counts are geometric, not engineering judgements.** A clash is an AABB intersection between a halo volume and another element. Bounding boxes overstate intersection for diagonal or non-convex members, so the counts are an upper bound.
- **Severity banding is a spatial heuristic, not a standards-derived threshold.** The bands are overlap fractions of the halo volume ($\geq 25\%$ critical, $\geq 5\%$ major). Table B.6 shows how sensitive the totals are to the clearance input.
- **One brace variant was evaluated.** The combined EN + DIN configuration assigns every variant the same clearance — a declared data gap in that configuration — so variant choice does not change these numbers, but a jurisdiction that differentiates would.
- **Corrosion coverage is measured on the recompute subset.** 116,006 piping elements across 24 models, not the full corpus.
- **The clearance sensitivity sweep covers five models.** Rows 5, 9, 12, 14 and 15 only. Its absolute totals are not comparable with Table B.2; read the ratios, not the counts.
- **Row 35 could not be read at all.** IFC2X2_FINAL is unsupported by IfcOpenShell, so the industrial category is one model short of an already-thin count.

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